Meeting of the Board
4 – 7 October 2021
Virtual meeting
Provisional agenda item 15

GCF/B.30/02/Add.07

14 September 2021

Consideration of funding proposals - Addendum VII

Funding proposal package for FP175

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Enhancing community resilience and water security in the Upper Athi River Catchment Area, Kenya";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat's assessment;
e) Independent Technical Advisory Panel's assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

Table of Contents

Funding Proposal

Project/Programme title:	**Enhancing community resilience and water security in the Upper Athi River Catchment Area, Kenya**
Country(ies):	Kenya
Accredited Entity:	National Environment Management Authority.
Date of first submission:	*[2016/10/12]*
Date of current submission	*[2021/07/21]*
Version number	*[V.4]*

List of Acronyms

ARCA	Athi River Catchment Area
ACA	Athi Catchment Area
NEMA	National Environment Management Authority
KMD	Kenya Meteorological Department
WRA	Water Resources Authority
WRUA	Water Resource Users Authority
AWS	Automatic Weather Stations
AWL	Automatic Water Level Station
MAM	March April May
OND	October November December
ITCZ	Intertropical Convergence Zone
ENSO	El Niño Southern Oscillation
ASAL	Arid and Semi-Arid Land
MoWS	Ministry of Water and Sanitation
UoN	University of Nairobi
KFS	Kenya Forest Service
SOE	State of the Environment
CEAP	County Environment Action Plan
SCMP	Sub Catchment Management Plans

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Contents

NOTE TO ACCREDITED ENTITIES ON THE USE OF THE FUNDING PROPOSAL TEMPLATE

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and Programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:
fundingproposal@gcfund.org
Please use the following name convention for the file name:
“FP-[NEMA]-[Country/Region - Kenya]- [2020/02/14]”

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Project	**A.2. Public or private sector**	Public
A.3. Request for Proposals (RFP)	Not applicable Not applicable Not applicable Not applicable		
A.4. Result area(s)	*Check the applicable GCF result area(s) that the overall proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of GCF budget devoted to it. The total of the percentages when summed should be 100%.*		
	Mitigation: Reduced emissions from: ☐ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☐ Forestry and land use: Adaptation: Increased resilience of: ☒ Most vulnerable people, communities and regions: ☒ Health and well-being, and food and water security: ☒ Infrastructure and built environment: ☐ Ecosystem and ecosystem services:		GCF contribution: Enter number% Enter number% Enter number% Enter number% 30% 35% 35% Enter number%
A.5. Expected mitigation impact	*Indicate t CO2eq over lifespan*	**A.6. Expected adaptation impact**	**1,156,620 direct beneficiaries** **3,693,380 indirect beneficiaries** 10.32% *of population*
A.7. Total financing (GCF + co-finance)	**9,999,983.26** USD	**A.9. Project size**	Micro (Upto USD 10 million)
A.8. Total GCF funding requested	______ Choose an item. *For multi-country proposals, please fill out annex 17.* 9,526,603.26 USD		
A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant **9,526,603.26** ☐ Equity Enter number ☐ Loan Enter number ☐ Results-based payment Enter number ☐ Guarantee Enter number		
A.11. Implémentation period	*4 years*	**A.12. Total lifespan**	30 years

A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ Programme, if available.* ***3/20/2020***	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* B
A.15. Has this FP been submitted as a CN before?	Yes ☐ No ☒	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☐ No ☒
A.17. Is this FP included in the entity work Programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country Programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	Does the project/Programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1. Yes ☒ No ☐		
A.20. Executing Entity information	*If not the Accredited Entity, please indicate the full legal name of the Executing Entity(ies) and provide its country of registration and ownership type. Note that there can be more than one Executing Entity. Also indicate if an Executing Entity is the National Designated Authority. Refer to the definition of Executing Entity in the Accreditation Master Agreement.* *The Executing Entities are as follows:* 1. *National Environment Management Authority (NEMA) – also the Accredited Entity* 2. Water Resources Authority – WRA 3. Government of Kenya represented by Kenya Meteorological Department - KMD of the Ministry of Environment and Forestry. The Executing Entities are all government agencies / institutions in the Republic of Kenya.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

A

1. Kenya is highly vulnerable to erratic climatic patterns and limited water availability due to its reliance on agriculture, tourism, hydro-energy, among other activities, that are susceptible to rainfall and water availability. **Kenya has experienced frequent floods and droughts resulting from climate change impacts, leading to national economic losses.** The 2008-2011 droughts are estimated to have slowed down the GDP by an average of 2.8 % per annum, with total damage and losses estimated at USD 12.1 billion. Additionally, climate change has adversely affected water availability especially during frequent drought conditions.

2. The effects of climate change and related disasters have the potential to adversely impact many Kenyans given that about 75% of the population depends directly on land and natural resources for their livelihoods. In recent years, there has been increased attention to climate change due to its impacts on the lives of Kenyans. This has been mainly due to an increase in the intensity and frequency of extreme climatic events such as severe droughts and flooding. These extreme events have had negative socio-economic impacts on almost all sectors of Kenyan society such as health, agriculture, livestock, environment, hydropower generation, and tourism. The seriousness of the problem has made it imperative for policy makers to focus on mainstreaming climate change in development policies and strategies[1]. Another important factor in determining Kenya's vulnerability to climate change is its social and economic inequalities. Forty-two percent of its population of 54 million lives below the poverty line. Access to basic facilities such as health care, education, clean water, and sanitation, is often a luxury for many people. Large segments of the population, including the burgeoning urban poor, are highly vulnerable to climatic, economic, and social shocks[2]

3. Climate change affects the global hydrological cycle and it has impacts on the quantity and quality of the water. Water security is a key issue given that Kenya´s people and economy are highly vulnerable to erratic patterns and limited water availability. In the past two decades, from 1992–2012, Kenya was ranked first among African countries in terms of people affected by droughts (roughly 46 million people) and fifth in terms of those affected by floods (about 2.8 million people) during the same period. Kenya has limited freshwater endowments and is classified as a chronically “water-scarce” country in absolute and relative terms. It faces the additional challenge of high inter-annual and intra-annual rainfall variability.

4. Along the Athi River Catchment Area, there are 12 counties. Historically, drought and flooding risks were the highest in Kiambu, Machakos, Nairobi, and Nyandarua (main beneficiaries), in that order. Droughts and floods are likely to rise in the future, especially in Kiambu and Machakos County. Consequently, the water infrastructure of the study area has suffered damage or has not had enough capacity to withstand these droughts and floods; for example, the water storage or potable water supply infrastructure have been insufficient. Irregularity of water supply due to climate change has threatened the sustainability of those counties and has also caused waterborne diseases. Impacts on the quantity and quality of water are also expected due to changes in temperature, precipitation, floods and droughts in the area.

5. With climate extremes expected to increase, climate-informed water management, climate-resilient water infrastructure will be critical in order to prepare for and respond to floods and droughts. The project therefore seeks to enhance **community resilience and water security in the Upper Athi River Catchment Area** through interventions across three Outputs:
   - Output 1: enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector.
   - Output 2: improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment.
   - Output 3: Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions.

6. The project would directly benefit 1,156,620 people in the project site are (Kiambu, Machakos, Nairobi, and Nyandarua), with indirect benefit reaching **3,693,380** people in/from 4 counties within the catchment.

7. The Project supports the implementation and operationalization of several key national policies strategies and plan, including the National Climate Change Response Strategy (NCCRS), National Climate Change Action Plan (NCCAP), National Adaptation Plan (NAP), the Nationally Determined Contribution (INDC) and the National Water Master Plan (NWMP) 2030.

8. Capacity building of national level institutions will be an upscale of the ongoing Kenya's approved GCF NAP Readiness that has focused on training county government officials on climate change and climate finance. The proposal plans to support capacity building and training for relevant water governance structures and communities

within the project (Water Resource Users Associations -WRUAs, County Environment Committees - CECs, County Adaptation Committees, Community Based Organizations -CBOs. The project area refers to the four targets counties Nairobi, Nyandarua, Kiambu and Machakos. Materials developed by GCF NAP Readiness will be utilized during mentioned trainings and the already trained personnel will play key roles in the training.

[1] Kenya National Human Development Report (2013) http://hdr.undp.org/en/content/climate-change-and-human-development
[2] https://tradingeconomics.com/kenya/gdp-growth-annual

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Kenya´s vulnerability to climate change

9. **Kenya was ranked 7th, on the 2018 Global Log Term Climate Risk Index published by Germanwatch and it is expected to be severely impacted by the negative effects of climate change in the future[3].** Much of the country`s vulnerability is linked to seasonal rains. Between March and July 2018, Kenya experienced almost twice the normal rainfall of the wet season. Kenya's most important rivers in the central highlands overflowed affecting 40 out of 47 counties, causing the death of 183 people, injury of 97 and the displacement of 321 630 people, as well as the loss of livelihoods and livestock.

10. Kenya located in the Africa Region, 581,309 km^2, 80% of which is arid and semi-arid, and its population is approximately 54 million. Kenya has a complex climate that varies significantly between its coastal, interior and highlands regions and from season to season, year to year, and decade to decade. This climatic variability is influenced by naturally occurring factors such as movement of the Intertropical Convergence Zone (ITCZ) and the El Niño Southern Oscillation (ENSO). The climate impacts of El Niño (and its counterpart, La Niña) are not uniform across the African continent, making the impacts of El Niño or La Niña on rainfall in Africa vary according to location and season. Additionally, Northeastern Africa generally becomes anomalously dry during its primary and longer rainy season of June – September. **Since 1960, Kenya's mean annual temperature has increased by 1.0°C, at an average rate of 0.21°C per decade. The rate of increase has been most rapid in March-May (0.29°C per decade) and slowest in June-September (0.19°C per decade). This is evidently shown by Machakos Meteorological station as shown below where temperature data was analyzed for a period of 37 years between 1978 and 2015. The temperature trend for Machakos station shows an increase in temperature at the rate of 0.00008 °C per day which reflects to 0.0292°C per year. This further shows that there has been an increase of 0.292 °C per decade. Annex 32 details the basin level rainfall and temperature trend analysis.**

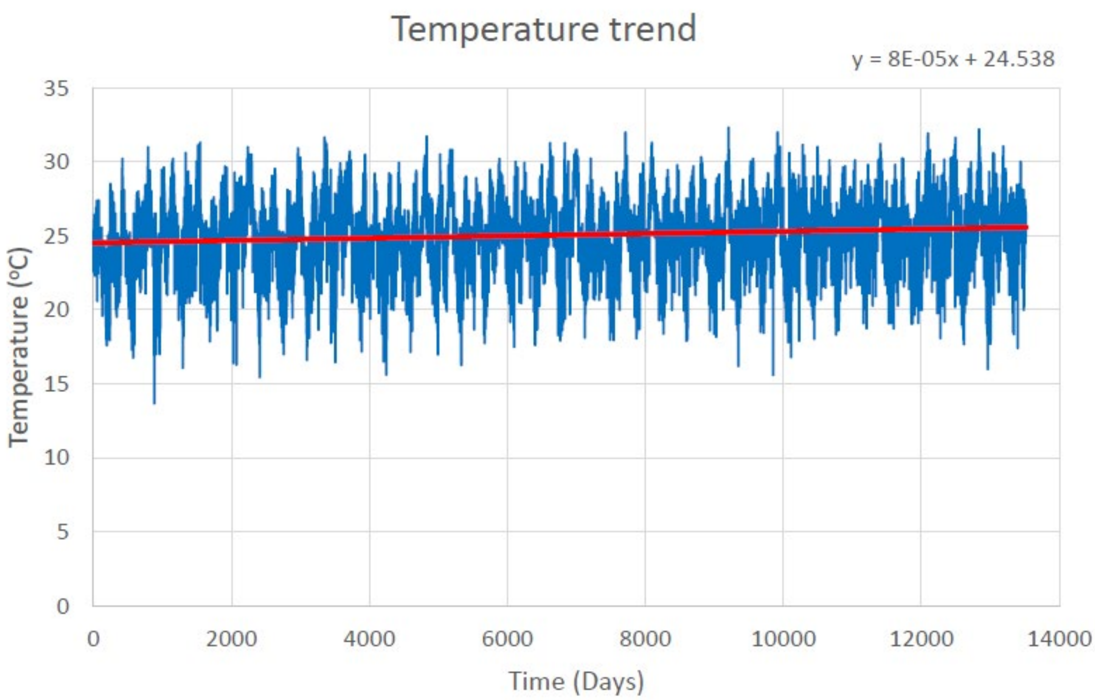

Figure 1: Time series chart for observed temperature

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C

Athi River Catchment Area (Project Location) baseline scenario

11. Athi River Catchment Area (ARCA) located in South and South West Kenya, is one of the major water resources in Kenya. Athi Catchment Area has varied topographical characteristics, from the highland in the Aberdare Ranges of around 2,600M above mean sea level (amyl) to the coastal area at the sea level. Even though the water catchment area is relatively small compared to other water catchments, water demand in this catchment is large and will be even larger (see Figure 1). According to the current monitoring situation of water resource, the quality of surface and ground water attains the target level by 84–100%. However, the quantity of water resource is highly insufficient. **The ARCA has present water demands of 1,145 MCM/year which represents 76% of the available 2010 water resources of 1,503 MCM/year.**

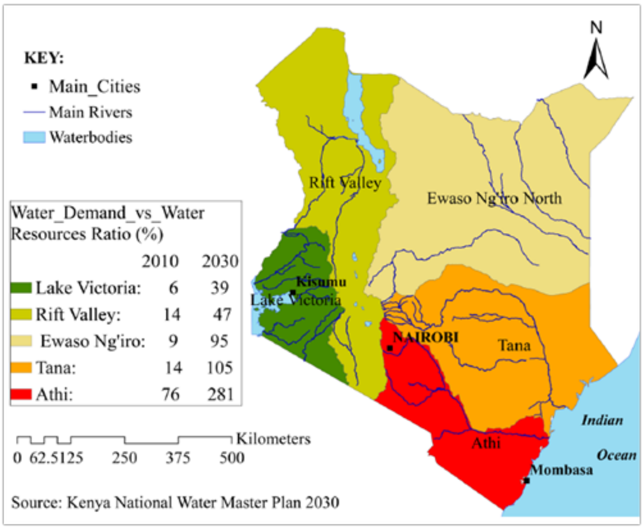


Figure 1. Water basins and water balance in Kenya

12. ARCA is classified as a semi-arid land except in the upstream area of the Athi River which is classified as a humid land non-Arid and Semi-arid Land (non-ASAL). The catchment has an annual average rainfall of 810mm. The annual rainfall differs spatially within the catchment area, ranging from around 500mm in the southern part near the border with Tanzania to 1,200mm in the western mountainous area. Evapotranspiration ranges from 1200-2500 mm per year in the catchment. Daily temperatures in the catchment area range between 10°C in the Upper Zone to 30°C in the Coastal Zone.

13. Wetlands and riparian zones: In Nairobi, Upper Athi, and Coastal Athi sub-regions, most wetlands and riparian land zones are rapidly being reclaimed and converted into residential estates or commercial/industrial centers. These wetlands and riparian zones need protection from degradation and restoration of their functional capacities. Croplands and rangelands: Croplands and rangelands constitute the largest land use areas in the catchment. Rangelands in the Middle Athi and Nolturesh Lumi area support pastoralism. Unfortunately, these rangelands are increasingly being converted to agro-pastoralism and urbanization with consequent loss of the protective vegetation cover. Their fragile soil once disturbed by cultivation becomes susceptible to soil erosion during rainy seasons. Agriculture and pastoralism: The Upper Athi consist of the areas of Kiambu and some parts of Nyandarua, where farming of coffee and tea is predominant. Horticultural farming is also practiced in these areas. Food crops like legumes, maize and fruits are grown in upper as well as middle Athi (Nairobi, Kajiado, Kiambu, Makueni, Machakos Counties and some parts of Kitui County). Other crops are cotton in Machakos County in the east of Nairobi. Cashew nuts, coconut and mangoes are grown in the coastal region of Kwale, Kilifi and Mombasa. Bananas are grown in Kiambu, Kwale and Taita Taveta Counties. Sisal is also grown in Makueni, Taita Taveta and Kilifi Counties.

14. The ratio of 76% of water demand to water resources, which is called a water stress ratio, shows a very tight balance between water resources and demands compared with the ratio of 40% regarded to indicate severe water stress. **It is projected that the water demands for 2030 are expected to increase to about 281% against the**

[3] https://germanwatch.org/sites/germanwatch.org/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_10.pdf

2030 available water resources. Water users in the catchment include urban and rural populations, major industries, livestock and wildlife which use both surface water and ground water. Groundwater forms a substantial part of the available water. Annex 27 provides a Peizometric surface trend analysis for acquifers. In addition to this**, the targeted areas experience 7 dry months and 5 wet months in a year which means that for a number of months in a year, communities have to seek out water through travelling distances away from their homes. This work is mainly taken up by women and children in the communities, leaving them to spend a few hours each day fetching water.**

15. Currently, out of the 6 catchment areas, **the Athi catchment has a negative water balance and is the least water secure region. This situation is further exacerbated by climate change impacts. In the context of climate change, majority of the ARCA is ranked as highly vulnerable with certain areas being highest and moderately vulnerable**. In April 2016, the ARCA experienced massive floods escalating climate change impact. Flooding events in the catchment has an impact on the local economies of the towns within the area. In addition, poor drainage in major towns (e.g. Nairobi, Kiambu, and Mombasa) causes urban flooding events. In addition, **the availability of water resources is unevenly distributed spatially and temporally in the catchment, with large areas suffering from water deficit**. **With the per capita water availability being less than half of the global standard the catchment experiences water supply problems which become worse during drought events**. This is expected to get worse by 2030. The water deficits exacerbated by climate change suggest better planning and managing water resources.

16. **In a recent study on the impact of future climate on the upper Athi River water resources (Annex 29), 8 out 9 simulations indicate increased flows in the upper Athi river basin in future. This study used CHIRPS (Climate Hazards Group Infra-Red Precipitation with Station) data n (Funk *et al*., 2015) for precipitation and Rossby Centre regional atmospheric model (RCA4) for future climate data analysis. One scenario (rcp2.6 for the period 2011-2040) indicates that the flows will be below the baseline period of 1980-2010. However, the water demand deficit is likely to increase in future due to increased demand with a good proportion emanating from increased evapotranspiration, for which the increase in streamflow is unlikely to satisfy**. In particular, January-March and July-October are the months that the upper Athi will experience water shortage when comparing the supply and demands, with irrigation sector will being the worst affected. The study recommended development of new water storage facilities and rehabilitation of old ones to improve the storage capacities. It also recommended the improvement of the landscape by planting vegetation/trees in order to reduce bare land and retain as much rainfall in the soil. This will reduce incidence of flash floods as well as sediment transport that check the increase of operation and maintenance cost of water storage and treatment works.

Future Climate Data was based three on representative concentration pathways—RCP 8.5, 4.5 and 2.6. The model used for the study is the Rossby Centre regional atmospheric model (RCA4 with a horizontal grid spacing of the simulation is 0.44 degrees (about 50 kilometers). The choice of the RCA model for this analysis was based mainly on the availability of the model outputs for the three different scenarios (RCP2.6, RCP4.5, and RCP8.5), as well as recent study (Endris et al., 2015) showing that the RCA model run driven by MPI-ESM-LR better reproduces the large-scale signals, such as the El Niño-Southern Oscillation and the Indian Ocean Dipole, in the historical period over the Eastern Africa region than RCA model runs driven by the other global climate models. The Rossby Centre regional atmospheric model (RCA4) output for the baseline period was evaluated against the CHIRPS datasets. Figure 2 shows the areal average precipitation at monthly interval and the accumulated precipitation for the two datasets. As shown in Figure 2 the two datasets have good agreement over the baseline period

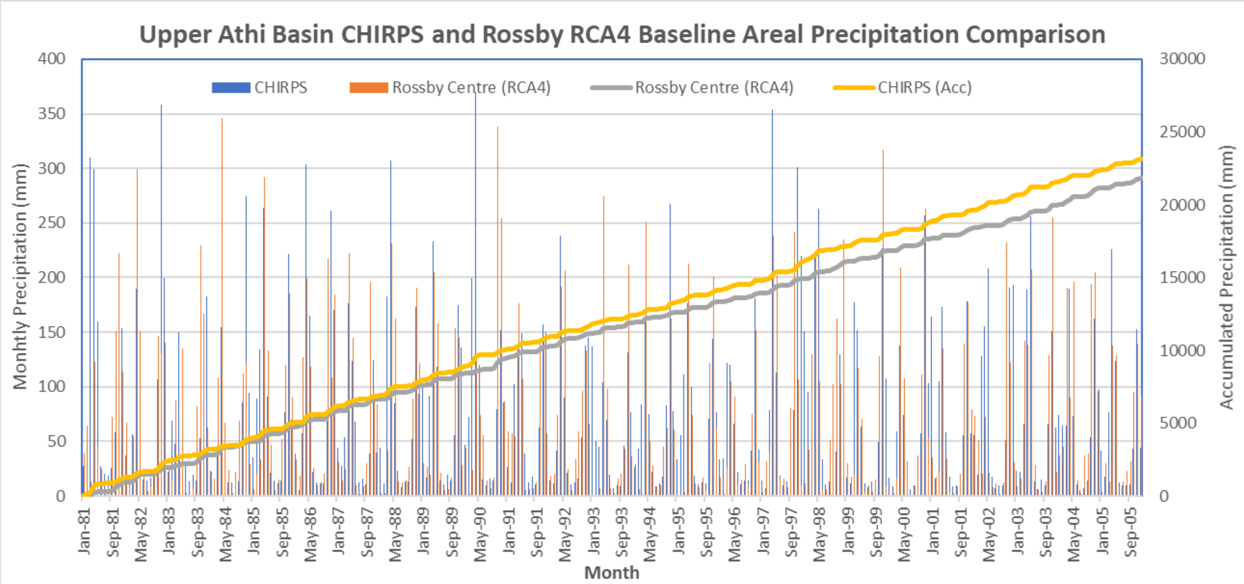


Figure 1 Comparison between Rossby Regional atmospheric Model (RCA4) and CHIRPS datasets

17. **The project is located in Southern Kenya and will cover the Athi River Catchment Area, an area of 68,900 km^2 with a population of approximately 9.79 million**[4]. Along the Athi River Catchment Area, there are 12 counties such as Nairobi, Makueni, Taita Taveta, Kwale, Mombasa, Kiambu, Machakos, Kajiado, Kilifi, Kitui and Nyandarua. **In order to ensure the effectiveness and efficiency of the proposed project activities within the limited financial resources and according the historical and projected drought and flood risk assessment and water scarcity, this project will be focused on these four counties: Kenya, which are Nairobi, Kiambu, Machakos and Nyandarua**. In addition, these counties have been selected for enhancement of Athi River Catchment Area through consideration, the consultative meetings with relevant government entities for this project, NEMA, WRA, MoWS – Ministry of water and Sanitation, KMD, UoN – University of Nairobi and the respective county governments.

Climate change vulnerability baseline, adaptive capacity and impacts

18. **The Upper ACA is classified as one of the most vulnerable area to the impact of climate change such as increasing temperatures and unpredictability of rainfall characterize. Climate change impacts and concerns in Upper Athi include water scarcity** (Nairobi, Machakos and Kiambu), **droughts** (Machakos) **urban flooding** (Nairobi and Kiambu) **and lack of awareness on climate change and its effects**. Additionally, the economy in Upper Athi just like in Kenya is dependent on climate sensitive sectors. Droughts and floods therefore have devastating socio-economic and environmental consequences in Upper Athi. Additionally, small holder farmer communities lack the necessary resources to invest in water harvesting structures such as pans and tanks. Investments into these structures through the project will enable the communities to have alternative water sources for water supply for domestic, agricultural use during the drought periods.

19. **Although the rainfall is observed to decrease annually and seasonally within ARCA, the rainfall fluctuates around a mean value thereby having enhanced rainfall in some seasons while others remaining dry or with depressed rainfall. These scenarios lead some areas especially Kiambu and Nairobi to experience urban flooding while areas like Machakos experience drought due to depressed rainfall. Both the intensity and the frequency of rainfall contribute to the above scenarios.** East Africa's seasonal rainfall is predicted to be strongly influenced by ENSO, which contributes to uncertainty in climate projections, particularly in future inter-annual variability.

[4] http://www.wrma.or.ke/index.php/wrma-regional-offices/athi/catchment-status.html

20. **The population in Athi River Catchment is vulnerable to climate risks due to the high dependency on natural resources for food, fuel and shelter**. According to IISD (2012), climate risk and vulnerability in Kenya indicates that water availability is especially critical as they live in one of the most water scarce countries in Africa[5]. Access to this basic resource is likely to become more difficult due to population growth, economic expansion, unsustainable management of water and forest resources, and changes in rainfall patterns. At the same time water is the core input for most economic activities. The future water demand projections as of 2040 indicate growing water demand. This is shown in the provided sub basin water balance. Areas far away from perennial rivers use pans as a source of water, this water sources will be affected negatively in future due to increased evapotranspiration and therefore less likely to withstand prolonged drought. The net impact in the project area will be reduced water availability with increased irrigation demands. Climate change is thus anticipated to have a negative impact on the Upper Athi Basin from a water resources perspective. Although the frequencies of extreme drought (SPI less than -2.0) occurrences in the baseline climate are much lower than the baseline values for severe drought events, projected changes in extreme droughts have spatially more consistent and widespread signals. The rainfall-runoff/water resources analysis shows that the balance between future projected rainfall and ET to be positive leading to increased runoff. Hydrological modelling (Annex 29) used catchment areal average of future projected precipitation and may differ with past point/station rainfall trend analysis. The increase of stream gauging station measurements may not directly have related to increase of rainfall because other factors may come into play like ground water contribution to stream flow.

    A robust analysis based on ensemble mean of the best performing RCMs shows that the frequency and intensity of droughts are projected to increase in future climates under the RCP4.5 and RCP8.5 emission scenarios. Results are detailed Annex 26 -Analysis of projected rainfall changes. And Annex 34 -Basin Level-Analysis of projected rainfall changes

The country's inland areas are largely arid, with two-thirds of the country receiving less than 500 mm of rainfall per year, limiting the potential for agriculture. Increased temperatures and rainfall variability are likely to exacerbate the conditions already experienced and may have a significant impact on water availability and droughts in the future. The prolonged drought of 2008 - 2011 serves to highlight some of the devastating and pervasive socio-economic consequences resulting from such events: crop production losses arising from reduced yields of food crops and cash crops amounted to KSh69 billion and KSh52 billion, respectively; the livestock sector experienced the worst impacts, losing approximately KSh699 billion, with KSh56 billion in damages due to costs from veterinary care, water, feeds and production decline and KSh643 billion in losses due to animal deaths. The risk assessment to drought was conducted by estimating integrated risk index, comprising hazard, exposure and vulnerability. Anthropogenic and environmental indicators were used in this assessment. **As a result, the risk of drought is expected to increase in the future, and Kiambu and Machakos seem to suffer from the more pronounced risk of drought (Figure 2).**

5 International Institute for Sustainable Development. 2012. Climate Risks, Vulnerability and Governance in Kenya: A state of the art review. Pre-publication version. Winnipeg: IISD. page 24.

Figure 2. Historical drought risk index (left: 1976 – 2005) and estimated drought risk index (right: 2011-2040)

21. Floods have caused equally devastating consequences in recent years, including loss of lives and livelihoods, personal property damage and damage to infrastructure, with ramifications for the economy. Major events were recorded in 1997-98. The 1997/98 El Niño floods are estimated to have caused damage equivalent to at least 11 per cent of GDP, including KSh62 billion (USD 777 million) in damage to transport infrastructure and KSh3.6 billion (USD 45 million) to water supply infrastructure. **While floods are generally associated with higher damages on public infrastructure assets, the burden of droughts falls more heavily on people, communities and the private sector.** Flood risk assessment was also conducted, which followed a similar approach of the drought risk assessment. **The flood risk seems to increase in the future and a large increase in flood risk is expected in the Kiambu and Machakos areas (Figure3).**

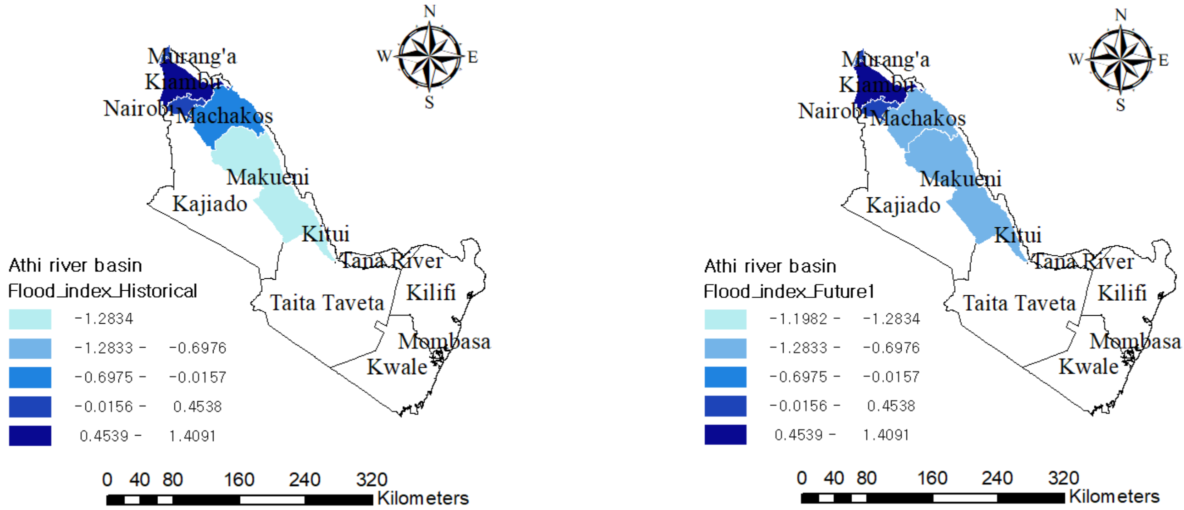
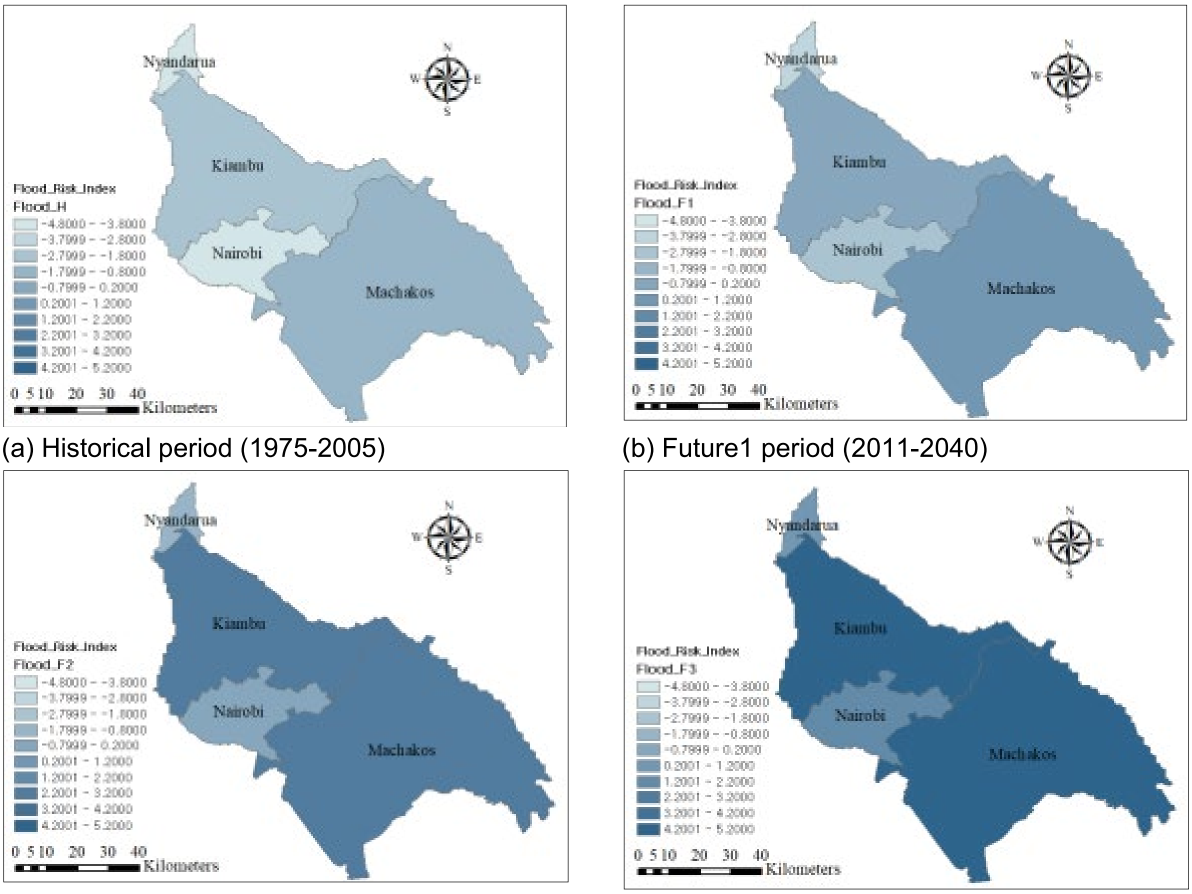


Figure 3. Historical flood risk index (left: 1976 – 2005) and estimated flood risk index (right: 2011-2040)

22. Climate change projections over the Athi River Catchment, modeling, downscaling and bias correction were done during the project preparation using 26 Global Circulation Models from CMIP5. **Results for annual rainfall across the 4 counties show an overall increasing trend, with the highest rainfall in Nyandarua and Kiambu[6]**. The Figures below show increased flood and drought risk in the Upper Athi Catchment Area across different time periods. Despite the projections on the Global Circulation Models, the local data results for annual and seasonal rainfall across the 4 counties show an overall decreasing trend, with the highest decrease in Machakos and JKIA Meteorological stations. The study was carried out from 3 Meteorological station within ARCA (Machakos, JKIA and Dagoretti stations) for a period between 1957 to 2015. The annual and the seasonal (JF, MAM, JJAS and OND) rainfall trends showed a decrease pattern ranging between 0.6 to 1.5 mm annually. The rainfall and temperature historical trend analysis was carried out at the basin level. 3 stations (Machakos, JKIA and Dagoretti stations) within the Athi basin were analyzed and their results presented with the analysis at the country level on Annex 28- Climatology of Kenya. Future projected rainfall changes for different season (MAM, OND) over the Athi River Basin based on ensemble averages on the top models is characterized by high temporal and spatial variations as described in Annex 34 - The basin level analysis of projected rainfall.

[6] More details on the methodology and model output results can be found in the Feasibility Study.



(a) Historical period (1975-2005) (b) Future1 period (2011-2040)

(c) Future2 period (2041-2070) (d) Future3 period (2071-2099)

Figure 4. Flood risk index of four counties under the RCP4.5 scenario

23. The trend analysis of rainfall in Machakos Meteorological station indicates a decrease of annual and seasonal rainfall in the area as depicted in the figure below. The annual rainfall as plotted from 1957 to 2015 indicates that the rainfall decreases by 1.5 mm annually. The same trend is observed in all the other seasons as rainfall seems to decrease seasonally. In January – February season the rainfall decreases by 0.1 mm annually. In March, April and May rainfall season, the rainfall was decreasing at the rate of 0.66 mm annually. In June, July, August and September, the trend is the same as rainfall was decreasing at the rate of 0.008 mm annually. The October, November and December season the rainfall was decreasing at the rate of 0.64 mm annually. If Machakos Station was taken as a representation of the ARCA, then it can be deduced that the rainfall within the Catchment decreases by a small margin annually

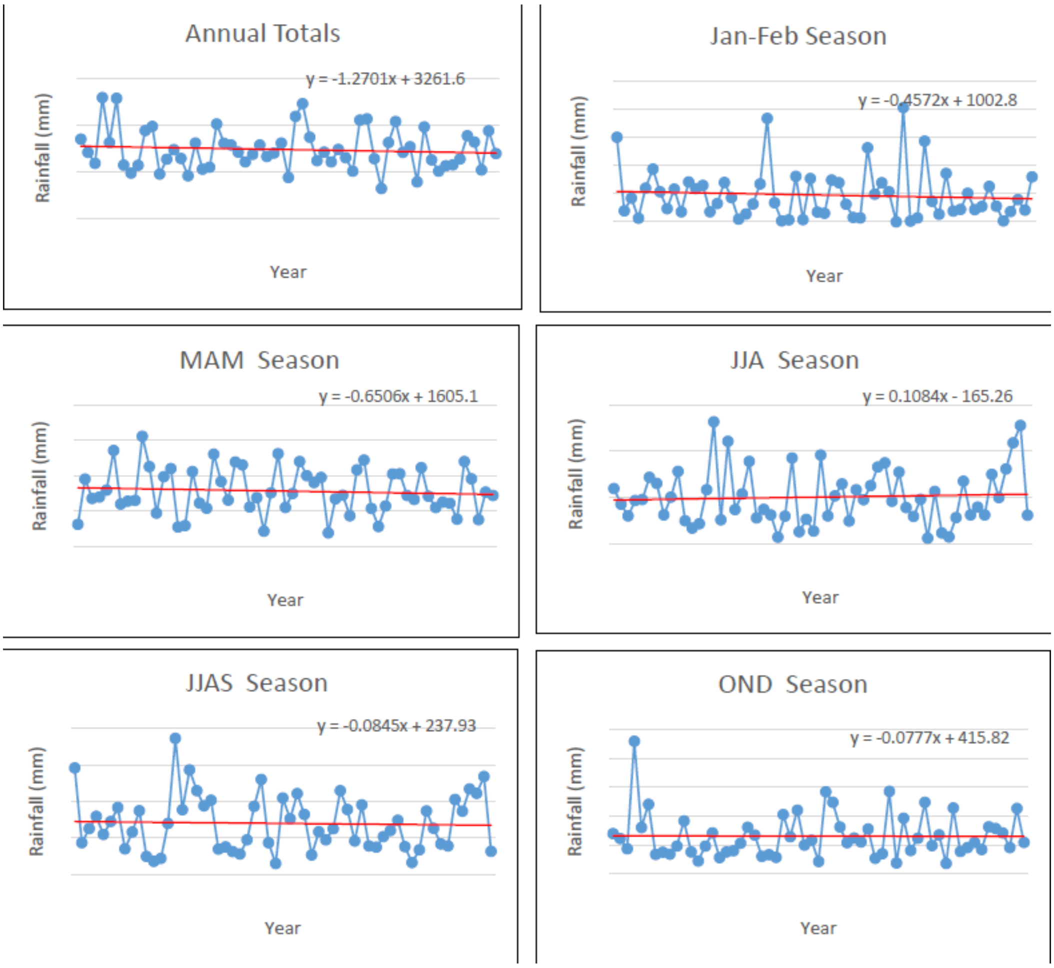
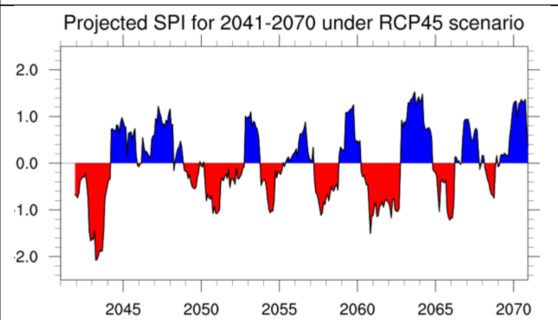
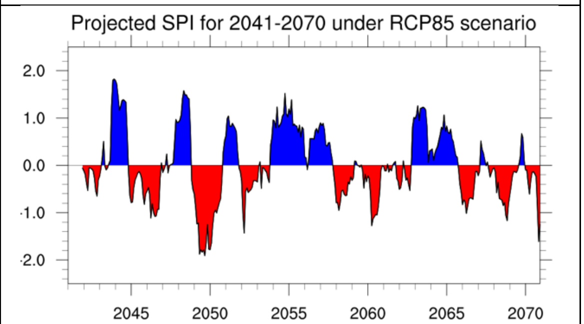


Figure 5 Rainfall Time series for Machakos Meteorological Station

24. The degree of dryness or wetness over Kenya was assessed using the "severe" and "extreme" categories and changes in projected SPI and frequency of dryness and wetness were examined for mid- and far-future climates under the RCP4.5 and RCP8.5 scenarios. Figure 5 provides the SPI over Kenya for different present and future periods and scenarios. Nationally averaged SPIs provide informative summary of precipitation differences between epochs and scenarios. SPI differences between the two scenarios in the present climate analysis are due to differences in the number of models used (three for RCP 4.5 and 4 for RCP 8.5). In addition to the response of climate to scenario forcing, temporal SPI variations for any epochs/periods depict the interannual variability of rainfall across Kenya. Overall, longer and/or more frequent drier events are projected in the far-future climate under the RCP 8.5 scenario.

	RCP 4.5	RCP 8.5
Mid-future (2041-2070)		

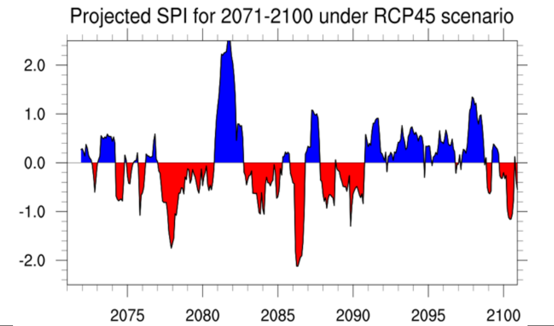
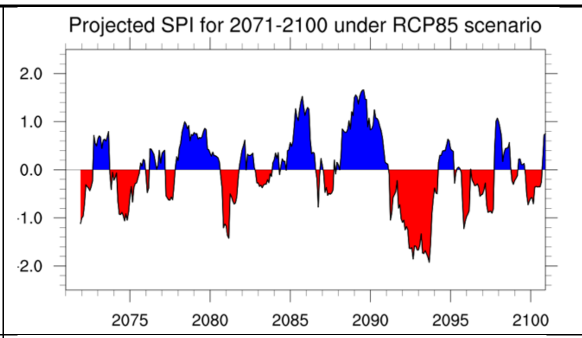
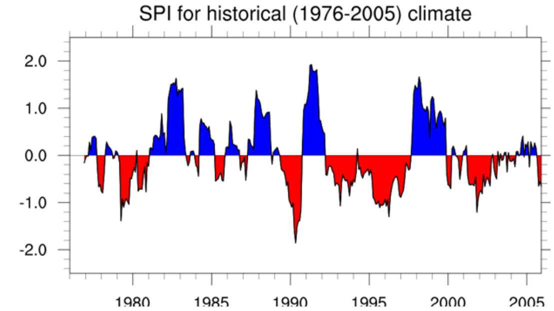
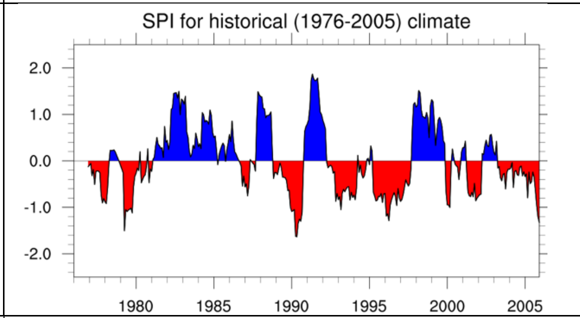
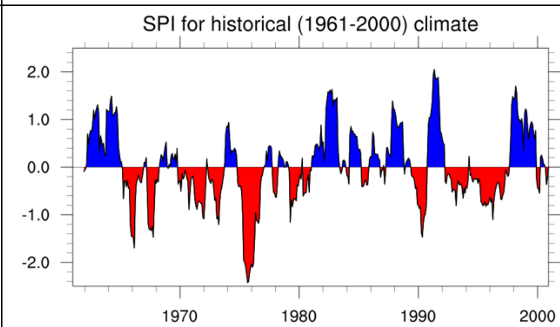
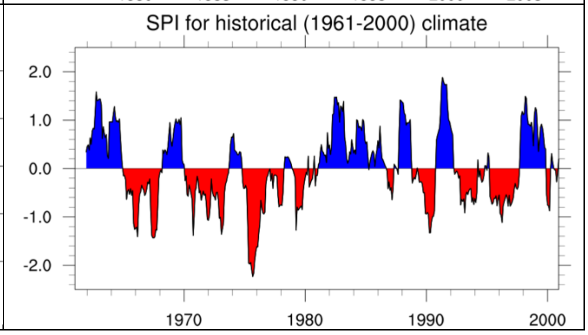
Far-future (2071-2100)		
Present Climate (1976-2005)		
Present Climate (1961-2000)		

Figure 6. Drought risk index under the RCP4.5 and RCP 8.5 scenario

The analysis of SPI indices is conducted at basin-level. As a standardized unbiased index, the SPI is designed to show departures from a baseline. To show projected changes in frequency and intensity of droughts, the number of cases exceeding two levels of drought severity thresholds in three 30-year projection periods (near, mid and far future climates) were compared with identically identified counterparts in present day climate (1976-2005) for RCP4.5 and RCP8.5 scenarios.

The analysis has demonstrated that;

- the frequency of **severe** droughts is projected to increase in mid- to far-future periods under RCP4.5 and RCP8.5 scenarios, especially over the coastal areas;
- the frequency of **extreme** droughts is projected to increase in future periods under RCP4.5 and RCP8.5 scenarios in the Upper Athi River Basin; and
- Projected changes in **extreme** droughts have spatially more consistent, widespread and intense signals over the Upper Athi River Basin compared to projected changes in **severe** droughts.

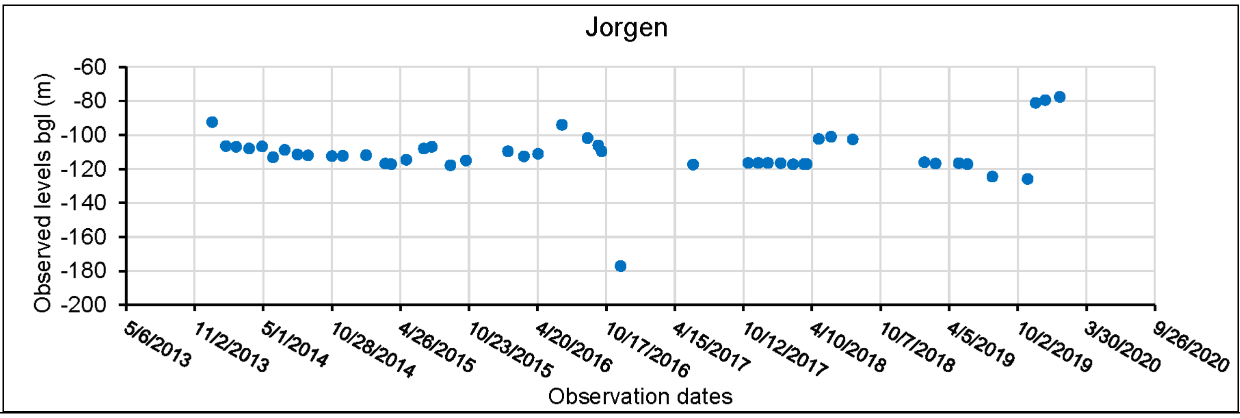
Results are detailed Annex 26 -Analysis of projected rainfall changes and Annex 34 -Basin Level-Analysis of projected rainfall changes

25. **From the above scenarios, the frequency of the floods and drought occurrence shows an increasing trend and becoming more severe in the upper Athi catchment.** The magnitude of flood risk in the historical period was substantial in order of Machakos, Kiambu, Nairobi, and Nyandarua, and similar trends were projected in the coming decades. Drought risk, on the other hand, was the highest in Machakos, while Nyandarua being the least prone area to both flood and drought events. Although the levels of flood and drought risks vary, all four counties

are projected to be increasingly vulnerable to the negative impacts of climate change, and therefore it is necessary to propose appropriate interventions to make the communities in the target areas more climate change resilient.

26. Kenyan rainfall is highly variable; it has a strong inter-annual and spatial variability, which is associated with extreme events (droughts and floods. Inter-annual variability of rainfall in Kenya is associated with perturbation of global Sea Surface Temperature (SST). In particular, El Niño/La Niña (ENSO) and IOD phenomenon play a critical role in this variability. El Niño and positive IOD are associated with heavy rainfall and La Niña and negative IOD are associated with depressed rainfall over Kenya. Inter annual rainfall variations in Kenya is highly linked to the El Niño Southern Oscillation (ENSO), with more rain and flooding during El Niño and droughts in La Niña years, both having severe impacts on human habitation and food security. Furthermore, to show projected changes in frequency and intensity of droughts relative to the present-day climate, the number of cases exceeding two levels of drought severity thresholds in three 30-year projection periods (near, mid and far future climates) were compared with identically identified counterparts in present day climate (1976-2005) for RCP4.5 and RCP8.5 scenarios. Results are detailed Annex 26 -Analysis of projected rainfall changes and Annex 34 - Basin Level-Analysis of projected rainfall changes

27. Along with the changes and projections of temperature and precipitation patterns in the study area, and its impacts that are shown as floods, droughts and reduced water availability; consequently, other challenges are expected. **One of the most important impacts of climate change on water systems is its influence on river flow, or discharge**. **In addition, climate change has affected groundwater levels through its influence on the amount of water available to aquifer recharge. Changes in annual rainfall as well as changes in the patterns of extreme precipitation events will affect availability of water for recharge**. Groundwater resources within the target areas have been significantly affected with some drying up especially in the drier part, a good illustration being Machakos county**. In Kenya climate change has caused higher temperatures thereby increasing evaporation rates, and also affecting recharge and groundwater discharge rates**. Due to decreased availability of surface water, there is increased reliance on groundwater to meet demand. The temperature trends of Machakos Meteorological station (project area) shows an increasing temperature since 1960.
28. The case study of Nairobi aquifer monitoring stations area demonstrate the contribution of rain events to groundwater recharge. The observation is a representative of what happens in other aquifers within the country and how rain periods manipulate aquifer replenishment. Depth to groundwater level from WRA monthly monitoring boreholes within Nairobi Aquifer Suite from 2013 to 2020 are displaying long-term declining trends and minimal influence of short-term pumping schedule. Recharge based recovery are evident as observed during periods of heavy rains around April – June every year and as experienced in late 2019 and January 2020 as shown in the figure below generated from monthly monitoring data. Annex 27 on Peizometric surface trend analysis further details the observations



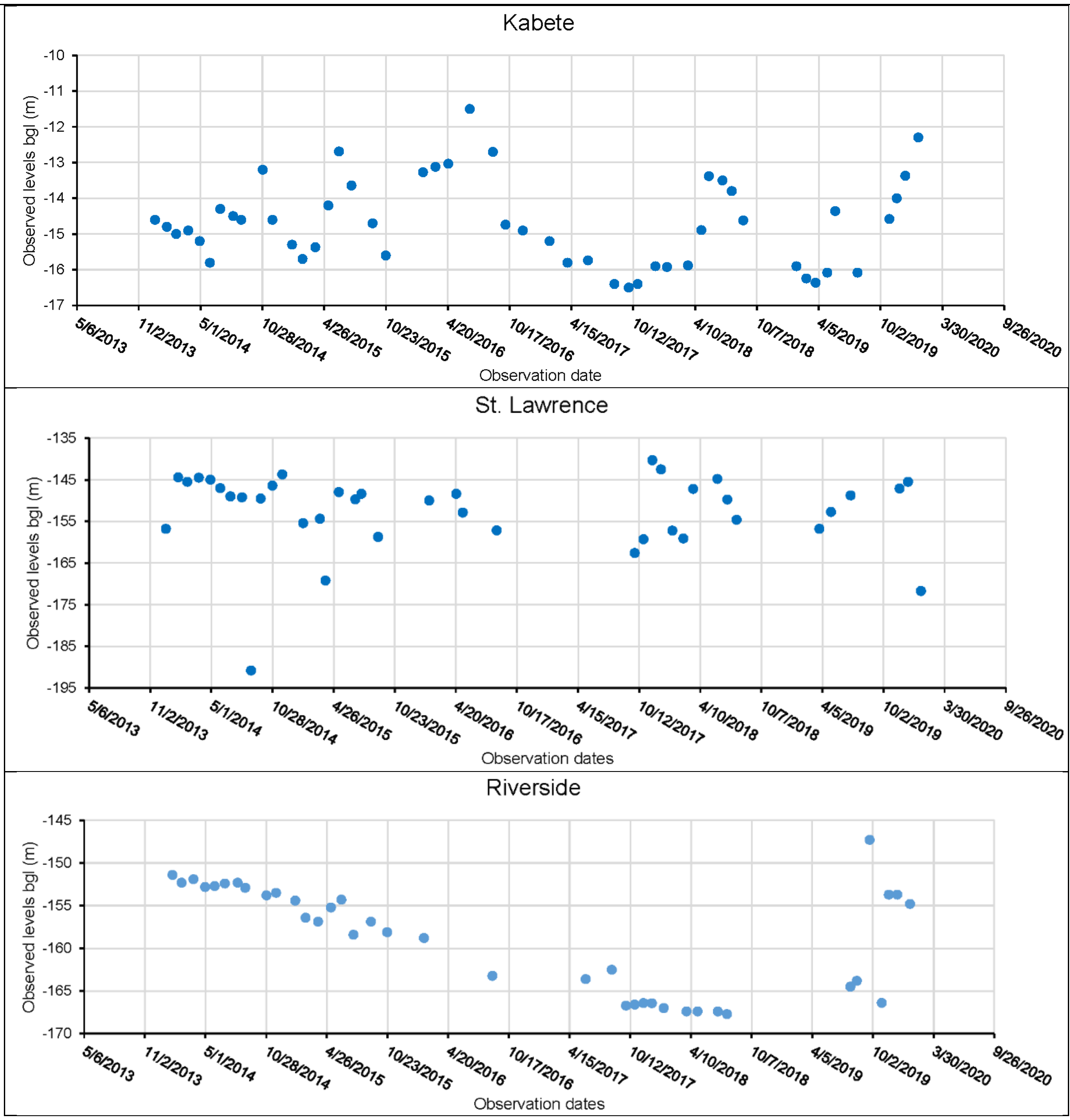


Figure 7: Observed groundwater levels (Nairobi sub-region)

29. **Climate change is also expected to exacerbate the water quality in the study area. Extreme weather events including typhoons, storms, and temperature jumps mainly result in water events, such as floods and droughts which may further affect the water quantity and quality.** Some studies have reported the impacts of droughts on the water quality, which mainly include increased pollutant concentrations, enhanced nitrogen mineralization, and delayed recovery from acidification. During the drought period, lower flows can weaken the dilution effects of some pollutants. Downstream populations depend on water from the river. However, increasing pollution can lead to health and sanitation related problems including waterborne diseases which make the population more vulnerable. There is a need to monitor the quality of the water in the river to ensure that communities without piped water are able to access safe and potable water for domestic use especially during drought conditions. In the Athi River Catchment area, the impacts of climate change on water quality are likely to be significant in several aspects: a) Water quality could suffer in areas experiencing increases in rainfall. Heavy

downpours can increase the amount of runoff into the river, washing sediment, nutrients, pollutants, trash, animal waste, and other materials into water supplies, making them unsafe, unusable and in need of treatment) b) Climate change is making heavy intense downpours, droughts and rising water temperatures more common. This can alter the quality of water. Bacteria and viruses thrive in these new conditions and when they come into contact with humans, can cause illness. Lack of water can also impact human health, especially in drought conditions. c) As water sources decline, the concentration of contaminants increases, making them more likely to affect human health. The monitoring of the water quality will therefore be important to institute measures to protect the water resources in the catchment.

30. Annex 28 - Climatology in Kenya, further details the Kenyan Climate context. The climatic changes are not uniform from one region to another. Future changes that are expected to occur are different from one climatological zone to another. Future rainfall variability is also expected to increase, therefore climate extremes of drought and floods are expected to increase its frequency and intensity toward the end of 21st Century. And these will result in economic losses to communities because of over dependency on rained agriculture. Temperature is also anticipated to increase, which cause might cause frequent heat wave especially in the low lying areas and also increase evapotranspiration and thus affecting agriculture. Crop production and livestock keeping will be greatly negatively impacted by decrease in the rangeland for livestock keeping these will lead to lose of livelihood for many households and will lead to over reliance on government relief food and increase conflict over resources. It was established high heat stress will increase heat related health risks more especially to the vulnerable. Due to projected increase in temperature and the rapid expansion of urban areas, urban heat island will affect the urban dweller, where night time temperature will generally increase.

31. From the climate models analysis, it has been established that temperature is increasing, the frequency of cold day and nights are reducing, while the frequency of hot day and warm nights are increasing. Depending on emission scenarios, the increase in temperature is likely to be between 0.3 and 5.4 0C by the end of the 21st century over Kenya. This is likely to change the hydrological cycle hence affecting precipitation intensity and extreme events and as a result affecting different sectors of the economy. Further, climate models indicate a reduction in precipitation during the long rains (MAM) and an increase in the short rains (OND). The models also show a reduction of annual precipitation. This exacerbates the drought condition within the area
32. **Another challenge related to its adaptive capacity to the impacts of climate change is the state of the water infrastructure, particularly water storage and water supply in the study area**. As a consequence of droughts, less water availability is expected in the 4 counties beneficiaries of the project. Proposed interventions in Nyandarua were primarily, water pans and springs. There are 12 water pans that require rehabilitation. In addition, two springs and boreholes will require rehabilitation. These structures are expected to serve about 65820 individuals with clean water supply. Proposed interventions in Kiambu were protection of springs, sinking of boreholes, rehabilitating existing ones and catchment protection to reduce erosion risks. In Kiambu County there are four springs that are proposed for protections, three water pans that are proposed for rehabilitation and protection and four boreholes are proposed for rehabilitation. These structures are expected to serve about 54000 individuals with clean water supply. Proposed interventions in Machakos County were primarily water pans. There are 9 water pans that were selected for rehabilitation in Machakos County. In addition, a spring and two boreholes were proposed for rehabilitation and establishment, respectively. These structures are expected to serve about 967,800 individuals with clean water supply.

33. **Other determining factors to reduce vulnerability (improve adaptive capacity) in the project area refer to the integral management of the catchment and to mainstream climate change into political instruments and management of water resources in Athi River Catchment Area**. Deforestation and forest degradation are rampant in the all the forested areas that feed the catchment. The National Water Master plan conducted a satellite image analysis with the forest area in ARCA in 2010 being 120,000 ha which corresponded to 2.0% of the forest cover in ARCA. The deforested areas during the last two decades were about 133,000 ha, which meant there was a decrease of 52.5% of the forest areas in 20 years since 1990. The small water sources (e.g., 20 springs and 17 wetlands), vulnerable to external factors, have deteriorated as reported in the interviews with stakeholders in the WRA and KFS in the ARCA. With a consideration of the condition of semi-aridity around the ARCA, the deterioration of these small water sources can negatively affect the water availability. The main environmental conservation activities proposed by the project are mainly reforestation under future climate intervention trees would play various

critical roles. This include increased forest cover, providing alternative source of fuelwood, contribution to biodiversity diversification and carbon.

34. The links between adaptation actions proposed by the project and the key climate change impacts & challenges outlined above are summarized in the table below.

Table 1. Key climate change impacts, associated challenges and immediate response measures needed

Key climate change (CC) impacts identified by the project	**Major challenges associated with key CC impacts (Vulnerability/adaptive capacity)**	**Immediate response measures needed**
The Upper ACA is classified as one of the most vulnerable area to the impact of climate change such as increasing temperatures and unpredictability of rainfall characterize**.** **Increased temperatures and variability in rainfall are likely to exacerbate prevailing conditions.**	Inadequate data collection, information, generation and analysis on water resources, water scarcity and variability, water pollution, enforcement of water laws, catchment degradation, and the impact of climate change (adaptive capacity) Inadequate maintenance of the hydrological and meteorological stations.	Enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector • Strengthening monitoring networks to enhance data collection and to improve climate and water information management system • Improving the use of water resources management tools for effective water resources planning • Strengthening stakeholder collaboration to enhance water storage and to address the impacts of climate change • Building staff capacity and improving the work environment
Water Scarcity **Droughts** **Flooding** **As a result, the risk of drought is expected to increase in the future, and Kiambu and Machakos seem to suffer from the more pronounced risk of drought** **Climate change is also expected to exacerbate the water quality in the study area**	Difficulty in accessing water resources due to the lack of water supply facilities and maintenance resource Significant impact on the availability of water. Crop and livestock production losses from reduced yields of food and cash crops	• Improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment. • Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions.

35. The project will support the implementation of the Kenya National Adaptation Plan (2015-2030) whose vision is enhanced climate resilience towards attainment of Vision 2030. Enhanced climate resilience includes strong economic growth, resilient ecosystems, sustainable livelihoods, reduced climate-induced losses and damages. On the other hand, the revised Environmental management and Coordination Act (EMCA) 2015, in the spirit of devolution has devolved environmental planning and governance to the Counties. Counties should develop and to establish County Environment Committees and prepare County Environment Action Plans (CEAPs) and participate in the preparation of State of Environment reports (SoEs). Under the County Government Act, Counties are also required to prepare County Integrated Development Plans (CIDPs) to guide sustainable development. **This project will support the Counties to integrate climate change adaptation and mitigation in their planning**

processes. It will also build the capacity of local level governance structures to address the impacts of climate change.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

36. The project targets to **enhance community resilience and water security in the Upper Athi River Catchment Area** (ARCA) **in Kenya,** particularly in the counties of Machakos, Kiambu, Nairobi, and Nyandarua to address the core problem that climate-induced droughts, floods, water scarcity threatening livelihoods and water infrastructure. The target catchment is key to Kenya`s water management and host some of the most vulnerable groups of society. Therefore, this project will address the following challenges and barriers to improve capacity to adapt in ARCA:

37. **Weak technical basis of monitoring and information system on hydrology and meteorology constrains the adaptive capacity against flood and drought within the ARCA.** Insufficient observatory facilities (rainfall and water level) can limit the quantity of monitoring data. In terms of a quality of the collected data by present monitoring systems, the hydrological data are not subject to systemic quality management due to the lack of standards and protocols and budget. Management of meteorological data is also barrier. Transmission of the measured data is compromised by the failure of equipment maintenance and transmission network. Importantly, with adaptation or change comes uncertainty. Decision makers and technical staff lack sufficient information on climate change impacts to make informed decisions on changes in water management.
38. **Limited technical and institutional capacity to cope with climate change in water sector**. In regard to the technical capacity in Kenya, the climate and water-related monitoring system and infrastructure is poor, many observation systems are not functioning, and data collection, analyzing and processing systems are unreliable. Furthermore, information and data sharing/dissemination and communication systems among weather stations, meteorological agency (center) and local communities are not properly linked. Investment in this area will lead to improved adaptation planning, thus building adaptive capacity of the targeted communities in the four Counties. Political and management instruments for water management in the beneficiary counties do not consider climate change as an important aspect for their planning.
39. **Climate change is expected to exacerbate water quality problems in the area.** The water resources and environment have been degraded by human activities such as industrial development, farming, informal settlements, deforestation, solid waste disposal and wetland and riparian encroachment. Pollution of Athi River is on the rise and has negatively impacted the downstream communities for whom the river is a lifeline. The Nyandarua watershed is highly crucial because it is a main water source of Kenya. In particular, the water flows of Athi and Tana Rivers are affected by this area. Unfortunately, the Aberdare forest, providing water resource with high quality, has been destroyed by illegal harvest and deforestation activities. Consequently, the quality and quantity of water from this forest has also deteriorated. Awareness of negative impacts due to these activities needs to be increased to villagers to prevent these damages in the future. In addition, monitoring about water quality and climate change impacts are necessary.
40. **Floods, droughts, and the reduced availability of water in the area have impacted the water infrastructure in the area, in particular the provision of drinking water and domestic use.** Poor drainage infrastructure has caused frequent floods in Nairobi. In particular, the upstream region of Athi River experiences these damages, compared to the downstream region. Irregularity of water supply due to climate change has threatened sustainability of this city, also causing waterborne diseases. With consideration to the dense population in this city (> 3 million), implementation of appropriate measures is urgently required. 7) The major water source of the upper Athi River passes through Kiambu. The geographical location of this region largely affects the quantity and quality of water resources in the upper Athi River. Failure to improve the water quality and quantity of this upper region can deteriorate the quality and quantity of the lower Athi River. Although this town is surrounded by hilly farmlands, rapid urbanization within limited land space is expected to disturb the water system. Furthermore, the risk management on drought and flood also shows the higher vulnerability of Kiambu.

41. **The Theory of change proposed use a results chain of inputs, activities, outputs, outcomes. The main outputs are:**
    - Output 1: enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector.
    - Output 2: improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment.

- Output 3: Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions.

42. The Fund level impacts of the project are an increased resilience of: i) infrastructure and the built environment to climate change threats; ii) most vulnerable people and communities; and iii) health and well-being, and food and water security. The paradigm shift is to move towards enhance community resilience and water security in the Upper Athi River Catchment Area (ARCA) in Kenya.

43. The first pathway to change (output 1) is development of monitoring networks and databases for change analysis as an adaptation to climate change impacts on freshwater resources in the upper Athi catchment. Hydro-meteorological Monitoring will avail methods for change detection, attribution and prediction of changes and vulnerability of groundwater, floods, low flows and droughts and also prediction of groundwater quality degradation and restoration needs. Further a network for analyzing hydrological data through the exchange of data, knowledge and techniques will be established that will assist on the understanding of hydrological variability in space and time in different counties within the upper Athi catchment which have different hydro-climatic and hydro-geological environments. Increased data, monitoring, knowledge, understanding, and predictive modelling of hydrological variables will be used to improve the management and design of water resource, agro-hydrological and eco-hydrological systems against the influence of global climate variability and change. Improved management will lead to improved adaptation planning, including policy-related interventions for adaptation whose outcome is resilient water systems in the catchment. This resilience will lead to improved access, thereby improving the wellbeing, food and water security of the area.

44. The second change pathway (output 2) premises that since availability of water is impacted by distorted patterns of rainfall, a systematic means of addressing water availability, is to avail infrastructure that collects natural sources of water. Harvesting rainwater directly responds to water scarcity. The infrastructure proposed will avail water to households that are currently struggling daily with their water demands. The increased availability of water will bring the ability to farm, improved health and wellbeing, and increase household social and economic value.

45. The third change pathway (output 3) is strengthening capacity governance on water resources. The government of Kenya has 3 main local level policy tools for water governance namely, sub catchment management Plan, County Environmental Action Plan and the water quality regulations. Sub catchment management plans define the stakeholders, their roles and responsibilities, and also define actual activities to be undertaken to manage the resource. The County Environmental Action plan defines a whole ecosystem approach to managing the environment. Water quality regulations assist in managing pollution levels. Development and implementation of these management tools will lead to improved management of water resources in the target area, thereby leading to increased access to water for better wellbeing, health and food security.

46. **Assumptions:**
    - The county government and the local community will support the project in order to address or reduce climate change impacts.
    - The Global System for Mobile communication will be operating at optimum levels
    - Staffs and personnel of NEMA, WRA, KMD, County governments and communities will use the information for decision making and better planning in climate change and water sector
    - County Governments will implement the recommendation of the planning documents – CEAPs, SCMPs and will mainstream climate change impacts on these public policy and management instruments.
    - The regulated stakeholders will be participating in the meetings and have enhanced compliance to the regulations and area aware of the climate change impacts in the area.

47. Below is the theory of change for the project:

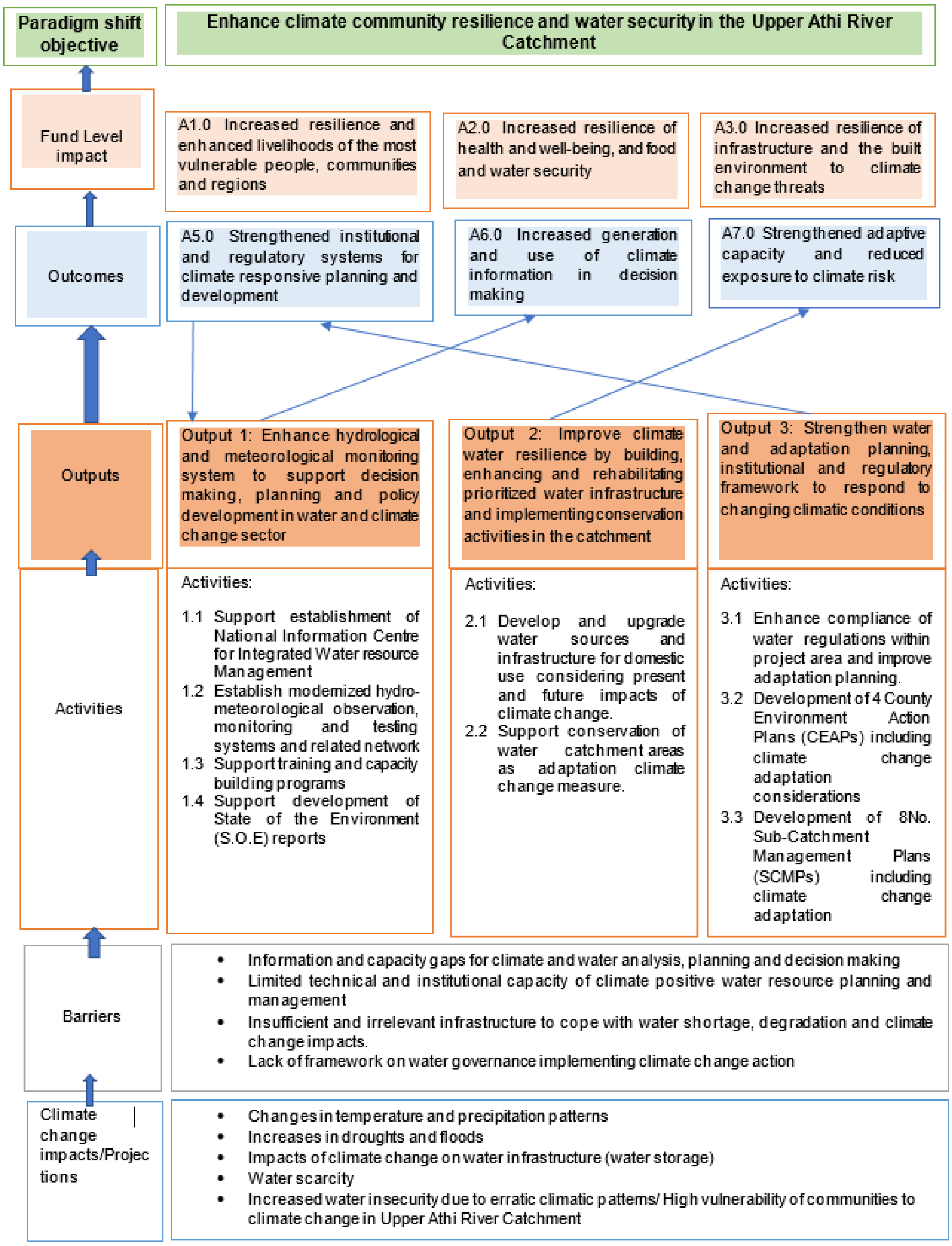


Figure 8: The Theory of Change of the project.

B.3. Project/programme description (max. 2000 words, approximately 4 pages)

Project objective, outcomes and impacts

48. The target of this proposed project is four counties: **Machakos, Kiambu, Nairobi, and Nyandarua** in the Upper Athi River Catchment Area. These counties were selected, given the counties´ high exposure to climate induced floods, droughts and water scarcity.

49. The project objective is to strengthen the resilience to climate change of communities and increase water security in the Upper Athi River Catchment in Kenya (Nairobi, Kiambu, Machakos and Nyandarua). This is expected to be achieved through three outputs:

- Output 1: enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector.
- Output 2: improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment.
- Output 3: Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions.

50. Lessons learned from past experiences targeted at strengthening water resource management and enhancing water availability indicate that support has been scattered and somewhat duplicated. Such actions in the past could not cover all of the Athi River Catchment to sustain service, so the required recognition from government and beneficiaries need to be provided systematically. While past projects and programs in Kenya and other developing countries have been infrastructure focused, those investments can only be sustained when the institutional capacity is reinforced, and service delivery has successfully improved.

Component 1: enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector.

51. **This output seeks to strengthen hydrological and meteorological information services to deliver relevant, accurate and timely climate information to local communities, and also to support decision making and policy development in the water sector.** This will be achieved by investments in optimized hydro-met monitoring networks, more effective management and exchange of hydro-met data; and improving the capacity to forecast future water and weather conditions. Ultimately, this information will be used to strengthen early warning systems.

52. Investing in hydro meteorological information systems will not only enhance protection from weather and climate shocks such as droughts and floods but will also improve our capability for gathering hydro meteorological data which is critical for monitoring and predicting extreme events. National Information Center for Integrated Water Resource Management to be established in WRA offices that will act as a hub of water resources monitoring information and will be equipped with proper ICT system for processing, analysis of information to produce integrated water resources information for informed decision making. Water resource monitoring systems in the catchment area are already established covering the main Athi River and its tributaries. However only 58% of surface water level, 35% of groundwater level and 66% of rainfall are actually being monitored currently by the WRA within the Athi River Basin. Therefore, it is necessary to increase the number of monitoring stations that include telecommunication abilities.

53. Hydro meteorological information is being collected by WRA as well as the Kenya Meteorological Department, but their analysis and dissemination is not coordinated. Also, WRA has signed MOUs with WRUAs in the catchment as required by the Water Act 2002. However, no MOUs have been signed with county governments. The Water Act 2016 has provision for Basin Water Resources Committees and County Governments' are represented in the committees with the role of advisory on water resources management issues within their respective basins; however, the Basin Water Resources Committees are yet to be established. Therefore, there is a clear need for building up coordinated governance for water resources management from the monitoring of water resources, sharing information, making decisions on the water at different levels. This project will aim to bring a coordinated approach to ensure that this information is collected, analyzed and disseminated for the different users including county governments, communities and industry. Necessary support materials and training activities are required to enhance capacity of government institutions as well as stakeholder groups to accomplish their respective role in coordinated water resources management.
54. A complete Hydro Met Station / Automatic Weather Station (AWS) comprises of a continuous monitoring of all meteorological and hydrological parameters at a specific location. These parameters include: Rainfall, Temperature (Max, Min and Actual), Wind direction and Speed, Humidity, Atmospheric pressure, Solar radiation and Soil Moisture. For RGS, water level is measured along the river channels. This information is very vital in the management of water resources by incorporating it in models to determine the future of state of water resources. Using this information some hydrological models will determine the future flood / drought situation. With the output

from the models, decisions on water harvesting and storage / drought risk mitigation will be reached. This will overall improve water security and hence food security

55. A "Radio Internet" – RANET station will be set us to broadcast in local dialect. The RANET Station will serve as learning institution for students on environmental and climate related studies. The radio station focus will mainly be on climate-related issues, market information, agriculture and emerging technologies in the climate change space. The direct beneficiaries will be the community settled within the project area in Machakos County. The station broadcast within a radius of approximately 50-75 km and mainly broadcast in local dialect and therefore will be able to reach all the target communities in the project area.

56. There is a clear need for building up coordinated governance for water resources management from the monitoring of water resources, sharing information, making decisions on the water at different levels. This output will also intend to bring a coordinated approach to ensure that this information is collected, analyzed and disseminated for the different users including county governments, communities and industry. Necessary support materials and training activities are required to enhance capacity of government institutions as well as stakeholder groups to accomplish their respective role in coordinated water resources management. Figure. 7 shows the reporting and monitoring framework that provide for coordinated governance of the WRUAs who are key water resources management stakeholders in Athi River Catchment.

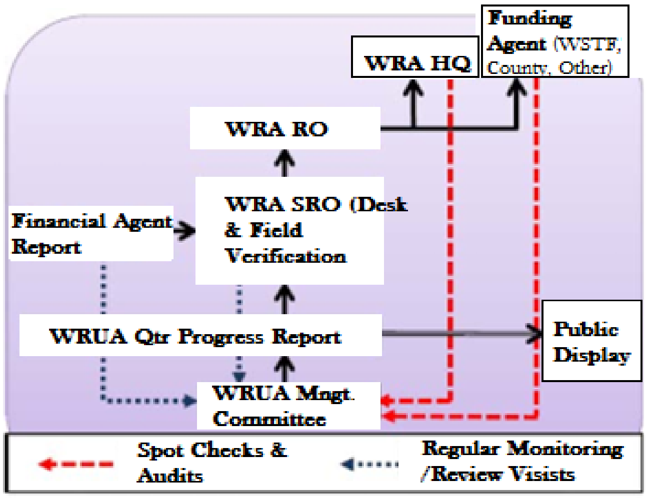


Figure 9: Reporting and Monitoring structure of WRUA Activities

57. This component also includes developing and implementing a capacity building, training and education program including formal training and retraining to build the capacity of NEMA, KMD and WRA staff on integrated water resource management and climate change for provision of accurate and timely information to Communities, County governments and relevant stakeholders in the Athi River Catchment for use in water resources management and adaptation planning.

58. The beneficiaries for output 1 include WRA and KMD for activity 1.1, 1.2 and 1.3, NEMA for activity 1.1, 1.3 and 1.4. KMD will acquire ownership of assets created under activity 1.1.3 and 1.2.4. WRA will acquire ownership of assets created under activity 12.1. and 1.2.3. NEMA will acquire ownership of assets created under activity 1.2.3. Hence WRA, KMD and NEMA will be direct beneficiaries of the mentioned activity lines. The county governments and the relevant water governance structures and communities within the project area are beneficiaries for all activities.

59. The eligibility criteria to apply in selection of beneficiaries under Activity 1.3.1 shall include; Geographical criteria: staff working within the project's geographical coverage; Job criteria: staff/members responsible for providing water/climate information to communities regardless of the institution; Project involvement: staff/members directly involved in implementing Project Activities either at the county or headquarters of the institutions and Equity criteria: priority should be given to gender balance, and consideration for staff with special needs e.g. disability.

60. The eligibility criteria to apply in the selection of beneficiaries under Activity 1.3.2 shall include; Geographical criteria: relevant organizations/committees must have mandates related to the project's objectives, and working within the geographical space of the project; Relevance: Organizations/committees/CBOs etc. should be involved in project-relevant work (i.e. work aligned with at least one of the project's activities) within Athi River catchment area; Equity: Selection of individual trainees should consider gender balance, persons with disabilities, and minorities and Priority given to WRUA members.

61. Table 2 explains in detail activities and sub-activities of output 1:

Table 2. Outputs and activities under Component 1

Component 1: enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector.	
Outputs	Activities
Output 1.1: Support the establishment of a National Information Center for Integrated Water Resource Management.	1.1.1 Set up an institutional mechanism for sharing information and getting feedback (data sharing protocols among players in the water sector e.g. through MOUs). • Policy dialogue Meetings aimed at developing data sharing protocols (between NEMA, WRA, KMD) and strategy for continuous communication between the user communities and county governments.
	1.1.2 Support the development of a National Information Centre for integrated water resource management (under the WRA HQ) by equipping it with necessary hardware and software facilities. • Acquire and install the centralized integrated water resource management information system (server, HDD 1TB, monitors, network facilities, UPS with additional batteries) with necessary software (operating systems, database, visualization, web server and other utilities).
	1.1.3 Support information analysis, forecasting and knowledge sharing platforms through an integrated database for weather and water resources-related information; • Develop an integrated analysis tools with visualization from monitoring information. The Decision Support System will assist in data entry, storage, analysis and reporting of collected data. The DSS is an information system that supports an organizations decision-making and facilitate its processes. • Develop a shared system of weather forecast information for development of early warning and install a flood and drought warning system in flood prone areas in Kiambu and Machakos counties • Develop web services for sharing information with stakeholders
Output 1.2: Establish modernized hydro-meteorological observation, monitoring, and testing systems and related networks	1.2.1 Installation of surface hydro-meteorological monitoring and network systems and rehabilitate existing monitoring systems (Nyandarua, Nairobi, Kiambu, Machakos county). • Install 25 Automatic Weather stations within the upper Athi catchment. all the stations will be equipped with Remote Terminal Units (RTUs), GSM/GPRS Modems, and the solar PV systems with solar battery, data protection equipment such as noise filters, surge protective devices. • Install 23 surface water level monitoring stations with multi-parameter water quality sensors for 20 stations. 5 rainfall monitoring stations. All the stations will be equipped with Remote Terminal Units (RTUs), GSM/GPRS Modems, and the solar PV systems with solar battery, data protection equipment such as noise filters, surge protective devices. 3 Water Level Stations will be manual hence will not require the sensors. The 5 rainfall monitoring stations will be distributed in specific sites in the project area and located within the Automatic Water Level Station (AWLs) compound.

	(Installation includes procurement of equipment and procurement of services for physical installation of that equipment)
	1.2.2 Upgrade water quality testing laboratory for WRA and NEMA • Acquire and calibrate devices for water quality test such as atomic absorption spectrophotometer, oxitops, Chemical Oxygen Demand (COD) digester, incubators, sediment analysis kit, High-performance liquid chromatography, Procure operational materials such as laboratory chemicals, reagents, and glassware. Both NEMA and WRA will be allocated funds for upgrading their water quality test laboratories
	1.2.3 Establish 10groundwater monitoring stations • Develop10 new monitoring wells in Nairobi, Kiambu and Machakos counties and Install necessary sensors to measure water level, conductivity, temperature, RTUs and other equipment for automated data transfer
	1.2.4 Setting up of a RANET broad casting station in Machakos County. • Installation of a RANET - "RAdio InterNET" broadcasting in approximately 50 – 75 Kilometers radius in the local dialect, Kiswahili and English for 24 hours, with an estimated audience of 1,000,000 listeners
Output 1.3 Support training and capacity building programs	1.3.1 Support capacity building and training for government institutions with water and climate change mandates within the project area. The executing entities will identity capacity building and trainings for their relevant institutions with water and climate change mandates within the project area and NEMA will approve and disburse fund accordingly 1.3.2 Support capacity building and training for relevant water governance structures and communities within the project area (Water Resource Users Associations -WRUAs, County Environment Committees - CECs, County Adaptation Committees, Community Based Organizations -CBOs among others) The executing entities will identify capacity building and training for relevant water governance structures and communities within the project area, develop training plan and manuals and NEMA will approve training plans and disburse fund accordingly
Output 1.4 Development of State of the Environment (S.O.E) reports	1.4.1 Support development of State of the Environment (S.O.E) reports • Consultative workshops to develop the S.O.Es.

Component 2: improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment.

62. The second output seeks to invest on improving and rehabilitating prioritized water infrastructure in the target counties considering climate change impacts and consequently increase access to potable water for domestic and improved livelihood. The component also covers catchment protection. Activities under this component will promote development of water harnessing and storage facilities for communities and local institutions in the counties considering historical and future climate change impacts. Athi catchment area has a number of degraded water storage facilities including water pans, sand dams and rock catchments that should provide critical access to water for communities. The project intends to rehabilitate these facilities to enhance water access for communities in close proximity to the structures. The water infrastructure assessed in Nyandarua, Kiambu and Machakos counties is ageing partly due to weather elements and effects of climate change. Water sources are affected by climate change through changes in annual rainfall and increased runoff resulting in decreased raw water quality. This causes poor water quality and scarcity and exerts stress on the water infra-structure affecting how communities can reliably access clean water. Through the consultation meetings and site assessments, rehabilitation and

construction of boreholes, water pans, springs, tanks, and sand dam in three counties (Kiambu, Nyandarua and Machakos) are proposed. Detailed list of intervention is available in Feasibility Study and the Technical study report.

63. Water storage and conservation allows communities to have water for a longer period of time especially during drought periods. Women and children particularly will benefit from increased access to water. In communities in the area, the roles of fetching water are left to women and children which in some cases can be quite difficult due to long distances covered to fetch water. Gender considerations will be considered when implementing the project to ensure that the benefits of access to water focus on the most vulnerable including women and children. It is expected that there will be a reduction in distances travelled to fetch water subsequently reducing the amount of time spent to fetch water. Reduction of costs of treatment for waterborne diseases and condition will also be one of the expected benefits of increasing access and availability of water.

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64. The restoration of degraded water catchment areas will be done through engagement of communities through existing water resources governance structures in the counties in rehabilitation/renewal/conservation and protection of degraded water catchment areas. Rehabilitation and reforestation activities within the selected counties will be carried out by WRA in collaboration with WRUA who will be engaged as service providers and eventually acquire the ownership of the established tree nurseries. Major water catchment degradation issues in Kiambu and Machakos counties are Sand harvesting, poor agricultural practices, wetland and forest area encroachment, untreated effluent discharges and vegetation clearing. The proposed safeguards for the structures to be constructed among other are as follows; Monitoring through the established integrated Water Quality &Pollution Control networks, Enforcement through waste disposal control plans, Issuance of effluent discharge permits / licenses, Enforcement of the set waste disposal standards and effluent discharge control plans. There exists no exclusion list, all entities shall benefit from the Funded Activity. The project will underscore on the successes of ongoing county led Climate Smart Agriculture- CSA practices. Additionally, the project will seek to network and build on the achievements with the ongoing Kenya Climate Smart Agriculture Project by World bank under implementation in 24 counties in Kenya; with the objective of increasing agricultural productivity and building resilience to climate change risks in targeted smallholder farming and pastoral communities in Machakos County.

65. Moreover, the project will liaise with a recent initiative launched by CIGAR and World Bank titled “Accelerating Impacts of CGIAR Climate Research for Africa (AICCRA).” This initiative aims to enhance access to knowledge, technologies, and practices to build the resilience of agriculture and food systems in Sub-Saharan Africa countries. The East Africa component is implemented in Ethiopia and Kenya through CGIAR Research Program on Climate Change, Agriculture and Food Security - CCAFS and Igad Climate Prediction & Applications Centre - ICPAC partnership. This project will partner with ICPAC and CCAFS EA to enhance use of agro-advisories and agricultural support decision support tools developed under the AICCRA and will promote climate-smart agriculture practices in the upper ARCA. The project will also participate in regular national and regional climate outlook forums (NCOF, GHACOF) for the co-development of tailored climate information services for targeted users thereby contributing to increased resilience of communities in the project area

66. The selection criteria of the infrastructure that will be rehabilitated (activity 2.2.1) was carried out through a stakeholder consultative process during the feasibility report development. The construction and technical standards that will be applied to the rehabilitation work will be in accordance to the guidelines provided for by the ministry of Water Sanitation and Irrigation in Kenya. For all the rehabilitated water structures, the management committees will ensure sustainability by actioning measures that ensure steady flow of funds for O&M.

67. The O&M post project implementation for those structures under water service providers in Kiambu, Nyandarua and Machakos will be done by the water companies. The structures under management by committees will require to register as water users’ associations under Water Sector Trust Fund - WSTF Community Project Cycle framework for community water supply and will entail development/review of by-laws to incorporate an aspect of water tariff to take care of O&M costs. During planning meetings at the point of project implementation, respective county governments and WSTF will be involved in order to incorporate their ‘pro-poor model’ for financing water structures.

68. The beneficiaries for output 2 are the relevant county governments, the relevant water governance structures and communities within the project area.

69. Table 3 explains in detail activities and sub-activities of output 2.

Table 3. Outputs and activities under Component 2

Component 2: improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment.	
Outputs	Activities
Output 2.1 Develop and upgrade water sources and infrastructure for domestic use considering present and future impacts of climate change.	2.1.1 Construct water storage and supply infrastructure • rehabilitate two boreholes, twelve water pans, two springs in Nyandarua county • build a new borehole, water pan, two springs and rehabilitate three boreholes, two water pans, two springs in Kiambu county • build two new boreholes, rainwater harvesting tanks and rehabilitate nine water pans, one spring and one sand dam in Machakos county
Output 2.2 Support conservation of water catchment areas as climate change adaption measure .	2.2.1 Rehabilitation and reforestation activities within the 4 target counties in collaboration with WRUAs • Establish tree nurseries for tree planting at new and rehabilitated water structures within the selected counties • Undertake planting of indigenous trees in two catchment areas with high risk of erosion in the target counties • Riparian marking and pegging (using indigenous tree) of 20km along Riara and Ndarugu Rivers • Construction of terraces (1000m) and 1500m gabions (gulley healing by planting vetiver grass and or installation of gabion boxes) in selected sites in Kiambu and Nairobi counties).

Component 3: Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions.

70. Regulations can constitute barriers or facilitate effective adaptation and therefore have a significant role on society and in driving the climate change adaptation agenda in Kenya. Ensuring compliance with the Water Act 2016 and Climate Change Act 2016 among regulated communities and enforcing the law is a critical means of contributing towards the realization of climate change adaptation agenda in the country. The Constitution of Kenya (COK) 2010 and the Climate Change Act No. 11 of 2016 require the public to be engaged in Climate change discourse in education, capacity building and awareness creation on actions to be undertaken relating to climate change.

71. This component seeks to strengthen institutional and regulatory framework by investing in capacities for enforcing the water regulations, supporting the four counties to develop their County Environment Action Plan (CEAPs) and Sub-Catchment Management Plans (SCMPs) for improved adaptation planning by incorporating climate change considerations into the plans. The Gazetted Water Quality regulation provides for protection of water resources, pollution control and has prohibition clauses for effluent discharge into the environment. These regulations apply to domestic industrial agricultural, recreational and any other purpose. In enforcing these regulations, the strategies used include water quality monitoring, restrictions and licensing.

72. Activities of this output will build the capacity of by supporting the review and implementation of sub-catchment plans with a focus on climate risk management in water sector. This is crucial to ensure that their catchment plans incorporate climate risk management approaches and activities. WRUAs will be supported in the implementation of sub-catchment plans in implementing climate smart activities.

73. The Activity 3.1.1 “Provide water quality monitoring systems to support enforcement of the water quality regulations” entails investing in capacities for enforcing the water regulations by providing equipment’s that will enable real time testing of water quality and enforcements of the regulations. Activity 3.1.2 “Sensitize the regulated communities to enhance compliance” entails having consultative forums with the regulated communities i.e. factories, filling stations, among others; whose industries emit effluents to the environment to inform and promote compliance to the existing good practices and regulations. Activity 3.1.3 “Conduct water quality pollution survey for point and non-

point sources of pollution" is a regulatory safeguard measure for sustainable protection of both surface and ground water resources. It aims at generating water quality and pollution information for planning, policy formulation and implementation; providing technological guidance for the waste emitters to develop and implement waste disposal control plans and enforcement of the prescribed regulatory waste disposal guidelines/standards and effluent discharge control plans.

74. The beneficiaries for output 3 are WRA, NEMA and the relevant county governments. The eligibility criteria to apply in selection of beneficiaries under Sub-activity 3.1.2 will include; Geographical criteria: The beneficiary individuals/institutions must be working within the geographical space of the project, Relevance: The beneficiary individuals/institutions" work must be directly related to the water sector, Equity: Selection of individual and institutions should consider gender balance, persons with disabilities, and minorities and Regulation Thresholds: The beneficiary institutions must be subject to or have a role enforcing water sector compliance requirements.

75. The project will create awareness and provide trainings to the beneficiaries drawing water directly form the water assets to be rehabilitated on proper hygiene / sanitation measures. All water assets will have draw pipes and disinfection dispenser will be availed on site as a hygiene measure. The disinfection dispensers are installed at the water draw off points to aid in the disinfection of remote potable water supplies (e.g. hand pump, springs etc.) 30% chlorine solution is contained in the dispensers. The measure is 1 drop per 20-liter container of water. The chlorine is allowed to mix for 15 to 30 min to allow for disinfection of the water making it potable. The disinfection dispensers are installed and refiled by the specific county public health office to all remote water access point i.e. access points not connected to central water supply system. Under Kenya's devolved governance system, County Governments are in charge of the provision of sanitation services as per Schedule 4 of the Constitution. All beneficiary counties in this proposal are implementing Community Lead Total Sanitation (CLTS) initiatives that seek to ensure all Kenyan communities are Open Defecation free among other Water and sanitation initiatives. The WASH strategies planned for implementation by the targeted counties are as reflected in the County Integrated Development Plan – CIDPs. There are adequate interventions on water and sanitation at the county level that will build synergy with the project. The project sits at the core of the respective county governments that have the legal mandate over water and sanitation issues in the county. This will ensure that developed water assets are protected from pollution that may arise from lack of water and sanitation interventions in the project areas. The USAID funded Kenya Integrated Water, Sanitation, and Hygiene Project (KIWASH) which seeks to improve access to improved water, sanitation, and hygiene covers Nairobi among 8 other counties. The Athi River project will seek to network and build on the achievements under this project in Nairobi County. The project will also work with other local level community based organizations that are engaged in WASH activities in the respective counties. Additionally, the project will work with partners implementing WASH activities in the targeted counties targeting schools, dispensaries and health centers for public awareness and behavior change, including borrowing lessons and building partnerships with UNICEF sponsored initiatives on hand washing and SATO pans toilet promotion activities in neighboring Kitui County that borders Machakos where the bulk of water assets are to be rehabilitated.

76. Table 4 explains in detail activities and sub-activities of output 3.

Table 4. Outputs and activities under Component 3

Component 3: Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions	
Outputs	Activities
Output 3.1 Enhance compliance of water regulations within the project area and improve adaptation planning	3.1.1 Provide water quality monitoring systems to support enforcement of the water quality regulations. These include; ***8 (eight)*** **Multi parameter sonde** (new hydrolab HL7) and ICT with ability to measure pH, temperature, nutrients (nitrates and phosphates), dissolved oxygen, total dissolved oxygen, salinity (conductivity), depth, chlorophyll, Rhodamine, blue-green algae, Ammonia, Chloride, etc. basically parameters that show effect of climate change on water quality; and its accessories; a ***Water testing kit for Microbiology analysis*** (POTALAB) with ability to detect Ecoli (Faecal contamination), Coliform (bacterial contamination), Salmonella (Typhoid), Pseudomonas (ENT

	infections); **3 (three) Potable Heavy metal detection meters** (Portable Metalyser for Heavy Metals) with the ability to measure Arsenic (III and Total), Antimony, Cadmium, Chromium, Cobalt, Gold, Mercury, Lead, Magnesium, Nickel, Selenium, Thallium, Tin, Zinc 3.1.2 Sensitize the regulated communities to enhance compliance: i) Promote WASH practices through hygiene awareness campaigns aimed at promoting water disinfection methods and hygiene practices in collaboration with the beneficiary county governments ii) Carry out Compliance Management Action Plan trainings / workshops to ensure compliance with the water sector laws and standards in Kenya (Both NEMA and WRA will identify stakeholder among the regulated communities to be sensitized, develop workshop schedules / plans and implement the activity. NEMA and WRA are government agencies with clear mandates in the water sector that require a coordinated approach to ensure compliance) Activity 3.1.3 Conduct water quality pollution survey for point and non-point sources of pollution
Output 3.2: Development of 4 County Environment Action Plans (CEAPs) including climate change adaptation considerations	3.2.1 Support development of 4 County Environment Action Plans (CEAPs)
Output 3.3: Development of 8 (eight) Sub-Catchment Management Plans (SCMPs) including climate change adaptation considerations	3.3 Facilitate development review and implementation of 8 Sub-Catchment Management Plans (SCMPs). i. Capacity building and SCMP development workshops

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Project implementation structure

77. The development of this project has been sponsored by several entities, the main one being NEMA, then Water Resources Authority - WRA, Kenya Meteorological Department - KMD and University of Nairobi -UoN. NEMA shall play the role of an Accredited Entity and partly in the implementation of some of the proposed activities in line with its accreditation credentials. NEMA has identified WRA and KMD as the executing entities for some proposed activities in line with their mandates. UoN has been identified as a service provider to develop decision support tool and formulate relevant training modules in the water and climate sector

78. The Figure 8 below illustrates the institutional arrangements to deliver the project. The institutions that will operate as the Executing Entities exist, however the committees will require to be formalized with guidance from the National Designated Authority - NDA before project implementation. The implementation will be conducted by the Executing Entities -EEs but will be guided by the National Project Steering Committee through quarterly meetings. The National GCF Steering committee will ensure that project implementation is conducted according to the country's priorities.

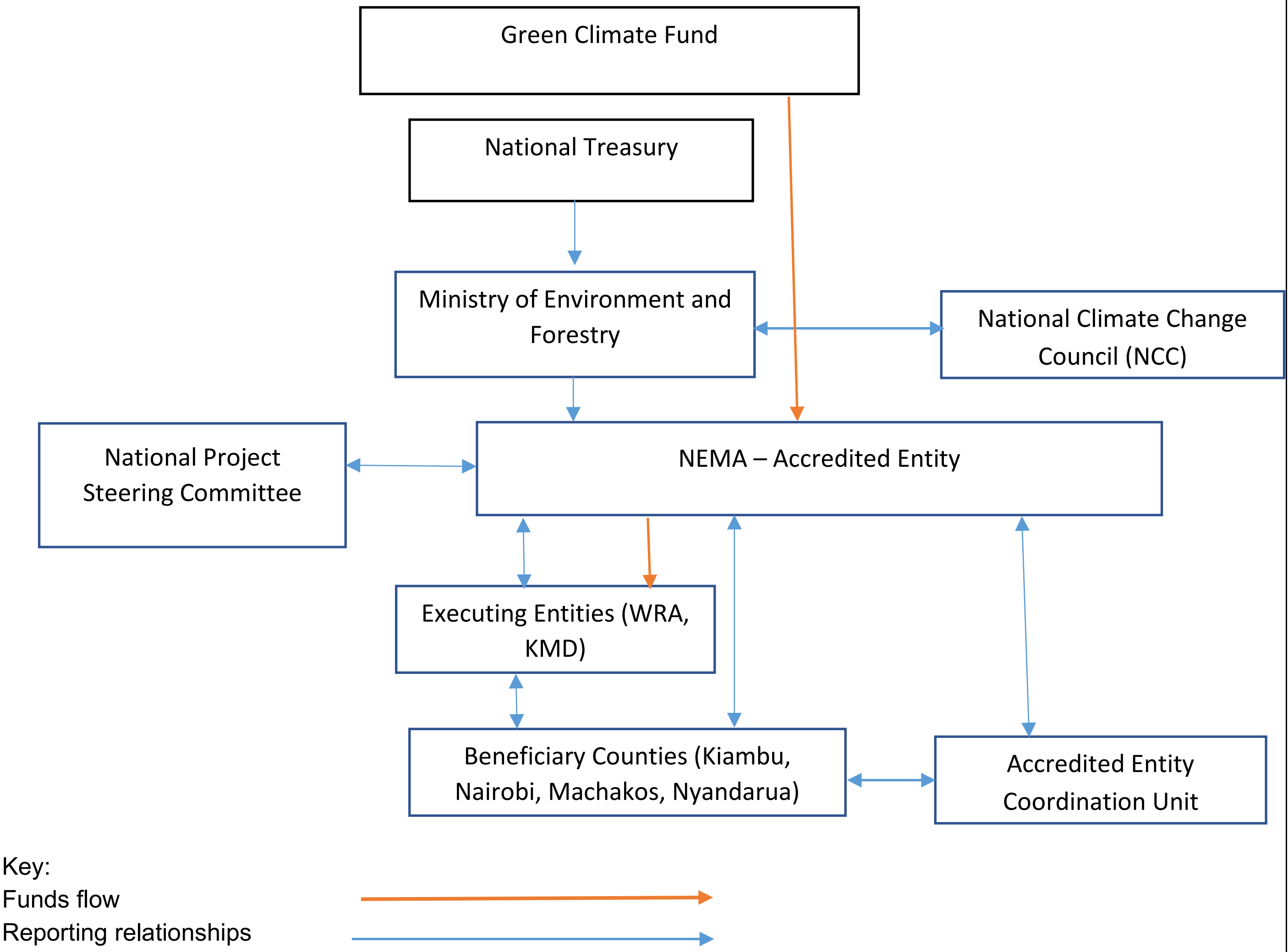

Figure 10: Project Implementation Arrangements

Roles and Responsibilities

79. National Treasury (NT)

The National Treasury (NT) is the National Designated Authority (NDA), the focal point for GCF in Kenya. The NT will ensure the Programme conformity with the national priorities. In addition, it shall ensure that PSC meets quarterly to review the progress and provide necessary guidance and direction to NEMA as the AE.

80. Project Steering Committee (PSC)
There will be an established PSC required to provide coordinated oversight to the project implementation. It shall be chaired by the Director General NEMA and will include representatives from the National Treasury, the Ministry of Water and Sanitation, and Ministry of Devolution and Planning, the Council of Governors and representatives of recipients' county governments. The PSC can appoint committee member from other relevant ministries based on necessity.

81. National Environment Management Authority (NEMA) as the Accredited Entity -A.E.
NEMA, will take the greatest responsibility for successful implementation of the project. As the Accredited Entity, NEMA shall submit requests for disbursement to GCF. NEMA will perform the following other roles:

- Project Management: NEMA shall be responsible for overall project planning, execution, coordination, monitoring, and evaluation and reporting.
- Reporting: NEMA shall prepare all necessary reports required by GCF.
- Budget and Auditing: NEMA shall be responsible for the overall project budgeting based on the agreed work plans prepared by the executing entities. NEMA will ensure project performance audits by the NEMA internal auditors and Kenya National Audit office.

82. National Environment Management Authority (NEMA) as the Executing Entity – EE
NEMA will be implementing the following project activities.

Output 1	Output 2	Output 3
Activity 1.4.1 Activity 1.2.2 Activity 1.3.1	Activity 1.3.2 Activity 2.1.1 Activity 2.1.2	Activity 3.1.1 Activity 3.1.2 Activity 3.1.3 Activity 3.2

The proposed activities will be integrated within the NEMA Work Programme and approvals to undertake the activities will be granted as per the existing institutional arrangements, approval and reporting requirements.

83. WRA as the Executing Entity – EE
WRA has demonstrated a strong framework on preparedness for project implementation. WRA will dedicate operational capacities for the implementation of the following project activities

Output 1	Output 2	Output 3
Activity 1.1.1 Activity 1.1.2 Activity 1.1.3	Activity 2.2.1 Activity 1.2.1 Activity 1.2.2 Activity 1.2.3	Activity 1.3.1 Activity 3.1.2 Activity 3.3

84. University of Nairobi (UoN) as a Service Provider
UoN is a training institution in Kenya and will be a Service Provider in the project that will work closely with WRA to develop decision support tool and formulate relevant training modules in the water and climate sector in the following project activities.
Activity 1.3.1 and Activity 1.3.2

85. Kenya Meteorological Department (KMD) as the Executing Entity – EE
The KMD is responsible for implementation of some of the activities; KMD will dedicate operational capacities for the Project execution of the following project activities
Activity 1.2.1, 1.2.4 and Activity 1.3.1

86. **County Governments:** The county government has two-fold mandates in the county, to provide leadership / coordination and to implement activities / service delivery. These roles are based on legal and institutional framework at national and county level mainly anchored on the EMCA 1999 and Water Act 2016. Water Resources Users Associations provided for in the Water Act has representation in the county both to implement project activities and operate them even after the project period. Among the devolved functions of the county government include implementation of specific national government policies on natural resources and environmental conservation, County planning and development. and Ensuring and coordinating the participation of communities and locations in governance at the local level and assisting communities and locations to develop the administrative

capacity for the effective exercise of the functions, powers and participation in governance level which are key to the successful implementation of the Programme. The county government will be the entry point in the initiation of the planned activities and collaborative working relations will be established. For this project, the County Governments are project beneficiaries and will perform operation and maintenance of the rehabilitated adaptation assets

87. The reporting line between GCF, National Treasury and MoEF are as follows: The Ministry of Environment and Forestry (MoEF)is in charge of the climate change policy frame work in the country including NCCAP, NAP among others and reports to the UNFCCC on the climate change initiatives and achievement of the NDC in the country. National Treasury (NT) is the National Designated Authority- NDA providing broad oversight of GCF activities in the country, NT provide reports relating to the financing of climate change activities in the country to MoEF and other government institutions. NT organizes stakeholders' consultations and provides a for No Objection letters for funding proposals to the GCF.

88. Funds Flow arrangements

The GCF will disburse funds to NEMA as per the approved funding proposal budget. NEMA will disburse funds to the Executing Entities (WRA, KMD) based on the approved work plans and requisite reports. NEMA will be implementing some of the Programme activities that will be integrated within the NEMA work Programme and approvals to undertake the activities granted as per the existing institutional arrangements, approval and reporting requirements.

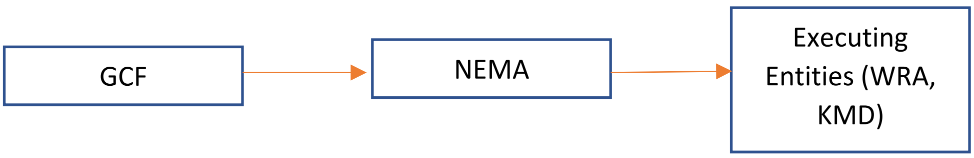

Figure 11: Funds Flow Diagram

Experience and track record of the AE and EEs

89. Accredited Entity (AE) - National Environment Management Authority (NEMA)

NEMA was established in 2004 as the principal government agency mandated to implement environmental policies, exert supervision and coordination on all matters environment. NEMA as the Accredited Entity (AE), takes the greatest responsibility for successful implementation of the project and will perform the roles of Project management, Reporting Project supervision and project budgeting. NEMA has robust technical team in the 8 core departments in NEMA, namely, Directorate, Environment Education, Information and Public Participation Department, Legal Services Department, Environmental Planning and Research Coordination Department (EPRC), Compliance and Enforcement department (C&E), Finance and Administration Department and Coastal, Marine and Fresh Waters Sub-Department. Hence a complementary relationship that allows for efficiency and effectiveness. NEMA has operational county offices in all the 47counties in Kenya. NEMA is also the National Implementing Entity with the Adaptation Fund Programme and is currently implementing a USD 10,000,000 Programme aimed at building resilience to climate change and adaptive capacity of vulnerable communities in Kenya, undertaken in 14 counties in Kenya. NEMA will implement some project activities under this project in line with its accreditation credentials. NEMA will be implementing activity The Construction of the water infrastructures will be executed by NEMA under a civil works contract. The O&M will be undertaken by the respective county governments and the management committees of the water structures established

90. Water Resources Authority - WRA

Water Resources Authority (WRA) is a state corporation under the Ministry of Water and Irrigation. The organization has been in existence for the past 12 years following its establishment under the Water Act (2002) that has now been replaced by the new Water Act 2016. WRA's mandate is to regulate, monitor, assess and allocate the water resources; and to collaborate with other institutions as required for better water resource management.
WRA has established six regional offices and 26 sub-regional offices across the country that report to the national office based in Nairobi. WRA is a technical institution and has competent staff to implement its various roles and mandates.

WRA has a capacity of over 750 staff. The Authority is governed by a Board of Directors consisting of 10 members and a senior management staff composed of 3 staff that include the Chief Executive Officer, the Technical Coordination Manager and the Finance and Administration Manager.

WRA uses a basin-based approach in accordance with the National Water Management Strategy in management of water resources countrywide. Accordingly, WRA has developed six Catchment Management Strategies (CMS) for the Country's five drainage basins namely: Lake Victoria Basin which is divided into North and South, Rift Valley, Athi, Tana and Ewaso Ng'iro North. The CMS are implemented at community level through development of Sub Catchment Management Plans by WRUAs. The Senior Management is supported by various component heads based at the head office under the Technical and Support Services. Since 2013 WRA's is currently managing an average of USD 20 Million per year and sufficiently equipped to manage other projects. The projects that WRA conducted are listed in section E5.2

	Kshs Budget	USD equivalent	Kshs Expenditure	USD equivalent
2013/2014	1,732,153,992	17,321,539.22	2,022,508,375	20,225,083.75
2014/2015	1,489,602,657	20,225,083.75	1,748,749,975	17,487,499.75

91. WRA Budget 2013/14 & 2014/15.

Section 69 of the Water Act, 2016 provides for handover of completed waterworks to the county government jointly with the Water Service Provider for use. The County owned water service provider is established as a public institution and will manage public water services assets on behalf of the public. WRUAs can serve as agents of WRA to monitor and ensure rational water use and allocation. Section of the Water Act 83 provides for a County owned water service provider established as a public institution and operating and providing water services to hold the county or national public water services assets on behalf of the public. WRUAs can serve as agents of WRA to monitor and ensure rational water use and allocation.
The water quality monitoring systems will be received by both WRA and NEMA county offices

The University of Nairobi (UoN) as a service provider will work closely with WRA to provide specialized training as part of the capacity building efforts proposed in this project. UoN is a premier training institution with a track record of excellence. Established in 1956 as the only institution of higher learning in Kenya for a long time, the University is credited for training regional and Africa's high-level manpower. The University has diversified academic programmes and specializations in basic sciences, applied sciences, technology, humanities, social sciences and the arts.

92. Kenya Meteorological Department (KMD)

The Kenya Meteorological Department (KMD) is a department under the Ministry of Environment and Forestry with the overall function of provision of meteorological and climatological services to various development sectors for better utilization of natural resources for national development. The KMD as the National Meteorological and Hydrological Service (NMHS) is the authoritative source of weather and hydrological warnings and also responsible for climate, air quality, and tsunami warnings. The KMD continuously monitors the earth's environment, develops predictions on potential changes related to weather, climate and water, and issue as timely and accurate warnings as possible of most hydro meteorological hazards. The Department has been operating on Presidential Executive Order No. 1 of 5th June 2018 which places the Department under the Ministry of Environment and Forestry.

KMD is a department within the Ministry of Environment and Forestry and therefore it doesn't have a legal personality as an independent entity. Kenya Meteorological Department as an executing entity will be representing the Ministry of Environment and Forestry.

Kenya Meteorological Department is headed by the Director of Meteorological services supported by 6 Deputy Directors. Other senior staffs are 15 Senior Assistant Directors and 35 Assistant Directors who are in charge of various divisions in the Department. KMD has decentralized its services according to the devolved government in Kenya. It has a County Director of Meteorological Services (CDMS) in each county who reports to the Director of Meteorological services. Further, KMD has 39 manual synoptic stations which are manned by trained KMD staff. Other stations are Automatic Weather Stations (AWS) distributed in the whole country that collect weather information and send to the headquarter automatically. KMD has a capacity of over 500 staff based in Nairobi, all county headquarters and in 39 manual synoptic stations. KMD hosts the National Flood Forecasting and Early Warning Centre (NFFEWC) based at the headquarters in Nairobi which is charged with collection of data, analysis, modelling and forecasting of flood. The projects that KMD has conducted are as listed below:

	Kshs Budget	USD equivalent	Kshs Expenditure	USD equivalent
2013/2014	102,298,345	1,022,983.45	90,345,456	903,454.56
2014/2015	435,398,098	4,353,980.98	401,764,915	4,017,649.15
2015/2016	1,286,762,629	12,867,626.29	1,124,275,197	11,242,751.97

93. Table 4 summarizes the activities for each executing entities:

Table 4. Summary of project activities and executing entities

Executing Entities	Activities to be implemented	
Water Resources Authority - WRA has demonstrated a strong framework on preparedness for project implementation. WRA will dedicate operational capacities for Project execution of their proposed activities	Activity 1.1.1 Activity 1.1.2 Activity 1.1.3 Activity 2.2.1 Activity 1.2.1	Activity 1.2.2 Activity 1.2.3 Activity 1.3.1 Activity 1.3.2 Activity 3.1.2 Activity 3.3
Kenya Meteorological Department (KMD) is responsible for implementation of some of the activities; KMD will dedicate operational capacities for the Project execution of their proposed activities.	Activity 1.2.1 Activity 1.3.1 Activity 1.2.4	
NEMA NEMA will act in its role as an Accredited Entity that allows for implementation of the proposed activities in line with its accreditation credentials and work in close collaboration with the Executing Entity and the County governments that are key stakeholders in project activities implemented within their jurisdiction. In this project, the beneficiary county governments will be as outlined: Nyandarua, Kiambu, Nairobi and Machakos.	Activity 1.4.1 Activity 1.2.2 Activity 1.3.1 Activity 1.3.2 Activity 2.1.1	Activity 2.1.2 Activity 3.1.1 Activity 3.1.2 Activity 3.1.3 Activity 3.2

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

94. Grant financing is requested from the GCF to enhance adaptive capacity of the communities in the upper Athi catchment, an area with high risk on water resources due to climate change. Although Government of Kenya develops and implement climate change adaptation measures based on National Water Master Plan, it is not enough to resolve the issues associated with frequent water deficit and floods, the limited technical and institutional capacity to cope with climate change, limited range of products, services for climate resilient development, low awareness and inadequate regulatory framework for water resource management are key barriers to overcome.

95. The proposed project purpose is to create a best practice of climate change adaptation practice of the water sector in Kenya to overcome those barriers by developing climate-informed decision support capability of institutions and communities with ICT based information products based on modernized hydro-meteorological monitoring systems. Upgraded water infrastructure and strengthened regulatory framework with updated county-based action plans will enhance awareness and its impact on community level. The County Environment Action Plan (CEAP) is futuristic and helps in better planning of both relevant interventions and available resources to combat the effects of climate change. GCF funding is sought to fill the gap that exists in the developing the statutory document that is key in planning the measures to address climate change impacts and allocating available resources to the same cause.

96. The project aims to enhance institutional capacity by building up systems and infrastructures that directly contribute to national adaptive capability; it is not a readiness project which concerns preparatory capacity of institutions for full project. Stakeholder engagement in this project is has been extensive during the planning stage and public participation is embedded in the Kenyan constitution and hence will be upheld during the implementation of the project.

Without GCF Funding.

97. The National Water Master Plan (NWMP) was launched in 2014 to enhance a capacity of water sector in Kenya. However, a lack of financing has limited the efficiency of this Plan. [7]The costs of adapting existing urban water infrastructure in Africa have been estimated at between US$ 1,050 million and US$ 2,650 million annually, implying this budget might be not affordable to developing countries in Africa.

98. Information system, comprising from monitoring climatic and hydrological data to distributing useful information to local communities, has not been working properly due to the lack of equipment and facilities. Furthermore, the capacity of water supply and storage service systems is still not enough to respond to variability in precipitation. However, a substantial investment is essential to improve these systems.

99. The institutional capacity cannot be improved. Furthermore, unawareness of local communities on the issues on the water sector limits a success of NWMP at local scales. Appropriate financing for capacity-building of institutions and raising awareness is required.

100. Low compliance to water quality regulations due to inadequate capacity to enforce and consistent monitoring. Inadequate information system and lack of institutional capacity might constrain adaptive measures to disasters. As risks of disasters (drought and flood) are expected to increase in the future, according to the feasibility study report, solutions on these circumstances are urgently required.

With GCF Funding:

101. The measures proposed in this project will increase to water security, resilience to people, ecosystems and economies in the four target counties
102. GCF funding will strengthen hydrological and meteorological information services to deliver relevant, accurate and timely climate information to local communities, and also to support decision making and policy development in the water sector. This will be achieved by investments in optimized hydro-met monitoring networks, more effective management and exchange of hydro-met data; and improving the capacity to forecast future water and weather conditions. Ultimately, this information will be used to strengthen early warning systems. Requisite training programmes will run along the proposed investments
103. GCF Funding will support investment improving and rehabilitating prioritized water infrastructure in the target counties so as to increase water security.
104. GCF funding will improve compliance to water quality regulations through strengthened capacity and consistent monitoring.
105. Capacity-enhancement of the water related governmental institutions becomes possible by GCF funding. This will increase effective management of water resources, thus decreasing the impacts of climate change. Training and Raising awareness of local communities derives change in behavior against disasters.
106. Furthermore, once this project is implemented successfully with GCF funding, it can play as an exemplary case with which the Kenyan government can leverage other funding sources from a variety of other funding providers such as MDBs and bilateral aid agencies. The success of the project that focuses on water catchment will be replicated in other catchments in Kenya and it will serve a point of reference and the documentation. The lessons learnt will be informative to the future processes hence ensuring high success of future implementation of a similar project.
107. The project activities are subject to the VAT as the proposed activity items are not zero-rated products in Kenya.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

[7] Adapting to climate change: water management for urban resilience, Mike Muller

108. The project will be led and implemented by local institutions including the Water Resource Authority and the Kenya Meteorological Department to ensure ownership from existing relevant institutions and for capacity building. It is expected that the project will further the work that NEMA, WRA and KMD are currently executing and are mandated to do. This will ensure that project activities continue beyond the lifetime of the project. The institutional mandates of the executing entities and the involvement of the county governments coupled with the public participation involving the community members which is a mandatory requirement in the Kenyan Constitution will ensure that the sense of ownership of the proposed interventions are inculcated early in the project development and planned implementation. This will ensure sustainability of the project assets and processes. The management committees established that will constitute of the representation of the WRUA members as part of the Operation and Maintenance will provide for sustainability of the project interventions.

109. KMD and WRA have put across appropriate structures to ensure sustainability beyond the project period. The institutions have decentralized its services to the county government in line with the devolved government. Therefore, the budget for Operation and Maintenance is going to be factored in when funding County offices after the project closure.

110. Additionally, Water Resource Users Associations -WRUAs has established by WRA are a vehicle for community participation to enhance ownership and sustainability of interventions on the ground. They are an association of water users, riparian landowners, or other stakeholders who have formally and voluntarily associated for the purposes of co-operatively sharing, managing and conserving a common water resource. With WRUAs in place as an organized group sustainability is also assured.

Close engagement with local governance structures at community level

111. Implementation will also involve local/community level water resource governance structures, in this case WRUAs, CECs among others. Engagement and collaboration will lead to enhanced participation from communities in water resource management and planning. It will also allow communities to address their present needs with regards to water resources.

Strengthened institutional capacity and awareness

112. The institutional capacity of NEMA, WRA and KMD will be strengthened to ensure appropriate water resource management for climate change adaptation planning. Capacity building will be a key activity with regards to use of climate data for resource planning and will also be supported by participation in training for a for expert courses in the areas of integrated water resource management and climate change. This will further enable for transfer of knowledge and expertise to communities and other stakeholders.

Establish feedback mechanism systems

113. To ensure that the climate information coming into the catchment area meets the users demands it will be important to create a feedback mechanism. This feedback mechanism will contribute to long-term sustainability through keeping the relevance of hydromet services.

Operation and Maintenance

114. During the installation and initial operation for the first half of the project period, comprehensive operation and maintenance work will be assessed. Through legislation process of such community plans, the assessed roles and responsibility on the operation and maintenance will be included and distributed to the different stakeholders and government institution. Contribution from government of Kenya during the project implementation will be in-kind support through commitment of staff time, office space and part of travel expenses.

115. To ensure sustainability of the new stations and structures beyond the project period, WRA and KMD will request the government to allocate more O&M funds for their sustainability beyond the project period and the county government will carry out the O&M of the rehabilitated / established structures after project hand over. WRA has an average annual budget for ground water and surface water O&M as 300,000 USD and 500,000 USD respectively. Additionally, the beneficiary County Governments have confirmed and committed to conduct O&M of the project adaptation investments through the project life span and beyond as they have an annual allocation for the O&M of water works. The average annual allocation for O&M in the counties is 830,000 USD, 480,000 USD and 400,000 USD for Machakos, Kiambu and Nyandarua county respectively. Annex 21 (O&M plan) has annexed the commitment letters from the respective beneficiary County Government

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		**Total amount**			**Currency**			
		9,526,603.26			million USD ($)million USD ($)million USD ($)million USD ($)million USD ($)million USD ($)			
GCF financial instrument		**Amount**		**Tenor**	**Grace period**		**Pricing**	
(i)	Senior loans	Enter amount		Enter years	Enter years		Enter %	
(ii)	Subordinated loans	Enter amount		Enter years	Enter years		Enter %	
(iii)	Equity	Enter amount					Enter % equity return	
(iv)	Guarantees	Enter amount		Enter years				
(v)	Reimbursable grants	Enter amount						
(vi)	Grants	**9,526,603.26 Million**						
(vii)	Results-based payments	Enter amount						
(b) Co-financing information		**Total amount**			**Currency**			
		473, 380			million USD ($)million USD ($)million USD ($)million USD ($)million USD ($)million USD ($)			
Name of institution		**Financial instrument**	**Amount**	**Currency**	**Tenor & grace**	**Pricing**	**Seniority**	
National Environment Management Authority (NEMA)		In kind	242,560.00Gov. of Kenya	Options USD ($)	Enter years	Enter%	Options	
Water Resources Authority (WRA)		In kind	220,820.00	Options USD ($)	Enter years	Enter%	Options	
Kenya Meteorological Department (KMD)		In kind	10,000.00	Options USD ($)	Enter years	Enter%	Options	
Click here to enter text.		Options	Enter amount	Options	Enter years	Enter%	Options	
(c) Total financing (c) = (a)+(b)		**Amount**			**Currency**			
		9,999,983.26			million USD ($)			
(d) Other financing arrangements and contributions (max. 250		*Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee).*						

words, approximately 0.5 page)	*Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets.* *Please also include parallel financing associated with this project or programme.* *N/A*

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

Component	Output	Indicative cost OptionsUSD($)	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount OptionsUSD($)	Financial Instrument	Name of Institutions
Component 1. Enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector	1.1 Support establishment of National Information Centre for Integrated Water Resource Management	349,310.00	179,050.00	Grants	170,260.00	In Kind	WRA
	1.2 Establish modernized hydro-meteorological observation, monitoring, and testing systems and related networks	2,026,113	2000113.00	Grants	10,000.00 12,000.00 4,000.00	In Kind	KMD WRA NEMA
	1.3 Support training and capacity building programs	200,000.00	200,000.00	Grants	0		
	1.4 Support development of State of the Environment (S.O.E) reports from the information developed	169,325.00	149,325.00	Grants	20,000.00	In Kind	NEMA
Component 2. Improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation	2.1 Develop and upgrade water sources and infrastructure for domestic use considering present and future impacts of climate change.	4,866,540.26	4,770,540.26	Grants	96,000.00	In Kind	NEMA WRA

activities in the catchment							
	2.2 Support conservation of water catchment areas as climate change adaptation measure.	128,411.00	128,411.00	Grants			
Component 3. Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions	3.1 Enhance compliance of water regulations within project area and improve adaptation planning.	813,229.00	808,829.00	Grants	22,000.00 22,000,00	In Kind	WRA NEMA
	3.2Development of 4 County Environment Action Plans (CEAPs) including climate change adaptation considerations	226,950.00	178,950.00	Grants	48,000.00	In Kind	NEMA
	3.3 Development of 8 (eight) Sub-Catchment Management Plans (SCMPs) including climate change adaptation considerations	166,135.00	166,135.00	Grants	0		
Support to Programmes activities	Monitoring and Evaluation	414,000.00	414,000.00	Grants	0		
	Development of Knowledge products	150,000	150,000	Grants			
Project Management Unit	Programme operational costs	439,970.00	370,850.00	Grants	34560.00 34,560.00	In Kind	NEMA WRA
Contingency		50,000.00	50,000.00				
Indicative total cost (USD)		**9,999,983.26**	**9,526,603.26**		**473,380.00**		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☐ No ☐

116. The project aims at developing and implementing a capacity building, training and education program including formal training and retraining to build the capacity of NEMA, KMD and WRA staff on integrated water resource management and climate change for provision of accurate and timely information to Communities, County

governments and relevant stakeholders in the Athi River Catchment for use in water resources management and adaptation planning

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

Expected Increased resilience of health and well-being, and food and water security

117. Water security contributes to job creation, increased gross domestic product (GDP) and enhanced development across most sectors – health, energy, agriculture, environment, mining, industry and social protection. It also improves the adaptive capacity of communities to climate change effects. The project expects to target 1,156,620 beneficiaries directly (49% male and 51% female) through increased access to water sources i.e. water storage, rainwater harvesting and rehabilitation of degraded structures. This increased access to water will also ensure that water for agricultural production is available. In addition, it will reduce time taken to fetch water among communities. The project will ensure that water is accessible in good quality by enforcing existing water regulations thus reducing incidences of water borne diseases and overall increasing the community's wellbeing.

Expected strengthening of institutional and regulatory systems for climate-responsive planning and development (PMF- A 5.0 and related indicator(s):

118. The project will aim to strengthen the capacity of NEMA, WRA and KMD in data collection, synthesis and dissemination to a larger audience. The project will help provide the necessary equipment and training to WRA and KMD to set up telemetric hydromet stations that will offer timely data for decision making. At the grass-root level the project will work at equipping WRUAs to incorporate and implement climate change issues within their SCMPs and enforce policies related to water use.

Expected increase in generation and use of climate information in decision-making (PMF- A 6.0 and related indicator(s)):

119. The project aims to boost quality hydromet data collection in the selected catchment areas and ensure timely relaying of the data. The project will further expand the system to have a decision support system that is aimed at supporting key decision making with regards to water resource management at catchment level. The data shall be analyzed to provide useful information for all stakeholders. It is envisaged that the information will be utilized for decision making.

Expected strengthening of adaptive capacity and reduced exposure to climate risks (PMF- A 7.0 and related indicator(s)), focusing particularly on the most vulnerable population groups and applying a gender-sensitive approach:

120. The use of climate information in planning for water resources will allow development planners to consider climate risks with an aim to reduce exposure for communities. Additionally, the provision of warnings of flooding events will reduce the risk of disasters, loss of lives and economic losses leading to reduced exposure to climate risks for vulnerable communities.

Expected strengthening of awareness of climate threats and risk-reduction processes (PMF- A 8.0 and related indicator(s)):

121. A key aspect of the project is building the capacity of local institutions including NEMA, WRA, KMD and County Governments on climate change impacts and the potential risks that come with it. This approach will allow these institutions to strengthen their awareness of climate risk while understanding the options and interventions required for risk reduction. The information generated from the DSS will be relayed to the communities in order to create awareness on climate threats and risks and the requisite risk reduction mechanisms.

122. Direct beneficiaries are calculated as the number of persons who will directly participate in the project and draw direct benefits from improved access to water as determined during feasibility study. Indirect beneficiaries: Total number of persons in the 4 counties within the catchment who derive indirect benefits from the project – reduced flooding, water resource conflicts, information, ecological function of the ARC, agricultural and livestock value chains, among others.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

123. The proposed project aims to enhance the resilience of communities and ecosystems in the ARCA through provision of hydromet information, mainstreaming climate change adaptation into sub catchment management

planning, development of a decision support tool and rehabilitation of riparian areas and degraded water structures. Short term benefits will include reduced travel time to water points, improved health/reduction of waterborne disease cases and increased access to water whilst the long term benefits will include reduced economic losses from flooding events, improved water resource management and strengthened decision making and development planning.

124. The proposed project's approach on use of hydromet information has been established globally as best adaptation practice to reduce and manage climate risk. In Kenya it is yet untested, and the implementation of this project will be the first of its kind. The paradigm shift in this case will be local government involvement in use of climate information for development planning and resource management through the integration of meteorological, hydrological, hydrogeological and water quality data for improved water resources management at national and county levels. The country has yet to adequately mainstream climate change adaptation into its development planning to reduce climate risk. County governments working with WRA and KMD will establish mechanisms for sharing hydromet information to determine current and future climate risks and plan development interventions especially in the water sector with this information in mind. It is anticipated that the approach will be successful, and lessons learnt from implementation may influence scaling up to the other catchment areas in the country.

125. Lessons learnt from the implementation will also establish best practice and build the capacity of county governments in adaptation planning at local levels which will also assist them in enhancing climate risk management measures. The project will emphasize lesson and knowledge sharing between the EEs and other counties along the Athi river, to enhance sustainability and capacity of the regional and national offices and create awareness at the county level.

126. The executing entities (both KMD and WRA) are National Government Agencies hence implementing policies formulated at the National Government level. However, these agencies have decentralized their functions at the county government level. Therefore, the national policy and planning will be implemented at the county / local government through the decentralized functions and therefore easier to replicate these projects to other five catchments

D.2.2. Potential for knowledge and learning

127. All EEs, shall be required to institute a protocol on how to capture knowledge, aggregate it per audience and disseminate it. All project specific activities that relates to training on the proposed acquisition of modern hydromet equipment, shall be documented for future referencing. Knowledge products will be developed for the programme to document and disseminate the lessons learnt and the successes during project implementation. They will constitute of brochures, magazine, bi annual programme status reports, banners and information posters. Knowledge sharing workshops will be conducted.

128. The project presents several opportunities for learning ensuring that the knowledge and lessons generated through this project will be captured and disseminate through direct (RANET broadcasting) and indirect measures (knowledge products and adaptation planning tools). Knowledge sharing platforms will be employed and will include impact stories, videos, journal articles and others.

129. The proposed activities will entail hiring of a consultant to undertake content creation and development for the programmes knowledge products, data collection and validation workshops. The expected topics for sharing include; building climate change resilience for water resources, creating awareness of climate change and water resource management, women as drivers of change in managing climate-resilient water resources among others

D.2.3. Contribution to the creation of an enabling environment

130. During the project period, government entities, private sector, researchers and local communities will be involved. In order to ensure effectiveness and sustained participation, the project shall strongly focus on institutional strengthening and capacity building, communication enhancement as well as improvement in service provision. The activities will also include coordinated dissemination of hydromet information, development of county policies and incentives for innovative water conservation solutions. This will allow for participation of the county governments, private sector, water resource management institutions and communities to create an enabling environment for participatory water resource management. The project will also enable coordination between public and private institutions to ensure benefits reach the most vulnerable and are sustained.

131. The project activities are meant to catalyze private sector involvement for sustainability of the adaptation interventions that ensure their businesses adjust well to consequences of climate change. Stakeholder engagement

through compliance inspections, trainings, awareness creation, audits, and adherence to license conditions among other actions will encourage industries to promote integrated water resource management and climate change adaptation. This engagements are expected to contribute significantly to a paradigm shift in the attitudes and behavior of the various stakeholders towards better awareness on climate change and sustainable water use.

D.2.4. Contribution to regulatory framework and policies

132. The Project supports the implementation and operationalization of several key national policies strategies and plan, including the National Climate Change Response Strategy (NCCRS), National Climate Change Action Plan (NCCAP), National Adaptation Plan (NAP), the Intended Nationally Determined Contribution (INDC) and the National Water Master Plan (NWMP) 2030.

133. The NCCRS and NCCAP prioritize water resource management as a key strategy to address drought. The NCCAP expounds that water resource management is linked to Kenya's expected economic and social transformation and is directly linked to food security, health and GDP growth especially in the ASALs. The project will support the implementation of listed water sector priority adaptation actions to improve water management specifically increased domestic water supply and participatory water resource management.

134. The NWMP 2030 seeks to implement water policy to meet social water demand up to 81% % through 2030–2050, which only 14% was attained in 2010. The project will also support the implementation of the NWMP 2030 specifically with the actions focused on the Athi River Catchment area. The project components are anchored on the recommended actions in the plan for the catchment area. The project is expected to strengthen mainstream climate change adaptation into the water sector and allow for adaptation planning.

D.3. Sustainable Development Potential

Wider benefits and priorities

Environmental, social and economic co-benefits, including gender-sensitive development impact

135. Kenya is highly endangered by serious natural disasters; the country is extremely exposed to droughts and floods. Naturally the country is highly vulnerable to climate condition caused by movement of the Intertropical Convergence Zone (ITCZ) and the El Niño Southern Oscillation (ENSO). The problem faced in Kenya is that this rainfall pattern, duration of dipole rainy season, and the amount of precipitation has been changed due to global climate change. Since the 1960s, precipitation has also been changed; greater rainfall has been observed during the short rains, and the long rains have become increasingly unreliable in locations.

136. Kenya has been determined to be one of the highly vulnerable countries in the world and the country need practical supports to survive under this unpredictable condition. In terms of water sector, even though Kenya recently graduated from the list of Least Developed countries (LDCs) ranking by the UN, its water coverage as at 2015 (63% in the Joint Monitoring Programme report by the United Nations) was lower than both the LDC's average of 69%, and Sub-Saharan Africa average of 68%. According to the water supply statistics data presented by JMP for 2015, the Kenyan portion of people without safe drinking water is close to 3% of the global figure.

137. It is clear that challenges facing the water sector, range from water scarcity, water quality, population pressure, climate change and others. Traditionally, women bear the primary responsibility for providing water and ensuring its quality in developing countries like Africa region and women in Kenya are regularly confronted with violence during their chores, especially gathering water. Close to half of Kenya's population (45.9% in 2006) is considered poor and more than three-quarters of them (79% in 2006) live in rural areas. In Kenya, rural areas are characterized by dispersed settlements thus making it costly to invest in piped water systems. Thus, majority of the water sources are point sources. It is estimated that 40% of the households in rural areas of Kenya use more than 30 minutes for round trip to obtain drinking water from the source.

138. The improved hydromet services and decisions on the water resources management together with improved water infrastructure will enhance the access of the water resources in more sustainable manner. Community based planning and implementation of the management plan will increase ownership of climate-informed water resources management leading to better perception of project outcomes.

139. The projects would have co-benefits of Improved sanitation and hygiene from improved access to water as reflected in the decreased prevalence of water-borne diseases and Catchment rehabilitation of the riparian areas will improve biodiversity and air quality.

140. The project would have the following co-benefits: a) Improved sanitation and hygiene from improved access to water as reflected in the decreased prevalence of water-borne diseases and b) Catchment rehabilitation of the riparian areas will improve biodiversity and air quality. One of the beneficial impact of the project is on people's health. This is because the project will eliminate the currently high dependence of communities on contaminated water supplies for drinking and basic hygiene, and therefore the significant negative impact this has on people's health and wellbeing.

D.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

D.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

141. The table below illustrates the vulnerability of the counties within the Athi River Catchment Area with a focus on poverty, access to water and education rates. The counties are included as the focus area for the project. The project will focus on the most vulnerable areas and impact of the project.

County	Climate Change Related Impacts	Poverty[8]	Access to improved sources of water[9]	[10]Percentage of population with no education
Nairobi	• High temperature. • Resource based Conflict • Damage from floods, • Waterborne diseases	22.5	84%	11%
Kiambu	• Flooding and soil erosion • Extinction of some crops like plums • Drying of small streams and wells • Drying of wetlands e.g. Manguo in Limuru, Ng'arua in Ngesha • Mosquitoes have found new habitats due to increase in temperature, e.g. even in Limuru, consequently a rise in malaria and other vector-borne diseases • A rise in the incidences of water-borne diseases such as typhoid during floods • Landslides are a common occurrence e.g. in Gatundu South, Mundara • Recent: frost that affected tea in most parts of the country including Limuru • Low agricultural production-both livestock and crops leading to loss of livelihoods, and increase in poverty levels • Water rationing • Destruction of infrastructure, e.g. submergence of waste water treatment facility in Kiambu Town, roads, the power line with the Juja sub-station blowing up recently in April 2012	27.2	74.9%	12%

[8] County Fact Sheets, Kenya Integrated Household Budget Survey

[9] Exploring Kenya's Inequality, Pulling Apart or Pooling Together, National Report, Society for International Development, Kenya National Bureau of Statistics

[10] ibid

	• Displacement and deaths of people by floods, e.g. Thindigwa in Kiambu and Gatwanyaga in Thika • Loss of human and animal life, e.g. recently in Kabete, from floods			
Machakos	• Poorly performing crops/wilting, low crop yields and livestock deaths • Drying up of rivers, earth dams due to increased evaporation rates • Human-wildlife conflicts • Flooding, water logging during flash floods • Delayed onset of short rains • Soil erosion • Displacement of people • Reduced livelihood options, poverty increase and increase in crime rates • Scarcity of water for domestic consumption and watering of animals • Inter-clan and inter-community conflicts over diminishing resources • Electric power shortages, reduced industrial output, leading to unemployment • Water-borne diseases-malaria, asthma, etc • Outbreak of pests and parasites associated with changes in climatic conditions-caterpillars, army worms and coffee thrips	59.6	38.6%	15%
Nyandarua	• Destruction and degradation of infrastructure like roads with negative economic consequences, e.g. milk spoilage as it takes too long to get to the markets • Receding water bodies, e.g. Lake Olbolosat, Milangine dam, and Karia Ka Ndagi dam • Decline in biodiversity, e.g. the endangered Sharpes Long Claw of the Kinangop plateau and Maccoa duck of L. Olbolosat • Perennial rivers turning into seasonal ones, e.g. Rivers Kinja, Turasha, Mukungi and Olkaria • Crop failure due to droughts (and recently also due to frost), • food insecurity • Human wildlife conflicts	46.3	68%	16%

142. The percentage of the population with no education in relation to climate is high and hence the exposure to climate risk. This is due to lack of climate change information and awareness.

D.4.2. Financial, economic, social and institutional needs

Economic and social development level of the country and the affected population

143. The proposed project if financed will be contributing towards implementation of the National Water Master plan and the National Climate Change plan as well as the Vision 2030. The project will enhance the resilience of the Athi River Catchment area and water resources within it that supports economic development and livelihood improvement. Populations in the area will have increased access to water which will enhance their resilience and support their livelihoods. Absence of alternative sources of financing (e.g. fiscal or balance of payment gap that prevents from addressing the needs of the country, and lack of depth and history in the local capital market)

144. GCF contribution to the project is critical to ensuring the resilience of the communities to climate change. Without GCF support, this project as designed cannot proceed. The cost of ensuring the climate resilience of the water systems and communities among the target counties is significant. GCF support is essential due to Kenya's climate vulnerability and the severe economic and fiscal constraints it faces in addressing these climate change challenges.

145. Kenya has mainly been receiving funding for mitigation. However, there has been a huge gap in adaptation financing. This project will be contributing to increasing the amount of financing for climate change adaptation efforts in the country. Furthermore, investments in the water sector have been limited and this project will be providing financing for water sector interventions in the country. Additionally, Kenya is a third world country and climate change adaptation funds are scarce and inadequate, thereby the request for a grant from GCF.

146. The project's approach is to work with water governance structures, especially at local levels, who have limited capacity in climate change adaptation interventions. The project will work to enhance the capacity of these institutions as well as strengthen their coordination efforts across local and national levels. There is Need for strengthening institutions and implementation capacity.

D5. COUNTRY OWNERSHIP

Beneficiary country's ownership of, and capacity to implement, a funded project or programme

D.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

147. Kenya is a party to the UNFCCC and has committed to implementation of policies to adapt to climate change and effectively develop climate risk management measures. The country has also submitted its Intended Nationally Determined Contribution (INDC) to the UNFCCC that included both adaptation and mitigation measures. However, the country has placed a significant focus on adaptation. Kenya in these documents recognizes that participatory water resource management will strengthen the country's economic growth and address poverty reduction through improving food security and health.

148. The project is well aligned with national priorities and measures identified in the NCCAP, NAP and NDC. The prioritized water sector actions in these documents are being operationalized through the proposed project. The project is also linked with the NDC action on the water sector i.e. "Mainstream climate change adaptation in the water sector by implementing the National Water Master Plan (2014)". The project components are strongly linked to the National Water Master plan but additionally promote the mainstreaming of climate change adaptation into the water sector at the catchment area level.

DE.5.2. Capacity of accredited entities and executing entities to deliver

Accredited Entity (AE) - National Environment Management Authority (NEMA)

149. NEMA was established in 2004 as the principal government agency mandated to implement environmental policies, exert supervision and coordination on all matters environment. NEMA as the Accredited Entity (AE), takes the greatest responsibility for successful implementation of the project and will perform the roles of Project management, Reporting and project budgeting. NEMA has s robust technical team in the 8 core departments in NEMA, namely, Directorate, Environment Education, Information and Public Participation Department, Legal Services Department, Environmental Planning and Research Coordination Department (EPRC), Compliance and Enforcement department (C&E), Finance and Administration Department and Coastal, Marine and Fresh Waters Sub-Department. Hence a complementary relationship that allows for efficiency and effectiveness. NEMA has operational county offices in all the 47counties in Kenya.

150. NEMA is also the National Implementing Entity with the Adaptation Fund Programme and is currently implementing a USD 10,000,000 Programme aimed at building resilience to climate change and adaptive capacity of vulnerable communities in Kenya, undertaken in 14 counties in Kenya.

151. NEMA will implement some project activities under this project in line with its accreditation credentials.

Capacities of the Executing Entities (EEs)

Water Resources Authority (WRA)

152. Water Resources Authority (WRA) is a state corporation under the Ministry of Water and Sanitation, established as a corporate body under the new Water Act (2016) that was operationalized on the 21st of April 2017 vide Special Issue Kenya Gazette Notice No. 60. The organization has been in existence for the past 12 years following its establishment as Water Resources Management Authority (WRMA) under the Water Act (2002) that has now been replaced by the new Water Act 2016. WRA's mandate is to regulate, monitor, assess and allocate the water resources; and to collaborate with other institutions as required for better water resource management. WRA has adopted Integrated Water Resources Management (IWRM) approach in its operations as a means ensure stakeholder participation and to adopt a holistic view of management of the water resources incorporating the environmental, social and economic context of the water resources. WRA is a technical institution with the Headquarters is in Nairobi, 6 Regional offices and 26 Sub Regional offices with a total of 750 staff of different cadres (technical, legal, finance, accounts, supply chain & procurement, administration, communication, ICT, and

records) distributed across the country. WRA has governance structures at local level through the WRUAs, with capacity to implement local activities at county level, in collaboration with other stakeholders. WRA has in the past implemented various projects geared towards securing water resources and is currently implementing projects across the country with funding support from both GOK and development partners. WRA's combined project implementation portfolio for both past and current projects is approximated at Ksh. 6 Billion.

Kenya Meteorological Department - KMD

153. Kenya Meteorological Department hosts the National Flood Forecasting and Early Warning Centre (NFFEWC) based at the headquarter Nairobi which is charged with collection of data, analysis, modelling and forecasting of flood. KMD has the required facilities for monitoring and forecasting flood and drought hazards, such as real-time rainfall data, evaporation, and storm monitoring tools (satellite imagery, lightning and thunderstorm detectors), and quantitative rainfall forecasts (numerical weather prediction products) are available at KMD. The department also has high capital investment facilities connecting it to regional and Global Telecommunication System (GTS) hydro meteorological information sources necessary for forecasting of flood and drought hazards.

154. The department is charged with provision of meteorological information and early warning services for the safety of life, protection of property and conservation of the natural environment as a contribution to sustainable development of the nation and has a clear mandate as defined by the WMO Convention (1947 and revised in 2007), and Conventions of the World Hydromet. Observing Stations. It has trained manpower in various fields of Meteorology, Operational Hydrology, Numerical Weather Prediction and related services with requisite expertise to undertake data analysis, forecasting and delivery of services to the specific users. Staff in KMD have been trained on climate change modelling

D.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

155. The NDA has been widely consulted and engaged in the development of this proposal. The NDA was the chair of the Kenyan GCF Project Steering Committee whose membership comprised of, various government institutions, AE, and readiness service providers. The NDA also had representation in the Project Development Teams (PDT) that developed this proposal. The PDTs comprised of institutions with water related mandates, University of Nairobi, Observers from Pan African Climate Justice Alliance and Transparency International also were part of the process. The NDA also convened some of the meetings and consultations that were held for this proposal.

Steps taken in the development of this proposal are detailed in the following Table.

Table 4: Steps undertaken during proposal development

Date	Activity
February 2016	Establishment of the GCF Steering Committee (SC) by the NDA. Composition includes NIE, MENR, UNEP and CDKN. Terms of reference for the SC approved by the NDA. NDA convened an inception workshop to begin the process of writing GCF proposals. Over 50 stakeholders from ministries, agencies, departments, civil society, observers and private sector attended Work-plan to develop GCF proposals developed and approved by stakeholders.
March 2016	NIE convened a GCF training on prioritization of adaptation actions. The training was attended by: the NDA, members of the SC, and representatives from government entities, including Ministry of water, Ministries of Devolution and Planning; Agriculture, Livestock and Fisheries; Industry, Communications, Technology and Innovation; Environment and natural resources; Industrialisation and Enterprise; Kenya Association of Manufacturing; civil society actors.
April 2016	The NDA convened a training workshop on GCF proposal development and cost benefit analysis. Workshop was attended by over 20 people. Representation was from Ministries of Devolution and Planning; Agriculture, Livestock and Fisheries; Environment and natural resources; Industrialisation and Enterprise and civil society actors
April 2016	NDA convened a meeting Technical Working Group meeting to prioritise potential GCF concepts. Prioritisation is based on NCCAP, INDC and NAP priorities. NDA and NIE appoint 25 individuals (Project Development Teams) from various ministries, civil society and private sector to assist in the development of 3 GCF proposals
May 2016	NDA and the NIE conducted the induction of PDTs GCF proposal writing begun

June 2016	NDA convened planning meetings for the PSC GCF proposal writing continued
July 2016	GCF proposal writing continued NIE trained PDTs on environmental and social safeguards
August 2016	GCF proposal writing continued
September 2016	Letter of no objection
2017	NEMA secures technical support from Korea Environmental Industry and Technology Institute (KEITI) to undertake feasibility studies
2017/2018	County/Community level stakeholder consultations
2018	ESMF, Feasibility studies, Gender action plan
2018	Validation workshops for ESIA at the County level
November 2018	Finalization and submission of revised FP
22nd April 2019	1st Feedback on the Completeness check on the Water proposal submitted
30th May 2019	2nd Feedback on the Funding Proposal Review
August 2019	3rd Feedback on the Funding Proposal Review
December 2019	4th Feedback on the Funding Proposal Review
January 2020	GCF review team call with the AE to discuss the 5th review comments
February 2020	AE responds to the 5th review comments
March 2020	GCF introduces a consultant to the AE to assist in the finalization of the F.P in preparation for B.26
May 2020	AE receives ITAP Assessment findings of the submitted proposal
June 2020	AE has a call with ITAP to discuss the assessment findings
October 2020 to date	AE updates the FP and Annexes , GCF task team guides the process

156. The process of developing this proposal began with a high level tripartite meeting with the Director General of Budget, Fiscal and Economic Affairs from the National Treasury, the National Climate Change Secretariat (NCCS) and the National Environment Management Authority (NEMA) which approved the project activities, work-plan, roles and responsibilities of different project structures. Thereafter agreements were reached with United Nations Environment Program (UNEP), CARE, World Resources Institute (WRI) and the United Nations Development Program (UNDP) on how to offer complementary capacity building activities as opposed to duplication as they are all involved in assisting the government in GCF proposal development. The first Project Steering Committee (PSC) meeting endorsed the roles and responsibilities of the different project structures and the inception meeting endorsed the project work-plan. A stakeholder analysis was conducted, and various stakeholders were mapped against their power and influence and classified according to their level of interest in the project. In addition, a multi-stakeholder plan which includes negotiation, partnerships and participation procedure, grievance resolution mechanism and stakeholder involvement in project monitoring was developed. As detailed in section E 5.3 more than 50 institutions have been involved in the process of developing this process in various stages as shown in E 5.3. They have assisted in data collection, formulation of text and peer review of the project's content.

D.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

D.6.1. Cost-effectiveness and efficiency

157. The net economic benefits and costs in the analysis of the overall project were derived by comparing the 'with' and 'without' project scenarios. The project aims to strengthen the resilience to climate change of communities and increase water security in the Athi River Catchment Area in Kenya, which will lead to the resulting benefits: (i) reduction in flood related losses; (ii) reduction in water sanitation and supply related diseases incidents and costs thereof; and (iii) time savings by households in accessing water. The desired interventions and outcomes are purely public in nature and they will not be displacing any existing mechanisms present in the targeted areas to cope with the challenges.

158. The results of economic analysis are all positive despite the analysis having used conservative estimates with regards to the benefits expected and attributable to the project. The economic analysis was done over 30 years, over which the project assets will likely yield benefits assuming that operations and maintenance activities described in the proposal and feasibility study are undertaken.

159. The project uses a discount rate of 12% as base case scenario and accounts for flood protection, health, and avoidance of social costs (e.g. time for water collection). The economic analysis was done at 30, 20 and 15 years and the results are presented as below;

	30 years, 12% discount rate	20 years 12% discount rate	15 years 12% discount rate
NPV (USD)	9,747,248.48	7,585,142.92	5,551,275.52
IRR	13%	12%	11%
Discounted payback period	7.22	7.22	7.22
AEV (USD)	1,210,059.08	1,015,489.68	815,061.81
Benefits/Costs Ratio	1.78	1.63	1.48

A detailed economic analysis and financial analysis for the project are included as an annex 3 and 3ii to the funding proposal.

D.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

160. The Government of Kenya (GOK) is expected to offer support to the project from Year 1 and throughout the project in terms of office space, staff time, vehicles, and utilities. These figures have been estimated as at Year 0 and grown over time with the average inflation rate. The figures for Year 0 are presented in the table below. The analysis has also assumed that the Gok will support the project from Year 5 to a tune of USD 531,784 and this figure has been assumed will remain constant for a lack of a better estimate. This operations and maintenance (O&M) estimate supported by an O&M plan (attached)

Item	USD	Description
Office Space	**27,000.00**	USD 45,000million per month * 12 months * 5% of office space
Staff time	**30,000.00**	50 staff * Average salary of USD 1,000 per month * 12 months * 5% of staff time.
Gok Vehicles	**1,500.00**	3 vehicles, worth USD 50,000 each, depreciated over 5 years (20% pa), and used for the project only 5% of the time.
Utilities	**2,400.00**	Utilities estimated at USD 200 per month for 12 months.
Totals	**60,900.00**	

Gok support to the project in year 0

Please make a reference to E.6.5 (core indicator for the expected volume of finance to be leveraged).

D.6.3. Financial viability

161. An economic feasibility analysis is conducted to estimate the economic feasibility of a project by comparing and analyzing the costs and benefits of the projects in monetary terms. The economic feasibility of a project depends on the sustainability of the project and its impact. Unlike an economic feasibility analysis, the main purpose of financial validity analysis is to figure out whether the return on investment from the project achieves the expected profit. If not, the analysis makes basic data available to set up plans to make the project feasible.

162. Beyond the Fund intervention, the project is will still be viable as indicated by the positive at year 14. Results indicate that the benefits will be realized after the project period and hence the proposed investment of GCF resources can be seen to be a cost-effective investment which will create net economic benefits that are about two times greater than the initial GCF investment. The project assets handed over for operation and management to the county governments. The management committees of the established assets shall be trained on sustainability measures that will ensure project benefits are up scaled and sustained beyond the Programme cycle.

D.6.4. Application of best practices

163. KMD successfully implemented the Nzoia Flood Early Warning System (NFEWS) under Western Kenya Community Driven Development and Flood Mitigation Project (WKCDD&FMP) in Western Kenya funded by the World Bank and the Kenya Government. The broad objective of flood management within the project was to reduce vulnerability of the community to adverse outcomes associated with recurrent flooding. This was addressed through; Preparation of multi-purpose long term flood management, Enhanced flood plain management in the Budalang'i plains and Development and institutionalization of proactive mechanism for a community-based flood early warning system.

164. The project will take forward lessons from similar projects across the world including the Integrated Study Project on Hydro-Meteorological Prediction and Adaptation to Climate Change in Thailand (IMPAC-T) which has been operating in Thailand since 2009. IMPAC-T involved enhancing Thailand's decision-making support system for adaptation strategies that address water-related issues, specifically those caused or exacerbated by climate change in Thailand.

D.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO_2 eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)	
	(a) Total project financing	US$_______
	(b) Requested GCF amount	US$_______
	(c) Expected lifetime emission reductions overtime	_______ tCO_2eq
	(d) Estimated cost per tCO_2eq (d = a / c)	US$_______ **/ tCO_2eq**
	(e) Estimated GCF cost per tCO_2eq removed (e = b / c)	US$_______ **/ tCO_2eq**
	Describe the detailed methodology used for calculating the indicators (d) and (e) above. Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.	
	Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)	
	Describe the detailed methodology used for calculating the indicators above. Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.	
Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)	N/A	

**** The information can be drawn from the project/programme appraisal document.***

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E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's Performance Measurement Frameworks under the Results Management Framework to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

LOGICAL FRAMEWORK

This section refers to the project/programmes logical framework in accordance with the GCF's Performance Measurement Frameworks under the Results Management Framework to which the project/programme contributes as a whole, including in respect of any co-financing.

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO_2 eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	Click here to enter text. t CO_2 eq
	Lifetime	Click here to enter text. t CO_2 eq
E.2.2. Estimated cost per t CO_2 eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing (b) Requested GCF amount (c) Expected lifetime emission reductions **(d) Estimated cost per t CO_2eq (d = a / c)** **(e) Estimated GCF cost per t CO_2eq removed (e = b / c)**	______ Cho ______ Choo ______ t CO ______ C **CO_2eq** ______ C **CO_2eq**
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged (g) Public source co-financed (h) Private source finance leveraged **(i) Total Leverage ratio (i = f / b)** (j) Public source co-financing ratio (j = g / b) (k) Private source leverage ratio (k = h / b)	______ Cho ______ Cho ______ Cho ______ ______ ______

E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	1,156,620 51% of female
	Indirect	**3,693,380** 51% of female
	For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.	
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	0.4 (Expressed as %) of country(ies)
	Indirect	10.32 (Expressed as %) of country(ies)
	For a multi-country proposal, leave blank and provide the data per country in annex 17.	

E.3. Fund-level impacts

Select the appropriate impact(s) to be reported for the project/programme. Select key result areas and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected impact result. The result areas indicated in this section should match those selected in section A.4 above. Add rows as needed.

Expected Results	**Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions**
				Mid-term	Final	
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	*A1.1 Change in expected losses of lives and economic assets (US$) due to the impact of extreme climate-related disasters*	Third party monitoring	0US$	US$ 1,751,647 flood related benefits at end of year 2	US$ 1,836,018 flood related benefits at end of year 4	a) 10% of the population in the targeted counties will utilize the EWS to move property b) Per capita cost of floods at USD 2,276.85 in real terms will remain fixed over the years c) 5% of the total property that families could save from floods in responding to EWS d) Only 5% of the total cost saved could be attributable to the project
A2.0 Increased resilience of health and well-being, and food and water security	*A2.3 Number of males and females with yearround access to reliable and safe water supply despite climate shocks and stresses*	Annual Project M&E Progress reports	Male: 0 Female: 0	Total: 90,000 Male: 44,100 Female:45,900	Total: 1,156,620 Male: 566,744	a) Water supply from springs sources is more than adequate to benefit up to 1.6M

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					Female: 589,876	individuals over 40-year period assuming a 2.65% population growth rate b) Only 36,500 individuals will benefit from boreholes c) Only 10% of the total population of the catchment will significantly reduce the time to the water source
A3.0 Increased resilience of intrastructure and the built environment to climate change	*A3.1 Number and value of physical assets made more resilient to climate variability and change, considering human benefits*	Installation report Monitoring reports	Number: 0 Value: 0	Number:26 16 water pans 7 springs 2 Boreholes 1 catchment rehabilitated Value: USD 27,497,475.63	Number: 40 22 water pans 7 Springs 8 Boreholes 1 Sand dam 2 catchment rehabilitated Value: USD 54,994,951.26	Project activities will be on schedule and there will be no or insignificant cost overruns

E.4. Fund-level outcomes

Select the appropriate outcome(s) to be reported for the project/programme. Select key expected outcomes and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected outcome. Add rows as needed.

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term)	Final	
A5.0 Strengthened institutional and regulatory systems for climate-responsive planning and development	*A5.1 Institutional and regulatory systems that improve incentives for*	Expert review using Score Card approach will be used to	0	2 County governments have climate	4 County governments have climate	Institutional collaboration will be adequate.

	climate resilience and their effective implementation	measure improvement of the Institutional and regulatory systems (From level 0 (Non - existent), 1 Prepared/approved, 2 partially implemented, 3 substantially implemented, 4 (Fully Implemented) County Environment Action Plans (CEAPS) Sub Catchment Management Plans (SCMPs) County State of Environment Reports (SOEs)		responsive policies/plans and at level 1	responsive policies/plans and at level 3	Existing regulatory system will provide an enabling environment
A6.0 Increased generation and use of climate information in decision-making	*A6.2 Use of climate information products/services in decision making in climate sensitive sectors*	Assessment reports on use of Climate information products/service available at market - (Daily / monthly/ seasonal weather forecast, Water situation reports, Air quality data, Flood forecast report,) Installation report	Low use of climate information in decision making (due to lack of information products, capacity, and framework)	Moderate use (some information products are produced and used in some decision making) 1 RANET station established	Substantial use (majority of information products are produced and used in most decision-making)	Capacity to generate and use climate information will be developed and sustained beyond project period
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	*A7.1 Use by vulnerable households, communities, businesses and public-sector services of Fund-supported tools*	Midterm and end of project Survey of project	0	5% of the population in the targeted	10% of the population in the targeted	The county government s will support the Programme implementation and

	instruments, strategies and activities to respond to climate change and variability			counties will utilize the EWS information to avoid flood risks	counties will utilize the EWS information to avoid flood risks 3% of the diarrheal incidents could be averted due project interventions 10% as the time savings attributed to the project interventions	communities will have Strengthened adaptive capacity and reduced exposure to climate risks There will be Increased access to portable water Sufficient awareness to the communities on EWS will be created, and the system will be easily accessible
A8.0 Strengthened awareness of climate threats and risk-reduction processes	*A8.1 Number of males and females made aware of climate threats and related appropriate responses*	Assessment report	Male: 0 Female: 0	Total: 14,000 Male: 6,860 Female: 7,140	Total: 27,539 Male: 13,494 Female: 14,045	Community members are receptive to awareness raising activities and have Improved compliance to relevant climate related regulations

E.5. Project/programme performance indicators						
The performance indicators for progress reporting during implementation should seek to measure pre-existing conditions, progress and results at the most relevant level for ease of GCF monitoring and AE reporting. Add rows as needed.						
Expected Results	**Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions**
				Mid-term	Final	
1. **Water resources monitoring systems enhanced for improved adaptation planning to support**	No. of installed and operational water resources monitoring equipment/stations/laboratories	Report	10	69	70	The county government and the local community will support the project

	decision making and policy development in water and climate change sector	(Operational means the equipment are maintained and in use)					Adequate resources for O&M will be available. The security of the installed gadgets will be assured The Global System for Mobile communication will be operating at optimum levels The system will be an Integrated water resources information with Decision support capability There will be no turnover of trained laboratories staff
2.	**Improved decision making in water and climate change sector**	Operational IWR monitoring and Information system and Decision Support System (DSS)	IWR monitoring and Information system Reports	0	2	2	The system will be fully operational. Impact to be measured through survey on number of institutions and households using the DSS to reduce potential climate threats and risks.
		Operational Decision Support System customized to ACA - Athi Catchment Area	Installation report Data Analysis reports	0	1	1	Completely installed and functional system in place Data sharing protocols effected
		No. of developed State of the Environment (S.O.E.) reports	S.O.E reports Meeting reports	0	4	4	The County Environment Committees - CECs will support the process
3.	**Strengthened institutions in water and climate change sector**	Number of staff trained to operate and apply IWR information and DSS	Training, reports	0	40	80	Staffs and personnel of NEMA, WRA, KMD, County governments and communities will use the information for decision making
4.	**Increased access to potable water for domestic use considering present and future impacts of climate change.**	Access to potable water for domestic use all year round for project beneficiaries	Assessment report	14%	30% of direct project beneficiaries have access to portable water all year round	60% of direct project beneficiaries have access to portable water all year round	The county government and the local community will support the project Rain water harvesting policy and guidelines on rain water harvesting put in place.

5. **Conservation of water catchment areas as climate change adaptation measure.**	Km of riparian land rehabilitated	Reports, Trees/vegetative options grown on targeted riparian land	50km	100km	150km	The trees planted in the water catchment areas will survive The weather will be favorable The WRUAs will support the initiative
6. **Strengthened institutional and regulatory framework for climate change adaptation**	Increased enforcement actions to improve water quality	Enforcement and water quality reports	20%	40%	60%	Operational water quality monitoring systems Community compliance to regulations Water quality data available for decision making The regulated stakeholders will participate in the meetings and have enhanced compliance to the regulations
	No. of CEAPs developed and Sub-Catchment Management Plans (SCMP) updated	CEAP reports developed Meeting reports Sub-Catchment Management Plans (SCMP) updated	0	6	12	The County Assemblies will endorse the developed CEAPS WRUAs supported to implement relevant parts of SCMPs. County Governments will implement the recommendation of the planning documents – CEAPs, SCMPs
	No. of workshops Conducted with regulated stakeholders to enhance compliance Improved compliance among regulated stakeholders	Attendance list on workshops / meetings Workshops / meetings reports	-	6	12	measurement of the uptake of the trainings will be done to determine their effectiveness

E.6. Activities

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.

Activity	Description	Sub-activities	Deliverables

Activity 1.1.1 Set up an institutional mechanism for sharing information and getting feedback (data sharing protocol among players in the water sector e.g, through MOUs)	A National Information Centre for IWRM at WRA headquarters will act as a hub of water resources monitoring information and will be equipped with proper ICT system for processing, analysis of information to produce integrated water resources information for informed decision making	• Policy dialogue Meetings aimed at developing data sharing protocols (between NEMA, WRA, KMD) and strategy for continuous communication between the user communities and county governments.	Data sharing protocols and MoUs among the executing entities Functional water resource management information system Water Resource Information products
1.1.2 Support the development of a National Information Centre for integrated water resource management (under the WRA HQ) by equipping it with necessary hardware and software facilities.		• Acquire and install the centralized integrated water resource management information system (server, HDD 1TB, monitors, network facilities, UPS with additional batteries) with necessary software (operating systems, database, visualization, web server and other utilities).	
Activity 1.1.3 Support information analysis, forecasting and knowledge sharing platforms through an integrated database for weather and water resources-related information	In order to have sustainability for project inputs, capacity building and training programs will be provided.	• Develop an integrated analysis tools with visualization from monitoring information. The Decision Support System will assist in data entry, storage, analysis and reporting of collected data • Develop a shared system of weather forecast information for development of early warning and install a flood and drought warning system in flood prone areas in Kiambu and Machakos counties • Develop web services for sharing information with stakeholders	Enhanced staff capacity on integrated water resource management and climate change Enhanced staff capacity on use of data generation and modeling software
Activity 1.2.1 Install and rehabilitate surface hydro-meteorological monitoring and network systems and rehabilitate existing monitoring systems (Nyandarua, Nairobi, Kiambu, Machakos county)	Hydro meteorological infrastructure will enhance protection from weather and climate shocks such as droughts and floods and improve the capability for gathering hydro meteorological data which is critical for monitoring and predicting extreme events.	• Installation of surface hydro-meteorological monitoring and network systems and rehabilitate existing monitoring systems (Nyandarua, Nairobi, Kiambu, Machakos county). • Install 25 Automatic Weather stations within the upper Athi catchment. all the stations will be equipped with Remote Terminal Units (RTUs), GSM/GPRS Modems, and the solar PV systems with solar battery, data protection equipment such as noise filters, surge protective devices. • Install 23 surface water level monitoring stations with multi-parameter water quality	Necessary equipment procured and installed at designated sites. Operators of the network trained on use of new equipment

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		sensors for 20 stations. 5 rainfall monitoring stations. All the stations will be equipped with Remote Terminal Units (RTUs), GSM/GPRS Modems, and the solar PV systems with solar battery, data protection equipment such as noise filters, surge protective devices. 3 Water Level Stations will be manual hence will not require the sensors. The 5 rainfall monitoring stations will be distributed in specific sites in the project area and located within the Automatic Water Level Station (AWLs) compound.	
Activity 1.2.2 Upgrade water quality test laboratory at WRA and NEMA (purchase, install, calibrate, test laboratory equipment /stations)	Acquire and calibrate devices for water quality test such as atomic absorption spectrophotometer, oxitops, Chemical Oxygen Demand (COD) digester, incubators, sediment analysis kit, High-performance liquid chromatography, Procure operational materials such as laboratory chemicals, reagents, and glassware	• Acquire and calibrate devices for water quality test such as atomic absorption spectrophotometer, oxitops, Chemical Oxygen Demand (COD) digester, incubators, sediment analysis kit, High-performance liquid chromatography, Procure operational materials such as laboratory chemicals, reagents, and glassware. Both NEMA and WRA will be allocated funds for upgrading their water quality test laboratories	
Activity 1.2.3 Establish 10 groundwater monitoring stations	Develop 8 new monitoring wells in Nairobi, Kiambu and Machakos counties and Install necessary sensors to measure water level, conductivity, temperature, RTUs and other equipment for automated data transfer	• Develop10 new monitoring wells in Nairobi, Kiambu and Machakos counties and Install necessary sensors to measure water level, conductivity, temperature, RTUs and other equipment for automated data transfer	Ground water monitoring stations established
Activity 1.2.4 Setting up of a RANET broad casting station in Machakos County	Establish a RANET station	• Installation of a “RAdio InterNET” broadcasting in approximately 50 – 75 Kilometers radius in the local dialect, Kiswahili and English for 24 hours, with an estimated audience of 1,000,000 listeners	Necessary equipment procured and installed
Activity 1.3.1 Support capacity building and training for government institutions with water and climate change mandates within the project area	In order to have sustainability for project inputs, capacity building and training programs will be provided.	i. The executing entities will identity capacity building and trainings for their relevant institutions with water and climate change mandates within the project area and NEMA will approve and disburse fund accordingly	

Activity 1.3.2 Support capacity building and training for relevant water governance structures and communities within the project area (Water Resource Users Associations -WRUAs, County Environment Committees - CECs, County Adaptation Committees, Community Based Organizations -CBOs among others)	In order to have sustainability for project inputs, capacity building and training programs will be provided.	i. The executing entities will identify capacity building and training for relevant water governance structures and communities within the project area, develop training plan and manuals and NEMA will approve training plans and disburse fund accordingly	
Activity 1.4.1 Support development of State of the Environment (S.O.E) reports from the information developed	Availability of data and documentation of environment is critical in projecting future scenarios, thereby reducing susceptibility and improving disaster preparedness. Through the SOE process, baseline data, trends of drivers and pressures impacting on the environment will be established, future scenarios will be projected, and appropriate policy interventions recommended.	Consultative workshops to develop the S.O.Es.	County State of the Environment (S.O.E) reports developed
Activity 2.1.1: Construct water storage and supply infrastructure	In selected counties within the ARCA, appropriate water storage infrastructure will be constructed	• rehabilitate two boreholes, twelve water pans, two springs in Nyandarua county • build a new borehole, water pan, two springs and rehabilitate three boreholes, two water pans, two springs in Kiambu county • build two new boreholes, rainwater harvesting tanks and rehabilitate nine water pans, one spring and one sand dam in Machakos county	Appropriate water storage infrastructure constructed Improved access to water for domestic use
Activity 2.2.1: Rehabilitation and reforestation activities within the selected counties in collaboration with WRUAs	The restoration of degraded water catchment areas will be done through engagement of communities through existing water resources governance structures (WRUAs)in the counties in rehabilitation/renewal/conservation and protection of degraded water catchment areas. The activities will include Establishing tree nurseries for rehabilitation and reforestation within the selected counties, perform catchment protection through tree planting in two	Rehabilitation and reforestation activities within the 4 target counties in collaboration with WRUAs • Establish tree nurseries for tree planting at new and rehabilitated water structures within the selected counties • Undertake planting of indigenous trees in two catchment areas with high risk of erosion in the target counties • Riparian marking and pegging (using indigenous tree) of 20km along Riara and Ndarugu Rivers	Awareness on catchment conservation Improved Catchment areas rehabilitated and conserved

	catchment areas with high risk of erosion in the target counties,	• Construction of terraces (1000m) and 1500m gabions (gulley healing by planting vetiver grass and or installation of gabion boxes) in selected sites in Kiambu and Nairobi counties).	
Activity 3.1.1: Provide water quality monitoring systems to support enforcement of the water quality regulations Activity 3.1.2: Sensitize the regulated communities to enhance compliance	Implementation of the provisions of existing legislations through compliance and environment management tool s will enhance climate change adaptation. Degraded water and water quality treatment systems, if not monitored emit gases such as NH3 and CO2 which are greenhouse gases. In addition, water of poor quality could be a source of waterborne diseases that will compromise human health and their resilience to climate change.	3.1.1 Provide water quality monitoring systems to support enforcement of the water quality regulations. The tools will include; ***8 (eight)*** **Multi parameter sonde** (new hydrolab HL7) and ICT with ability to measure pH, temperature, nutrients (nitrates and phosphates), dissolved oxygen, total dissolved oxygen, salinity (conductivity), depth, chlorophyll, Rhodamine, blue-green algae, Ammonia, Chloride, etc. basically parameters that show effect of climate change on water quality; and its accessories; a ***Water testing kit for Microbiology analysis*** (POTALAB) with ability to detect Ecoli (Faecal contamination), Coliform (bacterial contamination), Salmonella (Typhoid), Pseudomonas (ENT infections); **3 (three) Potable Heavy metal detection meters** (Portable Metalyser for Heavy Metals) with the ability to measure Arsenic (III and Total), Antimony, Cadmium, Chromium, Cobalt, Gold, Mercury, Lead, Magnesium, Nickel, Selenium, Thallium, Tin, Zinc 3.1.2 Sensitize the regulated communities to enhance compliance: i) Promote WASH practices through hygiene awareness campaigns aimed at promoting water disinfection methods and hygiene practices in collaboration with the beneficiary county governments ii)Carry out Compliance Management Action Plan trainings / workshops to ensure compliance with the water sector laws and standards in Kenya (Both NEMA and WRA will identify stakeholder among the regulated communities to	Improved compliance to existing legislations on water quality Compliance licenses issued

		be sensitized, develop workshop schedules / plans and implement the activity. NEMA and WRA are government agencies with clear mandates in the water sector that require a coordinated approach to ensure compliance)	
Activity 3.1.3 Conduct water quality pollution survey for point and non-point sources of pollution	The survey is a regulatory safeguard measure for sustainable protection of both surface and ground water resources.	i. Field survey and Consultant meetings	Water quality pollution survey for point and non-point sources of pollution report
Activity 3.2: Support development of 4 County Environment Action Plans (CEAPs) as an adaptive action	Environmental Action planning at the county level is critical in prioritizing environmental issues including adaptation and mitigation measures to be adopted in addressing county environmental challenges including those associated with climate change.	Workshops and meetings	County Environment Action Plans (CEAPs) developed County Prioritized adaptation activities identified for action Enhanced awareness on environmental action planning at the county level
Activity 3.3 Development of 8 (eight) Sub-Catchment Management Plans (SCMPs) including climate change adaptation considerations	Water resources are at risk from the adverse impacts of climate change hence it's important to mainstream climate change related interventions in Sub-catchment management plans. Development, review and implementation of sub-catchment plans and climate risk management with regards to water resource management is crucial to ensure that catchment plans incorporate climate risk management approaches and activities. WRUAs will be supported in the implementation of activities in the sub-catchment management plans.	i. Capacity building and SCMP development workshop	Sub-catchment management plans reviewed / developed and implemented

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Monitoring and evaluation (M&E) is one of implementing measures that support successful delivery of this proposed project. Monitoring, evaluation, and reporting for the project will follow NEMA's Monitoring and Evaluation Policy. A key aspect of the project will be learning and documenting of lessons as well as developing capacities of players of Executing Entities (EE. The M&E function will be carried out by the accredited entity. NEMA as the AE has a Monitoring and Evaluation section that conducts post project monitoring over the life span of the Authority's projects. The project results monitoring process after final evaluation will be collaborative activity together with the county governments that will be the custodians of the adaptation assets rehabilitated / constructed. The EE that have specific mandates in the hydrological and meteorological monitoring systems will undertake the monitoring of the established assets and carry out routine operation and maintenance to ensure the investments are in proper working conditions throughout the predicted lifespan.

M&E process will be undertaken using participatory methods/process, especially with the EEs to enhance continuous learning and promoting iterative project implementation. In addition, M&E system of this project will be gender sensitive and will follow guidance from the GCF and comply with GCF M&E policy, ensuring that the project maintains a simple and interactive monitoring system allowing for regular collecting data for ensuring progress, reporting and evaluating at all levels The responsibility for day-to-day project monitoring and implementation will rests with the Executing Entities - EEs. The EEs will develop quarterly and annual work plans to ensure the efficient implementation of the project. The EEs will inform the AE of any delays or difficulties during implementation, including the implementation of the M&E plan, so that the appropriate support and corrective measures can be adopted. The EES will also ensure that all project staff maintain a high level of transparency, responsibility and accountability in monitoring and reporting project results. It is expected that it will be based on the following core activities:

- Activity Recording/Process Documentation: Progress monitoring and reporting will provide evidence on accomplishment of the planned activities under each Output and Activity, by assigning milestones and implementation timelines. This will help the strategic and operational managers to identify which activities are properly taking place or not. The accredited entity of Kenya, NEMA, and executing entities will be responsible for ensuring routine monitoring on the use of inputs (including finances) and implementation of activities.

- Annual Progress report/Annual Performance Assessment: The annual project report/Annual Performance Assessment will be included but are not limited to; quarterly project reports, Annual reports, Project Outputs delivered per project Outcome, financial report, lessons learnt/good practice, and annual Work Plan (for the following year).

- A mid-term review: A Formative Evaluation process will be carried out Year 2 in order to analyze the progress of the proposed project and to suggest possible adjustments and improvement for the sake of efficiency, effectiveness, and sustainability before the end of the project period.

- End of project review and final evaluation: A summative evaluation will take place no later than 6 months prior to operational closure of the project. The terms of reference, the review process and the final terminal evaluation report will follow the standard templates of the Government of Kenya.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1		
Category	Probability	Impact
Forex	High	Medium
Description		

Inflation and dollar fluctuation. Kenya shilling has not been stable against the dollar and due to inflation in the country most prices for services and products change.		
Mitigation Measure(s)		
The project has factored in 5% inflation change.		
Selected Risk Factor 2		
Category	Probability	Impact
Governance	Medium	Medium
Description		
Resistance from the riparian adjacent landowners to have some of the riparian areas marked and protected		
Mitigation Measure(s)		
The project will ensure proper sensitization of the community is done to have all stakeholders on board at the beginning of the project. Additionally, the community members will be made aware of benefits from riparian rehabilitation and protection at a community level.		
Selected Risk Factor 3		
Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Lack of County Government goodwill of the planned activities		
Mitigation Measure(s)		
To mitigate against this risk, the project implementation will involve a wide range of stakeholders including local political and administrative leadership to allow for participation and decision making. The county governments participated in the county stakeholder’s forum during the feasibility hence are well informed of the programme.		
Selected Risk Factor 4		

Category	Probability	Impact
Technical and operational	Low	High
Description		
There is possibility of vandalism and/or theft of installed equipment.		
Mitigation Measure(s)		
To avoid against this risk, the water monitoring equipment and other installations will be set up within institutions that have adequate security. Where this is not possible, the equipment will be put within reinforced burglar proof structures with arrangements for emergency security response and ensures regular checks on the equipment. An incident reporting mechanism will be developed to ensure community members can report to the relevant authorities.		
Selected Risk Factor 5		
Category	Probability	Impact
Technical and operational	Medium	High
Description		
Poor maintenance of equipment		
Mitigation Measure(s)		
The feasibility study includes details on operations and maintenance plans including training of staff, inclusion of after sales service in procurement contracts, and preventive maintenance practices such as calibration and regular checks. Operations and maintenance costs are included in the project budget for the duration of the project. Relevant government agencies will cover O&M costs beyond the project lifetime. The mitigation measures lower the level of impact to medium.		
Selected Risk Factor 6		
Category	Probability	Impact
Other	Low	High
Description		

Installation of equipment on private land and potential for expropriation		
Mitigation Measure(s)		
Whenever possible, the project will install equipment in public or government-owned land. In cases where observation or monitoring equipment will be installed in private land, relevant formal agreements will be signed to govern the protection of the equipment. The mitigation measures lower the impact to medium.		
Selected Risk Factor 7		
Category	Probability	Impact
Other	Medium	High
Description		
Extreme events such as heavy rains prevent timely implementation of activities, particularly infrastructure rehabilitation		
Mitigation Measure(s)		
Annual work plans will be carefully drafted to avoid construction activities during heavy rains. In case of damage during construction, any damages and reconstruction costs will be covered by contractors' insurance. The mitigation measures lower the risk impact and probability to low.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

165. The project seeks to strengthen environmental management through activities such as protection of riparian areas and strengthening water governance structures. The other activities revolve around rehabilitation of existing water infrastructure i.e. degraded water pans, sand dams and boreholes. However, an Environment Social Management Framework (ESMF) for the activities proposed under this project has been undertaken as part of Kenya's larger GCF project implementation strategy. Recommendations have been provided for in the report to ensure proper due diligence procedures are adhered to during project implementation. The project placed under risk category B. The project activities vis a vis the GCF environment and social safeguards are discussed below:

Performance Standards	Risk Level (A,B &C)[11]	Mitigation measures
Labor and Working Conditions The project will adhere to existing Kenyan laws on labour (Cap 226) in engagement of staff and consultants, which are also in tandem with GCF ESS standards. The project will source unskilled labour locally and will tap any skilled labour within the community, thereby providing locals with an opportunity to earn income.	**C**	In sourcing for any skill and unskilled labor, the project will seek to hire locals from the area of implementation. This will ensure buy in and provide opportunities for incomes to local people
Resource Efficiency and Pollution Prevention The project activities that will involve rehabilitation of water storage structures e.g. water pans and sand dams may include some aspects of construction and use of cement and building material.	**B**	The project will work to ensure resource efficiency such as efficient water utilization through promotion of water efficient technologies (eg drip irrigation, piping rather than furrows, flooding or canal use) as provided in the water act of 2002. In any construction of water storage structures, the use of local materials will be promoted as much as possible
Community Health, Safety, and Security It is expected that local community members may be used during WRUA activities as well as used as unskilled labour for some of the construction activities. .	**B**	The project will ensure adequate safety and security precautions in line with the Occupational health and safety act of 2007. During the construction and operational phases of the project, workers, project managers and all visiting the project site will be provided with relevant personal protective equipment. Adequate signage will be put in place to warn, inform and alert all on potential dangers and what is required of them. The rehabilitated water pans will be fenced as a security measure to ensure public safety and water security
Land Acquisition and Involuntary Resettlement To install monitoring stations and equipment, land may be required. It is not expected that any acquisition or resettlement will be required.	**C**	The projects will be implemented on public land and there will be no need for land acquisition
Biodiversity Conservation and Sustainable Management of Living Natural Resources A number of activities will be implemented by the WRUAs focusing that may be dealing directly with biodiversity and natural resources including riparian restoration.	**C**	Protection of riparian areas within the catchment will have a positive impact on biodiversity as it will enable rejuvenation of fauna and flora within the buffer zone thus established. The WRUAs under the project will be involved in tree nursery establishment under which indigenous tree species will be propagated, some of which have medicinal and cultural value to the community.
Cultural Heritage During implementation of WRUA activities there may be close contact with community members and cultural heritage sites.	C	The project will not have any negative impacts on the cultural heritage of the target beneficiaries. The implementation approach will seek to understand the cultural heritage of communities in the target counties and seek to gain guidance. The nomadic lifestyle of the Maasai will be enhanced through availability of water from rehabilitated springs.

[11] Risk Level has been categorized into three level: (A) Significant adverse risks that are irreversible, (B) Mild adverse risks that could be reversible, and (C) Minimal or no adverse risks

Table 6 Environmental and Social Management plan

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

166. The gender assessment identifies and proposes remedies to gender based inequalities that could exacerbate existing or generate new forms of disparities in the way the project impacts on men, women and other vulnerable populations within the catchment areas. A cross-sectional gender analysis using qualitative methods was employed and emphasis was on making the process participatory through focus group discussions, key informant interviews, observation and workshops in Nyandarua, Kiambu, Machakos and Nairobi counties.

167. The stakeholder consultations were conducted among community participants, service delivery officers and policy makers in the water and environment management sectors, and gender mainstreaming oversight institutions. The assessment focus was on various elements of gender equality and equity that include: Access to resources, Level of knowledge acquisition and skills, Cultural beliefs and perceptions, Participation and decision making, Legal rights and status, and Time availability and space concerning location of water points.

168. The planning and design of the program adhered to the constitutional principle of public participation that requires consultation with stakeholders affected by public policy/project. Consultation Meetings were held at inception with project beneficiaries in Nyandarua, Kiambu, Nairobi and Machakos. 35% of the participants during these stakeholder forums were women hence in line with the constitutional two thirds gender representation requirement. Further, the project technical design team comprising technical teams key lead agencies was gender balanced. The project has also carried out a gender analysis and further developed a gender action plan that will ensure 51% participation of women in project activities and decision making. The gender action plan will ensure a gender-sensitive approach during project implementation. Further, the gender action plan has integrated gender sensitive indicators that will be tracked during project implementation. The accredited entity (NEMA) through GCF readiness support is developing a Gender Mainstreaming action plan to enhance gender integration in projects and programs. The project will also be cognizant of the Access to Government Procurement Opportunities (AGPO) which focuses on the fair, equitable, transparent and cost effective procurement of goods and services (Article 55 of COK 2010 which focusses on affirmative action in public procurement). The contractors will also be compelled to have 51% women involvement in works by introducing a clause in the contract

169. The gender analysis established that women and girls are the vulnerable gender majorly because they are responsible for finding water resource that their families need to survive - for drinking, cooking, sanitation, and hygiene and by extension they are also responsible for watering the livestock. They walk long distances to collect water, they also brave long queues in water collection points and in drought seasons they are forced to pay exorbitant amounts of money to secure water which is often very time consuming and arduous. The impacts of climate change that results in water shortage exacerbates the vulnerability of women and girls leading to exploitation and abuse.

170. Based on the gender assessment, a gender action plan was developed highlighting gender responsive actions that will be implemented against each project activity such as Constitution of a gender inclusive committee to gather and disseminate relevant climate information from local water catchment communities and county governments, development of an integrated database for weather and water resources-related information by ensuring that men and women receive seasonal forecast in a timely manner, promotion of gender balanced staffing during the technical training on operation of monitoring and information systems, ensuring public awareness and inclusiveness in the participation of men and women in the water catchment protection and management processes among others.

171. Key outcomes to be achieved from implementation of the gender action plan is reduced gender inequality and beneficial contribution of men and women in the project leading to reduced vulnerability, especially of women and girls.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

172. NEMA as the accredited entity has the responsibility of financial management, accounting and reporting. NEMA may appoint the EEs to undertake procurement of any goods and services for this project in line with the activities executed and based in the government procurement procedures.

173. Procurement of goods and services shall be guided by the public procurement and disposal Act 2015. NEMA shall be responsible for preparation of a disbursements schedule and shall prepare financial requests to the GCF.

174. NEMA shall also coordinate with the national auditing framework to ensure the Programme is audited in accordance with the Kenya National Laws. The funds shall be provided in advance to the Executing Entities on a quarterly basis and against a criterion that will ensure prudent use of resources. The expenditure shall be audited by NEMA internal audit section as well as the Kenya National Audit Office – KENAO to ensure compliance with the Kenya National Laws

175. Additionally, NEMA as a fast-track accredited entity under the Adaptation Fund has the basic fiduciary standards concerning transparency and accountability. The application of these standards will reduce any risks of money laundering (ML) or terrorism financing (TF) in the project.

176. NEMA has the following basic fiduciary standards concerning transparency and accountability:

- The NEMA draft Corruption Prevention policy is anchored in Chapter six of the Kenyan Constitution 2010, Article 232 of the Kenyan Constitution 2010, Leadership and Integrity Act 2012, Public Officer Ethic Act 2016, Anti-Corruption and Economic Crimes 2016
- NEMA Code of Conduct and Ethics
- NEMA has an established mechanism of receiving complaints and the procedure requires that NEMA reports third party complaints to the Ethics and Anti-Corruption Commission - EACC that provides for a platform to address fraud issues with the third parties
- Kenya has a draft whistle blowing Act that allows another avenue for reporting fraud. Additionally, NEMA has draft Whistle Blowing Policy, 2017, that provides for recourse on NEMA dealing with third parties on corruption issues
- NEMA has a Corruption Risk and Mitigation Plan that is under implementation
- Draft NEMA Corruption Prevention Policy is implemented through the appointment of Corruption Prevention committee –CPC and Integrity Assurance Officers committee IAO
- NEMA as the NIE has draft Anti – Corruption Policy and Finance and Procurement Manual
- NEMA also has a police unit seconded from the National Police Service, who largely investigate environmental crime, but also play an advisories role in all investigative functions.

G.4. Disclosure of funding proposal

Indicate below whether or not the funding proposal includes confidential information.

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES		
H.1. Mandatory annexes		
☒	**ANNEX 1**	NDA no-objection letter(s) **(template provided)**
☒	**ANNEX 2**	Feasibility study - and a market study, if applicable
☒	**ANNEX 3**	Economic and/or financial analyses in spreadsheet format
☒	**ANNEX 4**	Detailed budget plan **(template provided)**
☒	**ANNEX 5**	Implementation timetable including key project/programme milestones **(template provided)**
☒	**ANNEX 6**	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ **OTHERS (PLEASE SPECIFY – E.G. RESETTLEMENT ACTION PLAN, RESETTLEMENT POLICY FRAMEWORK, INDIGENOUS PEOPLE'S PLAN, LAND ACQUISITION PLAN, ETC.)**
☒	**ANNEX 7**	Summary of consultations and stakeholder engagement plan
☒	**ANNEX 8**	Gender assessment and project/Programme-level action plan **(template provided)**
☒	**ANNEX 9**	**LEGAL DUE DILIGENCE (REGULATION, TAXATION AND INSURANCE)**
☒	**ANNEX 10**	Procurement plan **(template provided)**
☒	**ANNEX 11**	Monitoring and evaluation plan **(template provided)**
☒	**ANNEX 12**	AE fee request **(template provided) template**
☒	**ANNEX 13**	Co-financing commitment letter, if applicable **(template provided)**
☒	**ANNEX 14**	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule
H.2. Other annexes as applicable		
☒	**ANNEX 15**	**EVIDENCE OF INTERNAL APPROVAL** **(template provided)**
☐	**Annex 16**	Map(s) indicating the location of proposed interventions
☐	**Annex 17**	Multi-country project/Programme information **(template provided)**
☐	**Annex 18**	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	**Annex 19**	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	**Annex 20**	First level AML/CFT (KYC) assessment
☒	**Annex 21**	Operations manual (Operations and maintenance)
☒	**Annex 22**	Indigenous People and Chance Find Procedure
☒	**Annex 23**	Updated ARCA Technical report
☒	**Annex 24**	BoQ of proposed water structures
☒	**Annex 25**	Peizometric surface trend analysis
☒	**Annex 26**	Climatology of Kenya
☒	**Annex 27**	WASH in Athi River Catchment project Area
☒	**Annex 28**	No. of beneficiaries breakdown - ARCA funding proposal
☒	**Annex 29**	Impact of Future projected Climate on the Upper Athi River basin Water Resources Report
☒	**Annex 30**	ARCA Basin Level - Rainfall and temperature trend analysis
☒	**Annex 31**	Basin Level-Analysis of projected rainfall changes
☒	**Annex 32**	Potential evapotranspiration (PET) for the baseline (1981-2010) and for the 2011-2040 for the three emission pathway projections

H

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)

REPUBLIC OF KENYA
THE NATIONAL TREASURY AND PLANNING

Telegraphic Address: 22921
Finance-Nairobi
Fax No.: 310833
Telephone: 2252299
When Replying Please Quote;

THE NATIONAL TREASURY
P.O. Box 30007
NAIROBI

Ref. No: ZZ/257/05/C/ (48)

12th June, 2019

Yannick Glemarec
Executive Director
Green Climate Fund
G-Tower, 24-4 Songdo-dong
Yeonsu-gu
Incheon City, Republic of Korea

Dear Glemarec

Re: Funding proposal for the GCF by National Environment Management Authority (NEMA) regarding "Enhancing Community Resilience and Water Security in the Upper Athi River Catchment Area, Kenya"

We refer to the project "Enhancing Community Resilience and Water Security in the Upper Athi River Catchment Area, Kenya" as included in the funding proposal submitted by National Environment Management Authority (NEMA) to us on 7th June, 2019.

The undersigned is the duly authorized representative of The National Treasury, the National Designated Authority of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Kenya has no-objection to the project as included in the funding proposal;

(b) The project as included in the funding proposal is in conformity with Kenya's national priorities, strategies and plans;

(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely

Yours

DR. KAMAU THUGGE, CBS
PRINCIPAL SECRETARY/NATIONAL TREASURY

ISO 9001:2008
Certified.

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Enhancing community resilience and water security in the Upper Athi River Catchment Area, Kenya
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	National Environment Management Authority of Kenya (NEMA)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Country: Kenya 1. Kiambu County 2. Nairobi County 3. Nyandarua County 4. Machakos County
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, May 20, 2020
Language(s) of disclosure	English
Explanation on language	English is an official language of Kenya.
Link to disclosure	http://www.nema.go.ke/images/Docs/NIE%20Docss/Annex%206%20ESMF%20for%20GCF%20funding%20proposal.pdf
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the Environmental and Social Management Framework (ESMF).
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, May 20, 2020
Language(s) of disclosure	English
Explanation on language	English is an official language of Kenya.
Link to disclosure	http://www.nema.go.ke/images/Docs/NIE%20Docss/Annex%206%20ESMF%20for%20GCF%20funding%20proposal.pdf
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the Environmental and Social Management Framework (ESMF).
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A

Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	The Indigenous Peoples Framework and Chance Find Procedure/ Wednesday, May 20, 2020
Language(s) of disclosure	English
Explanation on language	English is an official language of Kenya.
Link to disclosure	http://www.nema.go.ke/images/Docs/NIE%20Docss/Annex%2023%20ARCA-%20Indigenous%20%20People%20and%20Chance%20find%20procedure%20.pdf
Other link(s)	N/A
Remarks	The framework is developed as a precautionary measure for mitigation of adverse impacts on the vulnerable and marginalized groups, should screening indicate the presence of Indigenous People.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Wednesday, May 20, 2020
Place	The Registry and NIE Section in NEMA Head Quarters: Popo Road, South C, off Mombasa Road National Environment Management Authority, P.O.BOX: 67839-00200, Nairobi. Fax: +(254)-020-6008997 Tel No: 020-2101370, 020-2183718, 020-2307281, 020-2103696 Mobile: 0724 253398, 0735 013046, 0723 363 010 Email: dgnema@nema.go.ke NEMA County Offices (Nyandarua, Machakos, Kiambu): District Commissioner's Complex. 1st Floor, RM 10 P. O. Box 1756 – 20300 Nyahururu KFS Machakos Office Complex Machakos- Kathiani Rd P. O. Box 2 -90100 MACHAKOS Mapa House, 5th FLR Kiambu Town P. O. Box 427-KIAMBU WRA County office: Athi Basin Area P.O Box 1159-90100 Machakos-Kitui Road - Machakos Office Upper Athi P.O Box 1864-00900 Hospital Road, Opposite Kiambu District Hospital KIAMBU

Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 4, 2021

Note: This form was prepared by the Accredited Entity stated above.

Secretariat's assessment of FP175

Proposal name:	Enhancing community resilience and water security in the Upper Athi River Catchment Area, Kenya
Accredited entity:	National Environment Management Authority (NEMA)
Country/(ies):	Kenya
Project/programme size:	Micro

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
Linking hydrological, meteorological and water quality monitoring systems across Kenya's Water Resources Authority (WRA), Kenya Meteorological Department (KMD) and Kenya's National Environment Management Authority (NEMA) presents a paradigm-shifting opportunity towards climate-informed water resources management at both national and county levels.	Operation and maintenance (O&M) of equipment and infrastructure, which will be critical to long-term impact, has previously been an issue with similar investments in the project area. A thorough O&M plan has been developed to mitigate this risk.
Providing alternative water sources for the most vulnerable inhabitants of the Upper Athi River catchment area will improve their well-being and livelihoods, while alleviating water stress in the catchment.	
Community participation in O&M of water infrastructure and sub-catchment management planning will foster a greater sense of ownership of and responsibility for the water resources.	
Building partnerships with ongoing water, sanitation and hygiene (WASH) initiatives into the proposal will ensures that the supply of water will not lead to an increase in waterborne diseases and related health and safety issues.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the respective term sheet and addendum XVIII, titled "List of proposed conditions and recommendations".

II. Summary of the Secretariat's assessment

2.1 Project background

3. Kenya´s people and economy are highly vulnerable to water availability, thus making water security a key issue for the country. Water security concerns in Kenya are compounded

by climate change, which directly impacts on the quantity and quality of the nation's water resources. Kenya has limited freshwater and faces chronic water scarcity along with additional challenges related to high inter-annual and intra-annual rainfall variability.

4. Among Kenya's six water catchment areas, the Athi River Catchment has a negative water balance, in terms of water demand versus availability, and is the least water-secure region. It is also the most vulnerable to the adverse impacts of climate change. Among the 12 counties that compose the Athi River Catchment Area, four counties have experienced the highest drought and flooding risks: Kiambu, Machakos, Nairobi and Nyandarua, in that order. These are the four counties targeted by the proposed project.

5. In the target project area, most tributaries have recorded significant reductions in river flows, with some drying up entirely. In addition, climate change has impacted groundwater levels with knock-on effects for the water available to recharge the underlying aquifer. Groundwater resources in the project area have also been negatively impacted by higher temperatures leading to increased evaporation rates. Due to the decreased availability of surface water, reliance on groundwater has increased to meet demand, further straining resources. Water supply infrastructure and water resources management are simply not adequate to cope with these changes, which are exacerbated by climate change.

6. The project's climate objective is to increase water security and strengthen communities' resilience to climate change in Kenya's Upper Athi River Catchment area in Kenya. These counties were selected due to their high exposure to climate-impacted floods and droughts and climate-exacerbated water scarcity. The objective will be addressed through three sets of interventions related to: hydrological and meteorological information management; installing and rehabilitating water infrastructure; and strengthening planning and regulatory frameworks for water resource management.

7. In terms of environmental and social safeguards (ESS), the project is categorized as B for a moderate level of risks. The Secretariat's assessment of ESS documentation, underlying project risks and mitigation measures can be found later in this document.

8. The project proposal seeks USD 9.53 million of GCF grant funding alongside a combined USD 0.47 million in-kind co-financing from NEMA, the accredited entity (AE) and the two executing entities (EEs), Kenya's Water Resources Authority (WRA) and the Kenya Meteorological Department (KMD).

2.2 Component-by-component analysis

Component 1: Enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector (total cost: USD 2.74 million; GCF cost: USD 2.53 million, or 92 per cent)

9. This component aims to facilitate climate-risk informed decision-making on water resources management in the upper Athi River sub-catchment. Establishment of a National Information Centre for Integrated Water Resource Management will provide a central repository for meteorological, hydrological and water quality data and facilitate processing of that data into information essential for water use planning. Modernisation of the hydrological and meteorological observation networks will ensure that real-time data is available to guide decision-making, for example water releases from dams to prevent disasters. Groundwater monitoring wells will also be set up to facilitate decision-making about groundwater abstraction and required treatment. Capacity-building of the relevant government institutions will ensure that the monitoring networks and information system are effectively used and maintained. The integration of surface water, groundwater, precipitation and water quality data will be a first for Kenya and the region, with good potential for replication in neighbouring catchments and

countries. A broadcasting station will also be installed to disseminate weather and climate information.

Component 2: Improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment (total cost: USD 4.99 million; GCF cost: USD 4.90 million, or 98 per cent)

10. It is predicted that the Upper Athi River sub-catchment will continue to experience increasing frequency of both droughts and floods as a result of climate change. Annual rainfall across the four counties shows an overall increasing trend, but the river catchment is highly stressed by multiple competing uses and a lack of water storage. Thus, it seems plausible that construction/rehabilitation of small-scale water storage infrastructure, as well as a limited number of boreholes and associated water supply networks, will improve water availability for the most vulnerable without exacerbating demand for water from the river.

11. Furthermore, to improve ecological integrity and functioning of the river, there is need to revegetate the headwater and banks. Tree nurseries will be established to provide saplings for restoration of degraded water catchment areas. The physical work required will be provided by the beneficiary communities, engaged through the local water resources user associations. Community participation in catchment restoration is likely to deter any future degradation of the catchment.

Component 3: Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions (total cost: USD 1.21 million; GCF cost: USD 1.15 million, or 95 per cent)

12. Water resources are at risk from the adverse impacts of climate change, hence the importance of mainstreaming climate change considerations into sub-catchment management plans. The development and implementation of climate risk informed sub-catchment management plans is crucial to ensuring that water resources are allocated and distributed so as ensure that priority uses, such as drinking water, are secured under a changing climate. The project will also conduct a water quality survey to further safeguard water quality.

13. Beyond water resources management, there are other ecosystem management decision-making frameworks that need to be redeveloped to factor in climate change responses. In Kenya, one such framework is county environmental action planning. With NEMA leading this process, the beneficiary counties will be better equipped to manage multiple climate related risks.

14. Community sensitisation to influence compliance with water quality regulations is considered likely to result in improved ambient water quality in the river and its tributaries, thereby reducing the need for water treatment by both domestic and commercial users. Though difficult to estimate these diffuse emissions reductions, it is evident that better water quality in the river would mean less energy use at water treatment plants and perhaps reduced need for water boiling at household level, resulting in reduced deforestation.

15. Thus, it is the Secretariat's view that this project's approach towards improving both water quality and quantity available in the Upper Athi River catchment through climate-informed water management is holistic and sustainable.

Monitoring and evaluation (total cost: USD 0.56 million; GCF cost: USD 0.56 million, or 100 per cent)

Project management costs (total cost: USD 0.44 million; GCF cost: USD 0.37 million, or 84 per cent)

III. Assessment of performance against investment criteria

3.1 Impact potential *Scale: N/A*

16. The standard of living and livelihoods of the communities within the Athi River Catchment are very sensitive to climate variability and change. The demand for water far exceeds supply in the Athi River Catchment due to its negative water balance, making it one of the least water-secure regions in the country. Four out of 12 counties (Kiambu, Machakos, Nairobi and Nyandarua) in the basin have experienced the highest drought and flooding risks. The proposal's feasibility study on the analysis of the climate and hydrological cycle of the Athi River Basin established that the frequency and intensity of drought will increase under the projected global warming. The feasibility study also shows that the projected increasing temperature will lead to increasing evapotranspiration and will further exacerbate the negative water balance situation of the Athi River Basin. The proposed interventions will lay the foundation for an integrated and comprehensive solution to addressing the Basin's water security risk associated with climate variability and change.

17. The project is expected to directly benefit 1,156,620 people and indirectly benefit another 3,693,380 people. The AE includes direct beneficiaries as those with increased access to water sources for domestic and agricultural uses. The project also emphasizes water quality and the integration of a broader WASH approach by enforcing existing water regulations thus reducing incidences of waterborne diseases and increasing the community's well-being overall.

18. The project calculates indirect beneficiaries as the total population of the counties of Kiambu, Machakos, Nairobi and Nyandarua in the Upper Athi River Catchment area who derive wider project benefits of reduced flooding, avoided water resource conflicts, improved hydrological information and improved ecological functioning of the catchment.

19. Overall impact will depend on how well the investments made in this project can be maintained over time. The project has developed an adequate operations and maintenance (O&M) plan, but previous similar investments have been undermaintained.

3.2 Paradigm shift potential *Scale: N/A*

20. The proposal adopts an integrated water resources management approach where the biophysical and socioeconomic determinants of climate variability and change within the Athi River Basin are considered. The proposal identifies three areas of intervention: (i) enhance hydrological and meteorological monitoring systems to support decision-making, planning and policy development in the water and climate change sector; (ii) improve climate resilience and water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment; and (iii) strengthen water and adaptation planning, and the institutional and regulatory framework to respond to changing climatic conditions.

21. Beyond the water infrastructure to be installed or rehabilitated, the project's strongest aspect of paradigm shift potential is in the shared hydrometeorological systems and strengthening of institutional capacity of the key institutions, county governments and communities. These shared systems for water resource management, which link NEMA, WRA, KMD and the counties, will be an innovation for Kenya. The integration of WASH as part of an integrated and comprehensive approach to addressing water demand and supply will ensure that the investments will not lead to an increase in water-related diseases, environmental and public health risks.

22. A closely related strength is county government use of climate information for improved water resource management. County governments will be better positioned to manage water resources through the integration of meteorological, hydrological, hydrogeological and water quality data. As noted in the funding proposal, county governments will work with WRA and

KMD to establish hydromet information-sharing mechanisms, which will in turn help them determine current and future climate risks. Based on those new climate risk assessments, county governments will be better positioned to plan development interventions, particularly in the water sector. The approach, if successful, could be replicated in the other catchment areas in Kenya.

3.3 Sustainable development potential *Scale: N/A*

23. Currently, water demand far exceeds supply, with demand projected to increase to about 281 per cent by 2030. Water users within the project site consist of urban and rural population, small and heavy industries, farming communities, livestock and wildlife that place significant demand on water resources (both surface and ground water aquifers). Drought and floods present significant challenges to the region. The targeted areas experience seven dry months and five wet months annually, which results in women and children in the communities spending hours walking long distances to fetch water daily.

24. The proposed interventions will significantly reduce the risk of climate variability and change to lives, assets and livelihoods. The environmental, social and economic co-benefits to be derived from the project will be significant in that it will allow substantial costs to be avoided via sustainable provision of potable water for its various users. It will also enable businesses to derive increased and predictable revenue as a result of enhanced water security to sustain their operations. Co-benefits will also take the form of improved sanitation and hygiene from improved access to water. In addition, the water is expected to be of higher quality as a result of the project's focus on water quality testing and control. Improved sanitation and hygiene should be reflected in the decreased prevalence of waterborne diseases. Additionally, co-benefits will also include improved biodiversity and air quality as a result of the rehabilitation of the riparian areas in the catchment. The proposal does not quantify or estimate these co-benefits, and future funding proposals would benefit from this type of quantification.

25. The risk management benefits to be derived from hydromet services would also be significant since climate variability and change will be predictable through early warning and early action systems and protocols that will be established across all scales. The proposal is aligned with the Sustainable Development Goals (SDGs), Kenya's nationally determined contribution (NDC) and Sendai Framework.

3.4 Needs of the recipient *Scale: N/A*

26. In terms of climate vulnerability, Kenya's Athi River Catchment area is the least water secure of Kenya's six water catchments. In the context of climate change, the analysis in the feasibility study demonstrates that the majority of the catchment is ranked as highly vulnerable. Droughts, which are projected to become more severe by 2030, threaten the catchment's residents with the per capita water availability being less than half of the global standard. Climate change worsens the catchment's ongoing water deficits, pointing to a need for better planning and management of water resources. This is aligned with needs expressed in the NDC, SDGs and national and county long-term development plans.

27. The proposal makes the case that government funding is not sufficient to resolve droughts and floods, limited technical and institutional capacity, low compliance and an inadequate regulatory framework for water resource management. GCF funding will contribute to water security (projected to become substantially less secure in the next 10 years), strengthen hydrological and meteorological information services to deliver better improved information to local communities, and support decision making and policy development in the water sector.

3.5 Country ownership *Scale: N/A*

28. The proposal is strongly supported within the country and is among the first prepared by NEMA for GCF funding. The process of issuing a no-objection letter in Kenya seeks to ensure a fit between the proposed project and the country's climate strategies and plans, including Kenya's National Climate Change Action Plan, national adaptation plan and NDC. Just as importantly, the project activities are based on climate change adaptation measures and needs consistent with the National Water Master Plan.

29. More importantly, the core components of the proposal have been developed using local expertise and resources, demonstrating the strong ownership the Government has in the design and subsequent implementation of the project if approved. It will be implemented mainly by national institutions, counties and local communities which will integrate local knowledge, best practices and lessons learned to ensure the uptake and sustainability of the investments.

30. This proposal is the first brought forward to the GCF Board for approval. NEMA does have experience with other multilateral-funded projects, as it is also the national implementing entity with the Adaptation Fund programme. NEMA is currently implementing a similarly sized programme aimed at building climate resilience and adaptive capacity in 14 counties in Kenya.

3.6 Efficiency and effectiveness *Scale: N/A*

31. Based on the public goods nature of the interventions, GCF grant financing is assessed to be an appropriate financial instrument. However, the requested GCF grant financing leverages a relatively low amount of co-financing in the form of in-kind contributions from NEMA, WRA and KMD.

32. The proposal has established that the economic and financial viability of the project are positive and robust. At 12 per cent discounting rate, the Net Present Values (NPV) are estimated to be approximately USD 9.75 million, USD 7.58 million and USD 5.55 million for 30 years, 20 years and 15 years respectively. The respective Internal Rate of Returns (IRRs) are 13 per cent, 12 per cent and 11 per cent. The Benefits-Costs-Ratios are 1.78, 1.63 and 1.48 for 30 years, 20 years and 15 years respectively. This demonstrates that the return on investment from the project will lead to the expected economic and financial benefits.

33. The proposal also provides socio-ecological benefits through the restoration of ecosystem functions and services, such as healthy soil-vegetation-atmosphere interactions, which improve the hydrological cycle and broader local climates leading to improved livelihoods. The hydromet services and institutional effectiveness will also be improved through capacity development and technology transfer interventions.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

34. **Environmental and social risk category**. The AE has assigned a risk category of the project as Category B due to the scale of the activities and the fact that most of them would only involve rehabilitation of existing facilities. The Secretariat confirms the Category B classification. Under Component 1, the project would install and/or rehabilitate surface hydro-meteorological monitoring and network equipment, including weather, surface water and rainfall monitoring stations, and upgrading of an existing water quality testing laboratory. Under Component 2, the project would: (i) rehabilitate existing water sources and facilities, consisting of 8 boreholes, 23 water pans, 5 springs, 2 weirs, and a sand dam; and (ii) develop 2

new springs and 2 new boreholes; and installation of 80 new water tanks. It will also implement 2 existing catchment protection schemes.

35. **Safeguards instruments and disclosure**. The AE has submitted an Environmental and Social Management Framework (ESMF), a Summary of Consultations and Stakeholder Engagement Plan and an Indigenous People's Framework (both as separate Annexes to the Funding Proposal) which were disclosed in the English language on the AE's and GCF's websites. The ESMF provides for a more detailed assessment and management planning to be undertaken at the subproject levels. The ESMF also provides for the processes and steps of screening, assessment and review of the subprojects. The levels of impacts and risk category of individual activities (or subprojects) dictates the required assessment instruments which would also depend on the scale and locations of the subprojects, among other considerations. An Environmental and Social Management Plan (ESMP) is likewise provided in the ESMF which indicates the management actions, responsible parties, monitoring indicators including an indicative budget for its implementation.

36. **Key Environmental and Social Issues**. The ESMF contains an extensive baseline characterization of the environmental and socioeconomic conditions of the Upper Athi Catchment Area (ACA). The key impacts identified include those related to hydrology, vegetation clearing, soil erosion and sedimentation, air quality as regards dust and emissions from construction equipment and earth moving, generation of noise and solid wastes. The project may have potential impacts during water abstraction on the water balance, possible increase in water-borne diseases due to establishment of water pans, risk of encroachment into protected areas in the development of new water sources and catchment protection, concerns on community health and safety, and possible displacement of informal settlers or farm occupants in the target areas.

37. Construction works of individual subprojects are expected to be small-scale but should workers from outside the locality be brought in, behaviours may clash with the local culture and traditions. To avoid this, the project intends to source skilled and unskilled labour from the locality and allow for opportunities for additional income to local people. Depending on the storage capacities and locations of the reservoirs (water pans), these could also pose safety risk to the communities. The rehabilitation and construction of new water infrastructure could affect people in the vicinity of the structures depending on the land use status of the sites. For the rehabilitation works, this would depend on whether it would entail expansion of the footprint of the facility or whether the construction and maintenance would need additional easements. While the ESMF indicated that all the identified water infrastructures including the monitoring stations and equipment will be located on public utility lands, the project has to ensure that indeed there are no settlers in the project sites that would be displaced, including in the sites of the two new springs to be developed during project implementation.

38. The AE's preliminary assessment indicates that it is unlikely that indigenous peoples will be affected in the project's area of influence. Nonetheless, the project has developed an Indigenous Peoples Framework as a guide to mitigate adverse impacts on vulnerable and marginalized groups and should further screening of any of the subproject indicates the presence of indigenous peoples. The AE notes that the project expects that the nomadic lifestyle of the Maasai will be enhanced through availability of water from rehabilitated springs. The project has also prepared a chance find procedure in the event of possibility of presence of any cultural heritage or sacred sites near or within the project areas.

39. **Stakeholder engagement and grievance redress**. The project conducted discussions with user groups, specifically with the Water Resource User Associations (WRUAs) which allows the project to gather issues and concerns of the stakeholders. This also included interviews with technical officers from various government representatives. The AE has also prepared a Summary of Consultations and Stakeholder Engagement Plan which summarized the stakeholder consultations and engagements undertaken so far and the planned stakeholder

engagements during project implementation. The ESMF also provides for establishment and designation of an Independent Redress Committee (IRC) which shall be composed of various representatives from the community, the government and from non-government organizations, among others. It also provides for a designation of a grievance officer within the Project Implementation Unit (PIU) who will handle concerns and grievances during the course of project implementation. It should be noted that the GCF Indigenous Peoples Policy also states that the GCF independent Redress Mechanism and the Secretariat's indigenous peoples' focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

40. The AE has provided a gender assessment and gender action plan and therefore complies with the requirements under the GCF Gender Policy.

41. The gender assessment provides information on the existing enabling environment for promoting gender equality and women's empowerment. Kenya's constitution and bill of rights recognizes the primacy of addressing inequality and discrimination and has provisions to remedy gender inequalities. The National Gender and Equality Commission was established to ensure that national and county development processes are responsive to alleviation of inequality. Despite these legal and institutional frameworks inequality between women and men persists across all facets of development in the country.

42. The gender assessment illustrates the gender gaps in the four project counties. Information for the assessment was gathered through focus group discussions held with men and women and community leaders. The assessment provided information on the disparity between women and men in accessing resources such as land, knowledge, practices and participation in economic activities and legal rights. Disparities also exist in relation to household chores, balance of power, decision-making, and social status, while commonly held beliefs and perceptions further impede opportunities for women, leaving them at a disadvantage and contributing towards their current status of inequality in Kenya.

43. The assessment recognizes the central role water has for development, health, livelihoods, food security, and poverty reduction. Furthermore, the assessment analyses the obstacles faced by women accessing affordable water and highlights the following findings:

(a) **Accessibility:** As a result of droughts which have led to water shortages, it is necessary to travel ever greater distances to access water. This in turns leads to time poverty directly impacting women's participation in the formal and informal economy and compromises women's security when travelling to and from water sources. In addition, limited control over land and land ownership by women directly impacts their control over water resources as access is based on a permit system linked to land rights under formal tenure. Women have limited land rights;

(b) **Cost:** The purchase of water due to its limited availability has become an important source of household spending directly impacting poor households, especially female-headed households;

(c) **Sufficiency:** Inadequate irrigation has a direct impact on crop production where women have a preponderant role; and

(d) **Safety:** There are also physical safety concerns regarding access to clean water.

44. In the counties of Kiambu, Machakos and Nyandarua access to clean water for domestic consumption and irrigation, and improved drainage systems, were identified as priorities for the communities. The project responds to these needs by investing in institutional governance and integrated water resource management and increased access to potable water for domestic use.

45. Contexts related to the project and areas of intervention are also discussed in terms of access to climate information, which is less available to women. The assessment also indicates that female-headed households are a vulnerable group due to burdens of domestic, communal and productive roles with limited assistance.

46. Though the assessment is based on consultations in the form of focus group discussions, key informant interviews, observations and workshops in Kiambu, Machakos and Nyandarua counties, in many cases men and women from the same area have different perceptions on the same issues, such as access to resources (land, extension and income-generating activities), membership participation, roles and responsibilities. The assessment also indicates that patriarchy prevails in the counties and considers men to have a superior role in society compared to women, thus defining and prescribing roles that are challenging for women. A survey conducted into the division of labour and time use indicates that women and girls are assigned tasks that are time consuming, burdensome, considered low status and are unremunerated. This is in addition to the challenge that women face in accessing economic opportunities. The assessment concludes that in the four project counties women and girls are responsible for water resources, entailing long distances to collect water, time spent queuing and exorbitant payments for water.

47. The AE, in fulfilling the requirement of the Fund's Gender Policy, has provided a gender action plan. The gender action plan submitted includes activities that provides space for women to participate in gender assessment and consultation process and other activities, performance indicators with some sex-disaggregated targets and timelines for all activities. During the first year of the project, the action plan will be updated based on the consultation findings and further updates will take place as needed.

48. The activities included in the action plan are expected to address the challenges to gender equality as identified through the gender assessment. The activities included in the action plan are expected to address the challenges to gender equality as identified through the gender assessment. The activities are expected to provide opportunities to meet women's employment needs through their engagement in the rehabilitation and construction of water storage and supply infrastructure; engaging women in consultations prior to rehabilitation/new construction of water points; providing employment opportunities for women in civil works; reviewing the existing legal and regulatory frameworks in the water sector to ensure gender issues are mainstreamed; and participation of women in the County Environment Action Plan and other consultations.

49. The gender action plan also assigned roles and responsibilities to a gender expert that will be hired for the project to support with the implementation of the activities included in the gender action plan. The AE has allocated budget for the implementation of the gender action plan as part of the project's monitoring and evaluation cost.

4.3 Risks

4.3.1. Overall project assessment (medium risk)

50. GCF is requested to provide a grant of USD 9.5 million to enhance community resilience and water security in the Athi River Catchment area in Kenya. The AE and two EEs (WRA and KMD) are providing a total of USD 0.4 million in-kind contribution to the project.

4.3.2. Accredited entity/executing entity capability to execute the current project (medium risk)

51. NEMA will play a role as both AE and co-EE for this project. As a direct access entity, NEMA was established in 2004 to implement environmental policies in Kenya. NEMA has

operational county offices in all 47 counties and experience in managing projects funded by external donors (e.g. USD 10 million from the Adaptation Fund).

52. In addition to NEMA, there are two co-EEs: WRA and KMD. WRA is a state corporation under the Ministry of Water and Irrigation and is currently managing an average of USD 20 million per year since 2013. KMD has been managing projects with a budget between USD 1 million to USD 12 million for the past seven years.

4.3.3. Project-specific execution risks (medium risk)

53. O&M risk: the funding proposal has identified risks related to O&M such as poor maintenance, potential vandalism and potential expropriation. The project budget includes O&M costs during the implementation period and the government agencies will cover the cost beyond the project implementation. To avoid the risk of vandalism, the equipment will be installed in places with adequate security and burglar proof structures. As for the potential of expropriation, formal agreements will be signed to protect the equipment in case this has to be installed in private land.

54. Water tariff: the funding proposal states that the water service provider that is owned by the county government is established as a public institution and will manage public water services and assets on behalf of the public. According to the AE response, water service providers have their jurisdiction and collect the tariffs, but the revenue is not necessarily re-invested in the water management system as each county government has its own priorities. It is necessary that for steady operation of the project, the tariff determined by the county government is commensurate with the ability of the end users to pay. Also, the project will benefit if county governments reinvest the tariffs collected to cover the O&M cost for water resource management.

55. Project viability/concessionality: the economic analysis accounts for project benefits as avoided costs from flood damage, health treatment and travel time for water collection. The analysis results in the economic internal rate of return of 21.7 per cent. The net present value remains positive in sensitivity scenarios of 10 per cent increase in cost and 10 per cent reduction in benefits, with corresponding economic internal rate of return of 17.9 per cent and 14.8 per cent respectively.

4.3.4. GCF portfolio concentration risk (low risk)

56. In case of approval, the impact of this proposal on the GCF portfolio risk remains non-material and within the risk appetite in terms of concentration level, results area or single proposal.

4.3.5. Compliance risk (medium risk)

57. The proposed activities do not indicate an unusually high risk for money laundering, terrorist financing, or prohibited practices in themselves. Nevertheless, the capability to install internal controls may vary. The AE has described its approach toward assessing risks for illicit practices and will monitor and mitigate such risks. Based on the information provided, compliance risk is assessed as medium, subject to modification should additional details evolve.

4.3.6. Summary risk assessment and recommendation

Summary Risk Assessment		Rationale
Overall project	Medium	

Accredited entity/executing entity capability	Medium	The project aims to strengthen the water management governance and monitoring system. The AE will implement the construction of the water infrastructure and coordinate with other EEs and county governments for soft interventions. Steady support for O&M of the project by county governments will be critical for the sustainability of the project.
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

58. NEMA have the role of both AE and co-EE to implement some of the proposed activities. As the AE, NEMA shall be responsible for overall project planning, execution, coordination, monitoring and evaluation, and reporting. It shall also be responsible for reporting and the overall project budgeting based on the agreed work plans prepared by the executing entities.

59. NEMA has identified WRA and KMD as the co-executing entities for some proposed activities in line with their mandates. The implementation will be conducted by the EEs but will be guided by the National Project Steering Committee through quarterly meetings. The committee will ensure that project implementation is conducted according to the country's priorities.

60. GCF will disburse funds to the AE while NEMA will disburse funds to the EEs (WRA and KMD) based on the approved work plans and requisite reports. The funds shall be provided in advance to the EEs on a quarterly basis and against a criterion that will ensure prudent use of resources.

61. NEMA as the AE has the responsibility of financial management, accounting and reporting. It will ensure project performance audits by the NEMA internal auditors and Kenya National Audit office.

62. In terms of the procurement implementation arrangement, it is noted that the multiple EEs are foreseen to be involved in procurement execution. It is noted that the EEs are government agencies with dedicated procurement, finance, and internal audit departments, guided by the Public Finance Management Act, 2012 and are subjected to regular and annual external audits by the Auditor General's Office as well as general oversight by the State Corporations Advisory Committee in order to reinforce transparency and accountability in their operations.

4.5 Results monitoring and reporting

63. This project seeks to build resilience in three ways. First, the project strengthens hydrological and meteorological information services to deliver relevant, accurate and timely climate information to local communities, and to support decision-making and policy development in the water sector. Secondly, this project seeks to increase water security by investments in improving and rehabilitating prioritized water infrastructure in the target counties so as to increase access to portable water for domestic use. Finally, the project seeks to improve adaptation planning in the target area.

64. This is an adaptation project and is meant to increase the resilience of 192,770 households, with 1,156,620 direct beneficiaries, 51 per cent of whom are female. The project also aims to strengthen 150 km of catchment areas, and develop more resilient assets with a total value of USD 45 million. Finally, the project will also support local communities and it is

expected that at least 60 per cent of the beneficiaries will have more than 10 per cent increase in income towards the end of the project.

65. The project will contribute to achieving three fund-level impacts, namely:

(a) Increased resilience and enhanced livelihoods for the most vulnerable people, communities and regions;

(b) Increased resilience of infrastructure and the built environment to climate change; and

(c) Increased resilience of health and well-being, and food and water security.

66. Regarding the theory of change, there is a clear causal linkage between the barriers and proposed interventions with clear reference to assumptions and risks are made.

67. Regarding the timetable of implementation, earlier comments have been addressed and the project now conforms with the standard GCF format.

68. For the logical framework, the revised funding proposal incorporated most of the comments and suggestions provided by the Secretariat and conforms with GCF performance measurement frameworks and risk management frameworks.

4.6 Legal assessment

69. The Accreditation Master Agreement was signed with the Accredited Entity on 20 June 2016 (the "AMA"), and it became effective on 22 April 2021.

70. The Accredited Entity has provided a legal opinion confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

71. The proposed project will be implemented in Republic of the Kenya, country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The Secretariat provided a draft agreement on privileges and immunities to the NDA on 11 December 2015. Discussions are still in progress and the last communication was on 05 October 2018.

72. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

73. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP175

Proposal name:	Enhancing community resilience and water security in the Upper Athi River Catchment Area, Kenya
Accredited entity:	National Environment Management Authority (NEMA)
Country/(ies):	Kenya
Project/programme size:	Micro

I. Assessment of the independent Technical Advisory Panel[1]

1.1 Impact potential *Scale: N/A*

1.1.1. Adaptation impact

1. Aimed at reducing the impact of drought in the Upper Athi River Catchment Area (ARCA), the project has three components: (1) supporting decision-making, planning and policy development in the water and climate change sectors through the enhancement of the hydro-meteorological monitoring and information management systems; (2) constructing and rehabilitating water supply and storage infrastructure to improve climate resilience; and (3) strengthening institutional capacity and regulatory frameworks.

2. The technical features of component 1 include installing 25 automatic weather stations and 23 surface water level monitoring stations with multi-parameter water quality sensors, upgrading the water quality testing laboratories for the National Environment Management Authority (NEMA) and the Water Resources Authority (WRA), and establishing 8 groundwater monitoring stations to measure water level, conductivity and temperature. To address the need for coordinated governance of water resources, this component would support the development of a "National Information Centre for integrated water resource management" (under the WRA) by: (i) equipping it with necessary hardware and software facilities; (ii) developing integrated data analysis tools, a system of weather forecast information and a flood warning system; and (iii) developing data sharing protocols for institutions in the water sector and the community. Capacity-building and training would be provided to relevant government institutions and to the Water Resource Users Associations (WRUAs), which are the local-level water governance structures. In a subsequent submission, the proposal added the installation of a RANET radio and internet broadcasting station in Machakos County, to serve as an improved method of disseminating weather and climate information to target communities, and as a learning institution for students in environmental and climate-related studies. The direct beneficiaries will be the community settled within the project area in Machakos County.

3. Component 2 would include the construction and rehabilitation of boreholes, water pans, springs, sand dams, rainwater-harvesting tanks, and pipelines. The previous submission of this proposal failed to provide schemes and main dimensions of the proposed water systems. In the previous submission, annex 8 showed the location in longitude and latitude of the points of intervention, while annex 9 showed sketches of the specific interventions (e.g. a layout of a dam,

[1] This assessment is based on the funding proposal submitted to the independent Technical Advisory Panel (TAP) on 7 July 2021 and the responses to the independent TAP's questions provided by the accredited entity.

or of a borehole), but there were no descriptions or schematics of the systems as a whole. The present submission includes an accurate description of all proposed infrastructure rehabilitation and construction works, including a detailed provisional bill of quantities.

4. The original proposal, submitted in previous rounds of reviews, did not include an adequate water balance (demand and supply) for each system, on a seasonal basis, and for the years of project life expectancy with the climate change variability included. Since the driving force for this project is responding to aggravated seasonal drought due to climate change, the water balance is crucial under future conditions with the climate change variable incorporated. Questioned by the independent Technical Advisory Panel (TAP) on this point, NEMA introduced in this latest funding proposal submittal annex 29, which has a water demand and supply-based analysis for the target catchments of the Upper Athi River. Taking into consideration the fact that the data density leaves no choice but to use CHIRPS gridded datasets for the hydrological model analysis, the methodologies used for the analyses are adequate and the results obtained are clearly explained. Indeed, the potential evapotranspiration (PET) will increase for the January–March dry season by 1.7 to 3.2 per cent over the baseline PET in the Upper Athi catchment area under different plausible future scenarios. If future expansions in irrigated agriculture in the same catchment areas are taken into account, the extent of change in climate-change-induced PET appears much higher, indicating a greater probability of occurrence of drought despite an increase in water flow. The apparent increase in water flow in the dry season does not fully compensate for the losses in topsoil moisture due to perhaps much greater effects of increasing temperatures for the 2011–2040 period.

5. The funding proposal sets the direct beneficiaries of the project, based on the beneficiaries of the water infrastructure of component 2, at 198,000 households, or approximately 1,150,000 people. A subsequent submission included an estimation of direct beneficiaries from each piece of water infrastructure rehabilitated or constructed by the project.

6. Although the project recognizes that sand harvesting, poor agricultural practices, wetland and forest area encroachment, untreated effluent discharges and vegetation clearing are among the main catchment degradation issues, the only proposed activities to address catchment degradation in the previous submission were limited to the establishment of tree nurseries and tree planting in two catchments.[2] A subsequent submission also included riparian marking and pegging (using indigenous trees) of 20 km along the Riara and Ndarugu rivers, and construction of terraces and gabions (gulley healing by planting vetiver grass and/or installation of gabion boxes) in selected sites in Kiambu and Nairobi counties. The budget for this activity was increased from USD 64,000 to USD 128,000.

7. Component 3 is almost exclusively focused on institutional strengthening with the objectives of enhancing compliance with water regulations (specifically effluent discharge) and developing County Environmental Action Plans and Sub-Catchment Management Plans.

8. With the objective of reducing the discharge of untreated effluents to water bodies, therefore enhancing water quality, activity 3.2.1 involves "consultative forums" with representatives from industries and other activities that discharge effluents to surface water bodies, to inform and promote compliance with the existing good practices and regulations.[3] Paragraph 56 of the funding proposal indicates that "…The proposed safeguards for the structures to be constructed among other are as follows; Monitoring through the established integrated Water Quality & Pollution Control networks, Enforcement through waste disposal control plans, Issuance of effluent discharge permits / licenses, Enforcement of the set waste disposal standards and effluent discharge control plans". However, in the previous submission of this proposal to GCF, the project was criticized for not incorporating sufficient measures to reduce contamination of the receiving water bodies. Given it included no funding to build solid

[2] Funding proposal, p 25.
[3] Funding proposal, p. 25.

waste systems and wastewater treatment infrastructure, it was not clear how the project would achieve significant pollution control on water sources. The proponent was asked for the project to be implemented in conjunction with a sanitation project aiming to solve water contamination issues or, at the least, to include a specific study assessing water quality in the water sources targeted by the project and guarantee that without adequate wastewater treatment for that targeted area, the project would still be able to use water of acceptable quality. In response, the accredited entity (AE) included a new activity (3.1.3) focused on conducting a "water quality pollution survey for point and non-point sources of pollution", with a total budget of USD 350,000. As described in the funding proposal, the objectives of this activity are to generate water quality and pollution information for planning, policy formulation and implementation, to provide technical guidance to develop and implement waste disposal control plans, and to enforce the prescribed regulatory waste disposal guidelines/standards and effluent discharge control plans. As per the detailed budget plan (annex 4), the funds allocated for this activity are for hiring consultancy services for the "assessment of water resources, field data and information gathering, identification and inventory for point and non-point sources of pollution". In addition, 23 surface water quality monitoring stations with multi-parameter water quality sensors will be installed to monitor water quality under activity 1.2.1 and a water quality testing laboratory will be established under activity 1.2.2.

9. The present proposal also adds that: (i) the project would raise awareness of the beneficiaries of improved water infrastructure through hygiene campaigns aimed at promoting water disinfection methods and hygiene practices; (ii) all water assets would have "draw pipes" and "disinfection dispensers"[4]; (iii) all beneficiary counties in this proposal are implementing community-led total sanitation initiatives that seek to ensure all Kenyan communities are 'open defecation free', among other water and sanitation initiatives; and (iv) the project would seek to network and build on the achievements of the USAID-funded Kenya Integrated Water, Sanitation, and Hygiene Project, which seeks to increase access to improved water, sanitation, and hygiene in Nairobi among eight other counties (only Nairobi being inside the project area).

10. Activity 3.1.1 includes funds for the procurement of water quality monitoring systems to support enforcement of the water quality regulations.[5]

1.2 Paradigm shift potential *Scale: N/A*

1.2.1. Potential for knowledge and learning

11. The project allocates a significant portion of the budget for institutional capacity-building, technical training, and data and knowledge generation and management. A quarter of the budget for component 1 (on enhanced hydro-meteorological monitoring) would be used for activities involving support for the establishment of a "National Information Centre for Integrated Water Resource Management", technical trainings, and the implementation of a system for data analysis, forecasting and knowledge-sharing platforms.

12. Component 3 would support data generation through the provision of equipment for water quality monitoring, focused on the control of effluent discharge.

13. The project monitoring and evaluation plan is adequately described in annex 11, including sources of data, collection tools, frequency, indicators and indicative budget. Annex 11 contains most indicators that should be monitored during the project implementation and afterwards, but some formulations provided are very general and without further clarification

[4] When questioned on this by the independent TAP, the AE explained that disinfection dispensers would be provided with a 30 per cent chlorine solution to be used at a rate of one drop per 20 litres (a dosage of approximately 2 mg/L).
[5] For details on the proposed equipment see the detailed budget plan (annex 4).

might not be effective (e.g. what will be monitored as "reliable/year-round water supply"? Will it be unlimited daily supply or litres per person? Thresholds should be established for indicators classified as high, medium and low). Logically, annex 11 should be more detailed than the same indicators are described in Log-Frame (sections E.3–E.5) because the funding proposal is limited in pages; however, in this case, sections E.3–E.5 in the funding proposal provide more detailed descriptions of indicators than annex 11. Some of the indicators from sections E.3–E.5 are oriented towards activities instead of results. Indicator A.7.1 is formulated correctly and is oriented towards results; indicator A.6.2 should be similar. Indicator A.1.1 is particularly important, but it is not consistent with annex 3ii, cost-benefit analysis (CBA). Indicator A.1.1 from the funding proposal reports that baseline, mid-term and final monitoring values will be established prior to the submission of the inception report; however, without the baseline losses and assumptions on improvements at mid-term and at the final stage the CBA could not be conducted. This indicator should be consistent with the assumptions made in annex 3ii.

14. The project includes a fund allocation of USD 150,000 for the development of knowledge products to document and disseminate the lessons learned during project implementation. Knowledge products would comprise brochures, magazines, biannual programme status reports, banners and information posters, and knowledge-sharing workshops.

1.2.2. Contribution to the creation of an enabling environment

15. The cost of operation and maintenance (O&M) for the new and rehabilitated water infrastructure is estimated at approximately USD 600,000 per year. When questioned by the independent TAP on this matter, the proponent responded that "…WRA is mandated with the management of the monitoring boreholes and this status will remain post project implementation. WRUAs are recognized by law as the managers of community water infrastructure, consequently, the respective WRUAs in the various project sites will be responsible for the day to day running of the water projects and will work together with the County Governments which are responsible for the maintenance of the water infrastructure which will be developed/rehabilitated. The Water Act 2016 gives the County Governments responsibility for water service prfovision. Additionally, upon completion of the project, the infrastructure will be handed over to the County Governments who will be responsible for their maintenance post project." The County Governments have provided letters to confirm that the lion's share of finance for implementing the O&M plan will be shared by the local government institutions. During one of the interviews with the independent TAP, the AE indicated that water pumps, rainwater harvesting tanks, water ponds and sand dams by design are expected to have very low to negligible O&M costs. Installation of silt traps in water ponds will allow community members to maintain these systems with locally available labour. The AE indicated that the O&M cost of boreholes (relatively higher than the rest) will be covered by fees charged for their use. In addition to this, and in response to questions raised by the independent TAP on the coverage of O&M costs, the AE stated: "To ensure sustainability of the new stations and structures beyond the project period, WRA and KMD will request the government to allocate more O&M funds for their sustainability beyond the project period and the county government will carry out the O&M of the rehabilitated / established structures after project hand over." The AE also added that post-project O&M for structures under water service providers in Kiambu, Nyandarua and Machakos would be the responsibility of the water companies. Water infrastructure under the management of committees would require that they register as water users' associations under the Water Sector Trust Fund - Community Project Cycle framework for community water supply, and would incorporate a water tariff to cover O&M costs.[6]

1.2.3. Contribution to the regulatory framework and policies

[6] Paragraph 115 of the funding proposal.

16. Although there are no specific activities aimed at modifying the regulatory framework, activity 3.1 would support strengthening the authority's ability to control compliance with existing water regulations, specifically effluent discharge regulations.

17. Regarding policies, the project would support the development of County Environmental Action Plans and Sub-Catchment Management Plans.

1.2.4. Scalability and replicability

18. The moderate scalability potential of the project resides in the capacity-building that would be provided at the national level for the NEMA, WRA and the Kenya Meteorological Department (KMD), that would partially facilitate upscaling the project to cover other counties.

19. With respect to replicability, the project will emphasize the sharing of knowledge and lessons learned between the executing entities and other counties along the Athi River. The proposal includes specific activities related to the development of knowledge products for the dissemination of the lessons learned during project implementation.

1.3 Sustainable development potential *Scale: N/A*

1.3.1. Environmental co-benefits

20. Environmental co-benefits of the project would arise from activity 2.2, which would involve the establishment of tree nurseries, planting 50,000 native trees at catchment riparian areas, especially in abandoned quarry sites at the Riara, Kamiti and Ndarugu rivers, the construction of terraces and gabions, and riparian marking and pegging of 20 km along the Riara and Ndarugu rivers using indigenous trees. The budget allocation for this activity in relation to the total budget is relatively small (1.3 per cent).[7]

21. Potential environmental co-benefits include increased biodiversity, reduced soil erosion and river siltation, and better water quality in downstream surface water sources. However, the target performance indicators[8] assume this activity would cover 150 km of riparian land which, at a total of 50,000 trees, gives a planting density of 333 trees/km. This represents one line of trees in each river margin, at one tree every 6 meters. This density seems to be too low to generate appreciable positive impacts.

1.3.2. Social co-benefits

22. The social co-benefits are the central objective of the proposal, and include enhanced water and food security. Water would be used for domestic and agricultural purposes. Reduced drought and flood impacts would be achieved through the rehabilitation of the water systems and the installation of a flood warning system for flood-prone areas.

23. Increased water availability would result in health benefits. Activity 3.1.2 includes hygiene awareness campaigns to promote, among other things, water disinfection methods and hygiene practices.

1.3.3. Economic co-benefits

24. The economic co-benefits include reduction in costs from flood damage, health expenditures and timesaving.

[7] Detailed budget plan, annex 4.
[8] Funding proposal, section E.5.

1.3.4. Gender-sensitive development impact

25. The participation of women in consultations during the project design was not described in the previous submission of this proposal. When questioned by the independent TAP on the level of women's participation at the stakeholder engagement meetings held with non-governmental organizations, community-based organizations and WRUAs during the project design phase, the proponent explained that "…The WRUA Development Cycle (WDC), which is the framework that guides WRUA participation and engagement allocates 30% representation of woman. Further, the framework also makes provision for women to be engaged in leadership positions within the WRUAs, aside from just membership. Towards this end, one leadership position (chairman, secretary, treasurer) is always reserved for a woman. The WDC framework will therefore ensure equal participation of the women in the WRUAs". This describes the framework that guides women's representation at WRUAs, but participation of women in the project inception remained unclear. The present submission clarified that 35 per cent of the participants of consultation meetings held at inception with project beneficiaries were women, that the project technical design team was gender balanced, and that the contractors would also be required to have a 51 per cent involvement of women in works by introducing a clause in the contract.

26. The equal participation of women in management and leadership positions is crucial for the sustainability of water systems. However, other than citing current regulations and laws, the funding proposal fails to show how equal participation will be achieved.

1.4 Needs of the recipient *Scale: N/A*

1.4.1. Vulnerability of the country and climate rationale

27. About 75 per cent of the population depends directly on land and natural resources for their livelihoods. The national economy is highly dependent on agriculture. Therefore, Kenya is vulnerable to the variability of rainfall patterns.

28. The AE indicates that the funding proposal is mainly focused on addressing drought impacts. Although the proposal states that "Kenya has experienced frequent floods and droughts resulting from climate change impacts" and that "climate change has adversely affected water availability especially during frequent drought conditions", the expressed causality between climate change and drought risk is not demonstrated in the proposal, failing to prove the observed and projected climate change impacts in water stress.

29. The analysis of historical data in the previous submission was carried out using gridded data (CRU TS 3.21). Upon the independent TAP's request, the present submission includes a trend analysis based on observed data from 33 stations. However, the trend analysis is made at a national scale, and only 3 of the 33 stations are representative of the project area. This indicates poor data density for the target region. The analysis at basin level seems to be limited to paragraph 23 of the funding proposal, which states that "Despite the projections on the Global Circulation Models, the local data results for annual and seasonal rainfall across the 4 counties show an overall decreasing trend". The inference is drawn from analyses involving gridded datasets, largely due to the non-availability of longitudinal time series meteorological datasets from stations around the target catchments. However, this statement is not supported by time series charts of historical data or projections at basin level.

30. The impacts of future climate on the water resources of the Upper Athi River Basin are assessed in annex 29. The study uses rainfall projections from the Rossby Centre regional atmospheric model (RCA4 with a horizontal grid spacing of approximately 50 km). The specific model variant (MPI-ESM-LR) under RCA4 is found to reproduce large-scale signals such as El

Nino Southern Oscillation (ENSO) and Indian Ocean Dipole, which generally have significant influence on the overall climatology of the Eastern Africa region in general and Kenya in particular. In the absence of long-term longitudinal climatological datasets for the specific catchments, the model is validated using CHIRPS gridded datasets, where accumulated (monthly) rainfall figures for the period 1981–2005 are found to have a high correlation involving the model output with respect to CHIRPS datasets. The AE has made efforts to use a climate model for future projections under Representative Concentration Pathway (RCP) 4.5 (also for RCP 2.6 and RCP 8.5 scenarios) for the period 2011–2040 using MPI-ESM-LR.

31. The models used to make climate projections described in annex 26 (Climatology of Kenya) are validated against gridded data in the absence of high-density observational datasets.

32. The results from hydrological modelling based on those climate projections conclude that eight out of nine simulations indicate increased flows in the Upper Athi River Basin in the future (2011–2040) and that seasonal flow patterns indicate an increase of flows in the generally drier months from December to March. The proposal argues that, in spite of this, the net water deficit would increase due to increased temperatures, among other factors. Data analysis provided in the present submission shows that the increase in PET at catchment level for the months of January–March (i.e. generally the drier months) is 1.7 to 3.2 per cent higher than the baseline PET for the period 2011–2040. This incremental change in PET is completely attributable to climate change, where the rise is temperatures will be the dominant cause, overpowering the beneficial effects of increase in rainfall during the drier months. However, when such changes in PET takes into consideration future plans for the expansion of irrigation in the catchment areas of the Upper Athi River, the projected impact becomes much bigger than the "no expansion" scenario. This leads to the inference that the system needs to enhance its water retention capacities so that the restored stream flows due to increased rainfall could be utilized for irrigation purposes.

33. The proponents indicate that access to water is likely to become more difficult due to population growth, economic expansion, unsustainable management of water and forest resources, and changes in rainfall patterns. Yet it fails to indicate how much of the growing demand may be attributed to change in rainfall pattern (or decline in water availability). Since rainfall would only increase, as indicated, in two seasons, net availability should not be reduced as such. The funding proposal also states that water is the core input for most economic activities: irrigated and rain-fed agriculture, hydroelectric power that constitutes over half the installed capacity of electric power, sanitation and the provision of drinking water. Yet neither rain-fed agriculture nor hydroelectric power would be impacted by an increase in seasonal rainfall patterns.

34. The drought risk assessment contained in the feasibility study (section 2.3.2.1) of previous submissions concluded that "It appears that the drought risk is likely to rise in the future, especially in Kiambu and Machakos County". However, this statement was not demonstrated by data on current and projected rainfall patterns, which should include data on monthly precipitations. The present submission includes a more thorough description of projected drought and flood risks. These inferences are based on the outputs of a model which is validated only by using CHIRPS datasets. Therefore, the reliance over projection data is poor. Nonetheless, due to poor data density involving catchment scale observational data, the independent TAP accepts the projection analysis.

35. As seen in figure 19 of the feasibility study, the difference in drought risk in the Upper Athi River Basin for the periods 1975–2005 and 2011–2040 is not very significant. Moreover, the modelling-based projection for 2011–2040 portrays a picture of water deficit, despite the trends of increasing water availability in the critically dry months, as observed in the recent past (i.e. 1975–2005). The rate of evapotranspiration under climate change due to rising temperature during the drier months plays a greater role than the increase in rainfall in

corresponding months, which is why there is a gradual build-up of drought during the peak dry season (January–March). This is also seen in the Standardized Precipitation Evapotranspiration Index (SPEI) analyses at catchment scale, presented in this submission, where the extreme drought condition (SPEI value >-2) will be increasingly observed in the catchment areas. In contrast, due to higher seasonal rainfall in future, the currently observed floods will occur with higher frequencies than in the baseline period.

36. The funding proposal states that changes in annual rainfall and extreme precipitation patterns will affect groundwater recharge and that climate change has already affected groundwater levels and river flows through its influence on precipitation and evapotranspiration. However, the proposal does not present trend analyses regarding the height of the piezometric surface of the aquifer or references to or descriptions of the studies to provide evidence to justify this claim.

37. Flood damage analysis is based on one extreme flood event; therefore, it does not tell whether or not climate change is responsible for the occurrence of such a flood event. There is no trend analysis, based on past/historical evidence, and globally acceptable indicators are not being used for such analysis.[9]

1.4.2. Economic and social development

38. Kenya is one the most developed countries in East Africa, as well as one of the most unequal countries in the sub-region: 42 per cent of its population of 44 million lives below the poverty line. Agriculture, forestry, and fishing are the largest sectors in the Kenyan economy, accounting for about 22 percent of the gross domestic product.

39. As per the Joint Monitoring Programme for Water Supply and Sanitation, 57 per cent of Kenyans in rural areas had access to improved drinking water sources in 2015 and 14 per cent of Kenyans in rural areas were reported as having access to piped water through a domestic connection. From 1990 to 2015, access to improved water sources in rural areas increased from 33 per cent to 57 per cent.

40. Specific climate change vulnerabilities for each targeted county are adequately defined in the proposal.

1.4.3. Absence of alternative sources of financing

41. The absence of alternative sources of financing is not expressly demonstrated in the proposal. However, given Kenya's economic development, it could be assumed that funds for climate change adaptation are scarce, thereby justifying the request for a grant from GCF.

1.5 Country ownership *Scale: N/A*

1.5.1. Alignment with national climate strategy and policies

42. The project is well aligned with all the national policies, strategies and plans related to climate change, including the National Climate Change Response Strategy, National Climate Change Action Plan, national adaptation plan, the intended nationally determined contribution and the National Water Master Plan 2030.

1.5.2. Capacity of accredited entities or executing entities to deliver

[9] Funding proposal, paragraph 21.

43. As the AE, NEMA would have the responsibility of project management, reporting and budgeting. NEMA was established in 2004 as the principal government agency mandated to implement environmental policies, and supervise and coordinate all environmental matters. NEMA has operational county offices in all the 47 counties in Kenya, and is currently implementing a USD 10 million programme aimed at building resilience to climate change and adaptive capacity of vulnerable communities in 14 Kenyan counties.

44. The executing entities would be the WRA and the KMD.

45. The WRA is a state corporation under the Ministry of Water and Sanitation, mandated to regulate, monitor, assess and allocate the water resources, in coordination with other relevant institutions. The WRA has 6 regional offices and 26 sub-regional offices, with a total of 750 staff distributed across the country. The Water Users Associations are WRA’s governance structures at the local level. WRA’s experience with projects related to water security represent a combined project implementation portfolio for both past and current projects of approximately USD 50 million.

46. The KMD hosts the National Flood Forecasting and Early Warning Centre, which is responsible for data collection, analysis, modelling and forecasting of floods, and has the required facilities for monitoring and forecasting flood and drought hazards, such as real-time rainfall data, evaporation, and storm monitoring tools, and quantitative rainfall forecasts. The department has staff trained in meteorology, operational hydrology, numerical weather prediction, data analysis, forecasting and delivery of services to specific users, and climate change modelling.

1.5.3. Engagement with civil society organizations and other relevant stakeholders

47. The Summary of Consultations and Stakeholder Engagement Plan describes the consultations with stakeholders carried out during the preparation phase, including engagement with the national designated authority, county and community representatives and public consultations. As previously expressed, the rate of participation of women in these consultations is unknown.

1.6 Efficiency and effectiveness *Scale: N/A*

1.6.1. Financial adequacy

48. For the economic analysis, previous submissions of this FP considered a period of 50 years. However, this assumption seemed excessive given that most of the proposed water infrastructure (such as water pans, rainwater harvesting systems, and spring protection works) would not last that long. This type of project generally uses a maximum period of 25 years for the economic analysis. The current submission presents an economic analysis for a period of 30 years, which is more reasonable.

49. The proposal estimates that monetary health benefits would be approximately USD 250,000–300,000 per year, arising from avoided diarrhoea treatment costs. The monetarized timesaving is estimated at USD 700,000–800,000 per year and the avoided costs from flood damage are estimated at approximately USD 2 million per year. Avoided flood damage was estimated based on the recorded per capita costs of the 2009 floods, assuming 5 per cent of the total damage could be saved, and from that, 5 per cent as a result of project activities.

50. The bill of quantities included in previous submissions segregated individual costs for each infrastructure work, but the cost estimation was not complemented by a scheme and general description of each system. The current submission includes a detailed description of each infrastructure work together with the bill of quantities.

1.6.2. Amount of co-financing

51. The USD 9.5 million grant from GCF would be complemented by an in-kind contribution by the Government of Kenya to the amount of USD 470,000.

1.6.3. Financial viability

52. The project would not generate direct revenues. For this reason, a financial analysis is not applicable. The economic analysis considers three benefits: reduced flood damage, reduced health expenditures and timesaving. The estimated reduction in health expenditures is minimal in comparison to the estimations for flood damage and timesaving.

1.6.4. Best practices

53. For the installation of the flood warning system, the KMD would build on the experience of the Nzoia Flood Early Warning System, successfully implemented under the Western Kenya Community Driven Development and Flood Mitigation Project in Western Kenya funded by the World Bank and the Government of Kenya.

II. Overall remarks from the independent Technical Advisory Panel

54. The analyses on climate relevance of the project are weak, mostly due to a lack of data density involving long-term longitudinal observation data for the target catchment areas. However, this is typical of an African country that has graduated from least developed country status only recently. Given the structural deficiencies in presenting appropriate climate rationale and also recognizing the nominal risks associated with proposed adaptation response measures to address droughts in an African country, the independent TAP accepts the weak climate rationale as the basis for the proposed climate responses.

55. The results from hydrological modelling based on climate projections prepared by the AE conclude that there will be increased flows in the Upper Athi River Basin in the future (2011–2040) in the drier months from December to March. The proposal argues that, in spite of this increase in water availability in the drier months, the net water deficit would increase due to increased temperatures and changes in rainfall patterns, among other factors. However, the proponents also indicate that access to water is likely to become more difficult due to population growth, economic expansion, and unsustainable management of water and forest resources. The AE indicates that the increase in water supply projected by their climate modelling for the dry months would be counterbalanced by a major increase in moisture loss from the topsoil, which would lead to drier conditions. Previous submissions of this funding proposal did not provide data to substantiate these claims. The revised annex 29 to the present submission includes projections on irrigation water demand based on current and projected PET and irrigation area, showing that the apparent increase in water flow in the dry season would not fully compensate for the losses in topsoil moisture due to perhaps much greater effects of increasing temperatures for the 2011–2040 period.

56. Based on this assessment, the independent TAP endorses this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP175)

Proposal name:	Enhancing community resilience and water security in the Upper Athi River Catchment Area, Kenya
Accredited entity:	National Environment Management Authority (NEMA)
Country/(ies):	Kenya
Project/programme size:	Micro

Impact potential
The AE notes of the iTAP comments
Paradigm shift potential
The AE notes of the iTAP comments
Sustainable development potential
The AE notes of the iTAP comments
Needs of the recipient
The AE notes of the iTAP comments
Country ownership
The AE notes of the iTAP comments
Efficiency and effectiveness
The AE notes of the iTAP comments
Overall remarks from the independent Technical Advisory Panel: The AE notes the iTAP endorsement

National Environmental Management Authority (NEMA) and the Korea Environmental Industry and Technology Institute (KEITI)

Gender Analysis for Enhancing the Resilience of Communities and Ecosystems in the Athi River Catchment Area

Fourteen Falls, Kiambu County

Tunay Africa, 2018

Acknowledgement
As a productive resource, water intersects with multiple other resources such as land and labour to determine the impact of development initiatives. Access to adequate and safe water mediates attainment of food security and health indices for households. Women's instrumental role in procurement of water and its use across various domestic and commercial platforms renders a gender sensitive approach imperative. We therefore unreservedly congratulate NEMA and KEITI for committing to undertake a gender analysis of the Athi River Catchment Area to enhance the resilience of communities and the ecosystem.

Leaders and policy makers in Nyandarua, Kiambu, Machakos and Nairobi counties participated in stakeholder forums and created ample opportunities for the research team to interact with community members which lead to generation of authentic data for the gender analysis. We recognize this exemplary contribution to the gender analysis as we express special commendation to the leaders and policy makers the professional and impressive dedication to the exercise.

Information sharing is a crucial ingredient of development as it shapes and directs policy formulation and service delivery. This exercise is thus pricelessly indebted to the women and men who volunteered information either as community members or experts in response to the gender analysis exercise. We are confident that this report attests to the reality embodied in the voices of all those who shared their views and affirms the validity of their aspirations.

Planning and executing the gender analysis exercise that spanned across four counties was a daunting task that called for exceptional people skills, imagination, dedication and resilience. Staff and relevant stakeholders of NEMA and KEITI respectively demonstrated these qualities that immensely inspired the TUNAY Africa team to deliver on the assignment. We profoundly appreciate the professional guidance provided and believe that this report resonates with their expectations.

It is incumbent upon all duty bearers interacting with this report to facilitate its transition from data to pragmatic actions and tangible results in the lives of women and men in communities who narrated their experiences and spell out their dreams. We believe that the recommendations proposed and endorsed by stakeholders will alleviate the challenges faced by women and men in the four counties in accessing water and being protected against the vagaries of climate change. It is time to acknowledge that the proof of the pudding is in the eating, and a journey of a thousand miles starts with one step!

Table of Contents

List of figures

Acronyms and Abbreviations

NEMA	National Environment Management Authority
KEITI	Korea Environmental Industry and Technology Institute
GCF	Global Climate Fund
WRA	Water Resource Authority
WRUAs	Water Resource Users Associations
GAP	Gender Action Plan
FGD	Focus Group Discussion
PRA	Participatory Rural Appraisal
KII	Key Informant Interview
IR	Inception Report
CG	County Government
ACA	Athi Catchment Area
TCA	Tana Catchment Area
ASAL	Arid and semi-arid land
ARCA	Athi River Catchment Area
LVNCA	Lake Victoria Norths Catchment Area
LVSCA	Lake Victoria South Catchment Area
RVCA	Rift Valley Catchment Area
ENNCA	Ewaso Nyiro North Catchment Area
KNBS	Kenya National Bureau of Statistics

Executive Summary

This report presents findings of a rapid gender analysis study commissioned by the National Environmental Management Authority (NEMA) in Kenya through partnership and technical assistance from the Korea Environmental Industry and Technology Institute (KEITI). The gender analysis was predicated on enhancing the resilience of communities and ecosystems in the Athi River Catchment Area focusing on Nyandarua, Kiambu, Machakos and Nairobi counties. Specifically, the gender analysis sought to identify and propose remedies to development gaps that manifest gender based inequalities and forestall possible outcomes of the project that could exacerbate existing, or generate new forms of disparities in the way the project impacts on men, women and other vulnerable populations within the catchment areas.

The gender analysis was cross-sectional, using qualitative methods. Emphasis was placed on making the process participatory through focus group discussions, key informant interviews, observation and workshops in Nyandarua, Kiambu, Machakos and Nairobi counties. The study was framed around access to resources; knowledge, beliefs and perceptions; practices and participation; balance of power and decision making; legal rights and status; and time and space as domains that manifest gendered disparities. Study respondents included community participants, service delivery officers and policy makers in the water and environment management sectors, and gender mainstreaming oversight institutions. Qualitative data was majorly collected through extensive desk reviews. The main findings of the study are were also coined around various elements of gender equality and equity that include: Access to resources, Level of knowledge acquisition and skills, Cultural beliefs and perceptions, Participation and decision making, Legal rights and status, and Time availability and space concerning location of water points.

The constraints encountered in undertaking the study were lack of data on gender issues at the county level attributed to the formative nature of devolved units in the country. National data was thus used as proxy data to inform the situation in counties. Most counties do not have dedicated gender officers which contributed to inadequate capacity to understand and articulate development issues through gender lens. The main limitation of the study is the contextual nature of qualitative data which may not be generalized across other different development settings.

This report is organized in six main areas as follows:

Chapter One: Introduction – This section contextualises development progress in Kenya and goes further to highlight the interface between water and gender issues.

Chapter Two: Gender Analysis – This section of the report briefly articulates the programming purpose of undertaking the gender analysis.

Chapter Three: Methodology – This part of the report summarises the research process followed in undertaking the gender analysis.

Chapter Four: Findings – This section of the reports articulates and analyses feedback received through various methods of undertaking the gender analysis, including literature reviews.

Chapter Five: Summary and Recommendations – This part of the report condenses the findings and outlines measures in response to the same.

Annex – This section of the report presents an action plan based on the findings and recommendations.

1.0 Introduction

1.1 Country Context

Kenya underwent re-basing in September 2014 that conferred on the country the lower middle income status with the fourth largest economy in sub-Saharan Africa after Nigeria, South Africa and Angola. Despite women's population standing slightly higher at 19, 417, 639 above men that was 19, 192, 458 out of the total 38, 610, 097 as per the 2009 Kenyan census, which is projected to reach 67.84 million people by 2030, they still remain the most vulnerable, including being victims of injustices. This demographic gallop has exerted pressure on the country's resources for sustenance, highlighting the critical and precarious interplay between environmental concerns and community needs.

According to the Kenya National Bureau of Statistics (KNBS, 2019) census results, there is an increment in population level. Men's number is 23, 548, 056 while women 24, 014, 716, indicating 50.5% increase of women, equivalent to 466,660 more numbers gap above men. Intersects were factored in the Government- led survey and recorded as 1, 524. The total National population as at current is 47, 564, 296. Household size has reduced from 4.2m in 2009 to 3.9 in 2019. It is a confirmation that women are more than men as per the trend. Unfortunately, the worry is that women still remain the most vulnerable, subordinated and discriminated against within socioeconomic and political set-ups.

Further analysis on cross- sectional surveys indicate that 72% of household related tasks are done by women and girls, justifying the power and roles relation gaps that still exist between men and women. Nairobi and Kiambu counties are also listed by the KNBS 2009 as the top most populated devolved units. Though Household headship in the four counties- Nairobi, Machakos, Kiambu and Nyandarua presents that males surpass at 67.6% against females 32.4%, women still form the bigger lot regarding water collection and use, and demonstration of consciousness towards its proper conservation and management. The Constitution of Kenya 2010 through the Bill of Rights and Affirmative Action, as well as mandates of the National Gender and Equality Commission is accorded preference as an instrument towards addressing and redressing the gender gaps. The CIDPs for the respective counties on water resource and services management will form part of guiding avenues on water intervention- access to, and control by women and men, similarly girls and boys.

Territorial area for Kenya is 582,646 km², divided into water area of 11,230 km² and land area of 571,416 km². The major part of the inland water surface area is covered by a portion of Lake Victoria and Lake Turkana. Of the land area, approximately 490,000 km² (more than 80% of the land area) is classified as arid and semi-arid land (ASAL). The remaining area of about 81,000 km² is classified as non-arid and profitably usable lands, sustaining a substantial portion of Kenyan economy and human population.

The main economic blue print and road map of the Kenyan State is the Kenya Vision 2030 (2008 – 2030) which aims to transform Kenya into a newly industrialized, "middle-income country providing a high quality of life to all its citizens by the year 2030". The Vision 2030 is founded on the triad of economic, social and political pillars. The economic pillar is designed to catalyze a GDP growth rate of 10% per annum as from 2012 while the social pillar has the architecture of promoting justice, equity, societal cohesion and observance of prudent environmental conservation practices. The political pillar seeks to secure the country's democratic traditions and human rights of all and sundry. The Kenya Vision 2030 has entered its Third Medium Term Phase (2018-2022) which has prioritized the "Big Four Agenda" that outlines agriculture, universal health coverage, manufacturing and housing as the main development milestones over the next five years. The 2010 Constitution is the foundation of legal, institutional and governance reforms in the country, including the advent of devolved units (counties).[1]

1.2 Water Resources in Kenya

Kenya is divided into six water catchment areas, namely Lake Victoria Norths Catchment Area (LVNCA), Lake Victoria South Catchment Area (LVSCA), Rift Valley Catchment Area (RVCA), Athi Catchment Area (ACA), Tana Catchment Area (TCA) and Ewaso Nyiro North Catchment Area (ENNCA). According to the Kenya Water Master Plan, the amount of water available in each of the catchment areas vis-à-vis the demand for the same resource between 2010 and 2030 is as shown in Table 1. The water deficit in the ACA (58,639 km²) is glaring due to large water use as demonstrated by high water demand/water resource ratio of more than 40%[2], partly driven by the comparatively high population density of the area.[3] The Kenya Hydro-Economic Analysis[4] concurs that the Athi catchment, as of 2010 water use is already exceeding environmentally sustainable limits by a substantial margin. Catchment degradation, which increases hydrological variability (i.e. increases the risk of droughts and floods) is the challenge most widely faced, and one that increases risks for all water users, including farmers, energy producers, industry, and pastoral groups.[5]

Table 1: Water Available vs Demand for Water (Unit: MCM/year)

	2010	2030	2050

[1] Kenya is divided into 47 counties according to the 2010 Constitution. The gender analysis will cover Kiambu, Nyandarua, Nairobi and Machakos counties.
[2] Ministry of Environment, Water and Natural Resources. (2013). The Project on the Development of the National Water Master Plan 2030.
[3] Population density is 167 persons/ km².
[4] 2030 Water Resources Group. (2015). Water Resources in Kenya: Closing the Gap.
[5] Ibid., pg.5

Catchment Area	Water Resources (a)	Water Demand (b)	(a)/(b)	Water Resources (c)	Water Demand (d)	(c)/(d)	Water Resources (e)	Water Demand (f)	(e)/(f)
LVNCA	4,742	228	5%	5,077	1,337	26%	5,595	1,573	28%
LVSCA	4,976	385	8%	5,937	2,953	50%	7,195	3,251	45%
RVCA	2,559	357	14%	3,147	1,494	47%	3,903	1,689	43%
ACA	1,503	1145	76%	1,634	4,586	281 %	2,043	5,202	255 %
TCA	6,533	891	14%	7,828	8,241	105 %	7,891	8,476	107 %
ENNCA	2,251	212	9%	3,011	2,857	95%	1,810	2,950	163 %
Total	22,564	3,218	14%	26,634	21,468	81%	28,437	23,141	81%

Source: Ministry of Environment, Water and Natural Resources, 2013

1.3 Gender and Inequality Highlights for Kenya

Kenya has made significant progress in promoting and sustaining equality as a human rights and development imperative through legal instruments, policy frameworks and institutional structures and processes. The Constitution of Kenya (2010)[6] recognizes the primacy of addressing inequality and discrimination as prerequisites for nurturing a progressive and stable society. The Bill of Rights in the Constitution and affirmative action provisions are enacted to remedy gender inequalities, among other forms of deprivation experienced by women, children and marginalized men. Equitable development in the country is the basis of devolved government where resources and opportunities are contextualized within counties as stipulated in the Constitution. Devolved government mechanisms are therefore expected to engage with and remedy barriers that generate inequality, including those that are gender based. The mandate of institutions like the National Gender and Equality Commission (NGEC)[7] is to ensure national and county development processes are responsive to alleviation of inequality. Yet, despite the robust normative frameworks inequality between men and women and boys and girls persists across all facets of development in the country (Table 2).

[6] Republic of Kenya. National Council for Law Reporting (NCLR). Constitution of Kenya 2010. Nairobi: NCLR, 2010.

[7] Republic of Kenya. National Council for Law Reporting (NCLR). National Gender and Equality Commission Act 2011. Nairobi: NCLR, 2011.

Table 2: Kenya Ten Point Fact Sheet

Subject	Performance	Source
Population	Male19,192,458; Female19,417,639	KNBS, 2016
Development Index	44%	NGEC, 2016
Health Index	67%	NGEC, 2016
Education Index	61%	NGEC, 2016
Social Justice and Civic Participation Index	71%	NGEC, 2016
Labour Index	64%	NGEC, 2016
Gender Equality Index	38%	NGEC, 2016
Link between Equality and Inclusion Index	59%	NGEC, 2016
HIV Prevalence	5.6%	NACC, 2016
Sexual Violence	Male, 3.8%; Female,13.3%	KDHS, 2014

1.4 Gender and Water Resource Intersection

Water is a critical factor of production that affects access to livelihoods for both men and women. Access to water underlies women's and men's participation in reproductive (water for domestic use) and productive work, including irrigation, fisheries, livestock and industrial use. UNICEF has reported that despite 66% of the urban population in Kenya has access to drinking water that is free from contamination in 2015 compared to 70% in 2000. This can be a severe constraint to women and girls who are majority of those involved in collection of water for domestic use. A meta-analysis of household surveys across 45 developing countries found that 72% of daily household water-related tasks were done by women and girls. Improved water provision and a more equal distribution of unpaid work between women and men would mean women have more opportunity to devote time to other aspects of life, such as livelihoods and education.[8] Extensive and balanced involvement of women, men, transgender, and intersex people in the use, enjoyment and valuation of ecosystem services is necessary to create just, legitimate and effective policies and institutional arrangements. Interventions for water environmental services.[9] Feminist political ecology (FPE), a growing field of political ecology treats gender as a critical variable in shaping resource access and control, interacting with class, caste, race, culture, and ethnicity to shape processes of ecological change (Cole, 2017).

1.5 Mainstreaming Gender in Global Climate Fund Programmes

The GCF's commitment to gender equality centers on gender-responsive country programmes and initiatives that benefit everyone, women and men. This requires gender analysis to understand the social, economic and political factors underlying climate change-exacerbated gender inequalities, and the potential contributions of women and men to mitigating and adapting to climate change, and building climate resilience. It further entails adopting methods and tools to promote gender equality and reduce gender disparities in climate funding by mainstreaming gender in the project cycle. Finally, it means measuring the outcomes and impacts of project activities on women's and men's resilience to climate change through gender-responsive monitoring and evaluation (M&E). Gender mainstreaming should shape the entire project cycle, from project identification to M&E.

[8] Australian Water Partnership. (2016). Gender & SDG 6: The Critical Connection.
[9] Ibid., pg 3.

2.0 Baseline survey on building resilience in the ACA

2.1 Baseline had four steps

2.1.1 Step 1: Inception

At this phase, preliminary consultations were conducted with NEMA, KEITI staff and other stakeholders. The purpose of inception phase was to discuss the proposed protocol, proposed methodology for undertaking the study-including data collection tools, work schedules as well as proposed support (to the research team). Following these initial consultation meetings the consultants submitted the proposed methodology as informed by the inception meeting, data collection tools, processes and analytical tools as well as a comprehensive list of key stakeholders and informants to be met during the course of the baseline survey.

2.1.2 Step 2: Desk study

Key project documents reviewed as part of the evaluation were:
The Constitution of Kenya 2010, the Beijing Platform of Action 1995, CARE International. Women's Empowerment SII Framework at
http://pqdl.care.org/sii/Pages/Women%27s%20Empowerment%20SII%20Framework.aspx and other documents that are used in connection with the strategic focus areas of NEMA.

Step 3: Field work

The gender analysis was cross-sectional, using mostly qualitative methods. Emphasis was placed on making the process participatory through focus group discussions, key informant interviews, observation and workshops in the four counties. The consultants used purposive sampling method. This sampling methodology was used in order to have a statistically acceptable and defensible distribution of respondents as a representation of the entire focus population within the project areas.

2.1.3.1 Focus group discussions

Focus group discussions (FGD) were used to obtain views of groups of stakeholders. In all the target areas visited, FGDs was conducted with a *minimum* of 8 participants. The participants were purposively selected to get the views of water sector stakeholders and the community (men, women and community leaders) in the target areas.

2.1.3.2 Key informant interviews

Detailed interviews using semi-structured checklists were conducted with key informants. The key informants included the following:

- Chief water and irrigation Officer (Machakos)
- Director of Gender and Culture (Kiambu)
- Deputy Director water Resources (Kiambu)
- Deputy Director Environment (Nyandarua)
- Gender Officer (Nyandarua)

2.1.3 Step 4: Data analysis

Qualitative data was analysed it involved coding into various categories in line with the focus area of enhancing the resilience of communities and ecosystems in the Athi river catchment area project

2.2. The Athi Catchment Area

ACA is located in the southern part of the country as shown in Figure 1.3.1 and borders on the Tana Catchment Area in the north, the Indian Ocean in the east, Tanzania in the south and the Rift Valley Catchment Area in the west. The Aberdare Range, one of the Five Water Towers, lies in the northern edge of the area. Total area of ACA is 58,639 km^2, corresponding to 10.2% of the country's total land area. Based on the Census 2009, population of the area in 2010 is estimated at 9.79 million, or 25.4% of the total population of Kenya. Population density is 167 persons/ km^2.

The topography of ACA varies, from the highland in the Aberdare Range at around 2,600 m amsl to the coastal area at sea level. ACA is divided into three zones, with the upper zone at 2600-1500 m amsl, middle zone at 1,500-500 m amsl, and coastal zone at 500-0 m amsl.

The Athi River flows from the southeast of Nairobi north-eastward in the upstream reaches and then turns to the southeast in the north of Ol Doinyo Sapuk National Park and flows along the catchment area boundary with Tana Catchment Area and pours into the Indian Ocean in the north of Malindi. The drainage area of the Athi River is 37,750 km^2, or 64.4% of the Athi Catchment Area. The Lumi River, Lake Jipe, and Lake Chala flow into Tanzania and the Umba River flows from Tanzania to Kenya. Other rivers such as the Rare, Mwachi, Pemba, and Ramisi rivers flow into the Indian Ocean and the total drainage area is 19, 493 km^2. There are several springs in ACA such as Mzima, Kikuyu, Njoro Kbwa, Nol Turesh springs, etc.

ACA is classified as a semi-arid land except the upstream area of the Athi River which is classified as non-ASAL. The mean annual rainfall ranges between 600 mm in the central part of the area to 1,200 mm in the upstream area of the Athi River. The catchment area average mean annual rainfall is 810 mm. The renewable water resources, which is defined as precipitation minus evapotranspiration, is estimated at 4.54 BCM/year in 2010 for ACA and the per capita renewable water resources is calculated at 464 m3/year/capita.

Source: Source: Ministry of Environment, Water and Natural Resources, 2013

2.3 Enhancing the resilience of communities and ecosystems in the Athi River Catchment Area

The National Environment Management Authority (NEMA), Kenya's nationally accredited entity to the Green Climate Fund (GCF), in collaboration with the Korea Environmental Industry and Technology Institute (KEITI), is developing a $ 10 million project titled '*Enhancing the resilience of communities and ecosystems in the Athi River Catchment Area*' for submission to the GCF.

NEMA submitted a draft proposal of the project to the GCF Secretariat in 2016 to acquire initial feedback and received a list of additional tasks that need to be carried out further to completion of the comprehensive proposal whose objectives were to increase water security through Integrated Water Resource Management (IWRM) and to enhance the health and well-being of the vulnerable populations within the Athi River Catchment Area (ARCA).

The ARCA is classified as semi-arid land and suffers from frequent drought and flooding events exacerbated by climate change. During the long rainy seasons, extreme heavy rains give rise to bursting of the river banks especially in the upper and lower catchments. The floods experienced in March, April and May 2018 lend credence to the climatic and environmental fragilities of the catchment areas. Apart from roads becoming impassable, thousands were displaced, farmlands destroyed and several people died during the floods.[10]

The project is therefore aimed at addressing environmental and water related challenges of the ARCA through modernized monitoring and utilization of water, and enhancing actions for sustainable livelihoods. Specifically, the gender analysis is designed to identify and propose remedies to development gaps that manifest gender based inequalities and forestall possible outcomes of the project that could exacerbate existing, or generate new forms of disparities in the way the project impacts on men, women and other vulnerable populations within the catchment areas. The focus on gender and inclusion demonstrates compliance with national statutes[11] on participatory development and ensures adherence to human rights concerns in the design and delivery of initiatives. Recognition that environment and gender[12] are sacredly intertwined is the fundamental principle that underpinned the analysis whose objective was to examine gender related socio economic dynamics and prepare a Gender Action Plan.

[10] Kenya Red Cross Reports
[11] The Constitution of Kenya 2010
[12] The Beijing Platform of Action 1995

3.0 Methodology

3.1 Gender Analysis Approach, Framework and Domains

The analysis adopted a gender and development approach whose main focus is the unequal relations of power between men and women that prevent equitable development and women's full participation. Gender and development strategies and projects seek to empower disadvantaged women and men to achieve more equitable relationships by addressing their practical needs and strategic interests. The analysis invoked the Women's Empowerment Framework[13] whose components are agency, structures and relations. This particular framework was useful in elevating the analysis to respond to the global mantra[14] of promoting gender equality and empowering women. The gender analysis domains that guided the study were a**ccess to resources; knowledge, beliefs and perceptions; practices and participation; balance of power and decision making; legal rights and status; and time and space.**

3.2 Study Design

The gender analysis was cross-sectional, using mostly qualitative methods. Emphasis was placed on making the process participatory through focus group discussions, key informant interviews, observation and workshops in Nyandarua, Kiambu, Machakos and Nairobi counties. Respondents included community participants, service delivery officers and policy makers in the water and environment management sectors, and gender mainstreaming oversight institutions. Qualitative data was majorly collected through extensive desk reviews.

[13] CARE International. Women's Empowerment SII Framework at http://pqdl.care.org/sii/Pages/Women%27s%20Empowerment%20SII%20Framework.aspx

[14] Inference to Sustainable Development Goal Number 5 at

3.3 Key Research Questions

The analysis responded to the following key research questions that were investigated through literature reviews and participatory methodologies Table 3.

Table 3: Research Questions

Broad Gender Analyis Questions	Specific Gender Analysis Questions
What is the context?	✓ What demographic data disaggregated by sex and income, including the percentage of women-headed households, are available?
	✓ What are the main sources of livelihoods and income for women and men?
	✓ What are the needs and priorities in the specific sector(s) to be addressed by the planned intervention? Are men's and women's needs and priorities different?
	✓ What impacts are men and women experiencing due to specific climate risks?
	✓ What is the legal status of women?
	✓ What are common beliefs, values, stereotypes related to gender?
Who has what?	✓ What are the levels of income and wages for women and men?
	✓ What are the levels of educational attainment for girls and boys?
	✓ What is the land tenure and resource use situation? Who controls access to or owns the land? Do women have rights to land, and other productive resources and assets?
	✓ What are the main areas of household spending?
	✓ Do men and women have bank accounts? Have they received loans?
	✓ Do men and women have mobile phones, access to radio, newspapers, TV?
	✓ Do women and men have access to extension services, training programmes, etc.?
Who does what?	✓ What is the division of labour between men and women, young and old, including in the specific sector(s) of intervention?
	✓ How do men and women participate in the formal and informal economy?
	✓ Who manages the household and takes care of children and/or the elderly?
	✓ How much time is spent on domestic and care work tasks?
	✓ What crops do men and women cultivate?
Who decides?	✓ Who controls/manages/makes decisions about household resources, assets and finances? Do women have a share in household decision-making?
	✓ How are men/women involved in community decision-making? In the broader political sphere?
	✓ Do men/women belong to cooperatives or other sorts of economic, political or social organizations?
Who benefits	✓ Will the services/products of the proposed interventions be accessible to and benefit men and women?
	✓ Will the proposed interventions increase the incomes of men/women?
	✓ Will the proposed interventions cause an increase/decrease in women's (and men's) workloads?
	✓ Are there provisions to support women's productive and reproductive tasks, including unpaid domestic and care work?

3.4 Organization of Gender Analysis and Quality Assurance

Workshops in Nyandarua, Kiambu, Machakos and Nairobi counties were used as entry points for data collection and identification of sites for project interventions. This was followed by community meetings with men, women, leaders and key people deemed to be having specific expertise. The data collection was undertaken by gender and community development specialist assisted by research assistants drawn from similar fields. The senior researchers and research assistants underwent comprehensive training on the gender analysis broadly and specifically on the research tools that entailed comprehensive peer review. The qualitative data obtained was transcribed and subjected to qualitative analysis strategies and techniques. The data was largely analysed using a comprehensive thematic matrix that facilitated identification of common patterns and trends arising from the narratives. Content analysis of the data was undertaken in order for inferences, judgments and conclusions to be as accurate as possible.

3.5 Compliance with Research Ethics

Participation in the gender analysis was voluntary and based on informed consent. It entailed providing study participants with information on the assessment and its approach, their role in the gender analysis, benefits of their participation (both directly and indirectly) and finally obtaining consent from each respondent willing to participate in the assessment. The participant's right to anonymity and confidentiality was given due attention. The interaction between the research team and the participants as well as among the study participants themselves was based on mutual respect and trust.

3.6 Research Constraints and Limitation

The main research constraints of the study were lack of data on gender issues at the county level attributed to the formative nature of devolved units in the country. National data was thus used as proxy data to inform the situation in counties. Most counties do not have dedicated gender officers which contributed to inadequate capacity to understand and articulate development issues through gender lens. The main limitation of the study is the contextual nature of qualitative data which may not be generalized across other different development settings.

3.7 Structure of the Gender Analysis Report

This report is organized in six main areas as follows

Chapter One: Introduction – This section contextualises development progress in Kenya and goes further to highlight the interface between water and gender issues.

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4.0 Findings

4.1 Kenya's Gender Profile

4.1.1 Population

Kenya sustained tremendous population growth in the last five decades after independence. During the first post-independence census in 1969 the country had a population of 10.9 million people compared to 38.6 million people enumerated during the 2009 Population and Housing Census, representing a 35% increase from the 1999 census figures. The 2009 census revealed marginal although demographically significant differences between males (19,192,458) and females (19,417,639). A youth bulge is another feature of the Kenyan population which promises demographic dividends in 2030. Most Kenyans (68%) live in rural areas as confirmed by the 2009 censuses.

Table 4: Rural and Urban Population Sizes

Residence	Male	Female	Total
Rural	12,913,647	13,209,075	26,122,722
Urban	6,278,811	6,208,564	12,487,375
Total	19,192,458	19,417,639	38,610,097

Source: Population and Housing Census 2009

4.1.2 Health

Maternal mortality ratio in Kenya is estimated at 362 maternal deaths per 100,000 live births (KDHS whereas the infant mortality rate is 39 deaths per 1,000 live births, and under-5 mortality is 60/1,000 for boys and 52/1000 for girls.[15] The latest Kenya AIDS Indicator Survey (KAIS) data on HIV prevalence in Kenya indicated that women and girls still bear the disease burden.[16] Accordingly, prevalence among females aged 15 to 64 years is 6.9% which is higher than the national average of 5.6% whereas prevalence among males within the same age category is 4.4%.[17] The KAIS report revealed that 21% of new HIV infections in Kenya occur among girls and young women aged 15-24 years.[18]

The prevalence of gender-based violence (GBV) in Kenya has been documented during stability and humanitarian emergencies. The National Crime Research Centre (NCRC) undertook a study on GBV across 13 counties in 2014 and established rape as being a prevalent violation in both rural and urban settings.[19] The Commission of Inquiry into Post-Election Violence (CIPEV) reported numerous cases of gender based violence during the PEV period in 2007-08.[20] Female genital mutilation / cutting (FGM/C) is declining slowly over time.

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[15] Republic of Kenya. Kenya National Bureau of Statistics (KNBS). Kenya Demographic and Health Survey 2014: Key Indicators. Nairobi: KNBS, April 2014.

[16] National AIDS and STI Control Programme (NASCOP), Kenya. Kenya AIDS Indicator Survey 2012: Final Report. Nairobi, NASCOP. June 2014.

[17] Ibid.

[18] Ibid.

[19] Republic of Kenya. National Crime Research Centre (NCRC). Gender Based Violence in Kenya. Nairobi: NCRC, 2014.

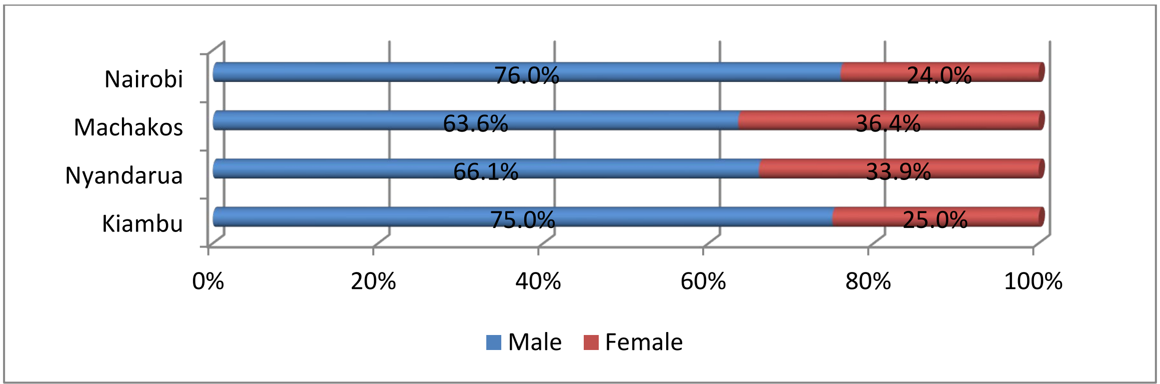
[20] Commission of Inquiry into Post Election Violence in Kenya (2008). Commission of Inquiry of Inquiry into Post Election Violence Report available at http://kenyastockholm.files.wordpress.com/2008/10/the-waki-report.pdf

The 1998 KDHS reported 38 % of women in Kenya having been circumcised, which declined to 32 % in 2003, 27 % in 2008-09, and 21 % in 2014.[21] This can be attributed to innovative and robust awareness creation programmes and strategies that target structural factors such as the alternative rites of passage initiatives and transformative dialogue with community gate keepers. Enactment and gradual strengthening in the enforcement of the Prohibition against Female Genital Mutilation Act (2011)[22] have also triggered the impetus against the practice.

4.1.4 Household Headship

Most households in Kenya are male-headed (67.6% male and 32.4% female) with an average size of four members.[23] Women constitute lower proportions of household headship in both in both rural and urban areas at 35.9% and 27.8%, respectively.[24] Majority of household heads (52.4%) fall within the 25-44 age bracket. The highest proportion of 14.7% of household heads was in the 30-34 years age group. Nationally, households headed by persons aged 65 years and above accounted for 11.6% of households.[25] Household headship in Kiambu, Nyandarua, Nairobi and Machakos counties is shown in Figure 1 which reveals a patriarchal pattern that is highest in the more cosmopolitan counties of Nairobi and Kiambu.

Figure 1: Household Headship



Nairobi leads in the male household headship and contrary bottoms in female side. It is followed by Kiambu, Nyandarua then Machakos. However, Machakos was found oo be the county with the highest number of female headed households, followed by Nyandarua, Kiambu and Nairobi consectively. The women are burdened with triple roles as they multitask on both domestic, communal and productive roles in order to get an income for sustaining their families. Some of the female household heads have difficulties in creating ample time away

[21] Republic of Kenya. Kenya National Bureau of Statistics (KNBS). Kenya Demographic and Health Survey 2014: Key Indicators. Nairobi: KNBS, April 2014.

[22] Republic of Kenya. National Council for Law Reporting (NCLR).Prohibition Against Female Genital Mutilation Act 2011. Nairobi: NCLR, 2011.

[23] Ibid.

[24] Ibid.

[25] Ibid.

from home duties to participate in forums regarding water resources management including right to access and usage.

Source: Kenya National Bureau of Statistics 2018

4.1.6 National Priorities

Kenya's development blue print is Vision 2030 which is premised on transformation of the country into a prosperous middle income nation. The Third Phase of the national vision is contextualized within sustainable development goals, zeroing in on universal health coverage, access to affordable and decent housing, growth in manufacturing and expansion in agriculture for subsistence and commercial income as the main priorities dubbed the "Big 4 Agenda."

4.2 Access to Resources

4.2.1 Educational Attainment for Men and Women

Access to both basic and tertiary education in Kenya has grown exponentially in the last 15 years since declaration of free primary education (FPE) in 2003 and free day secondary education (FDSE) in subsequent years. Policy reforms have also contributed to expansion in access to tertiary and university education in Kenya especially after promulgation of a new constitution in 2010 that re-emphasized the primacy of education as a human right.[26]

Nationally, the gross attendance ratio (GAR)[27] for pre-primary school is 94.4% (95.4% for males and 93.5% for females) while for primary school it rises to 107.2% (109.0% for males and 105.4% for females).[28] The same trend of higher school attendance for boys than girls is maintained at secondary school level with a GAR of 66.2% (67.2% for males and 65.2% for females).[29]

Table 5: Gross Attendance Ratio (%)

County	Pre-Primary			Primary			Secondary		
	Male	Female	Total	Male	Female	Total	Male	Female	Total
Nyandarua	110.2	81.4	**95.4**	107.8	112.6	110.1	81.7	81.1	81.4
Kiambu	106.2	74.8	92.9	105.4	113.5	109.2	76.2	74.5	75.3
Nairobi	74.5	86.4	80.8	100.5	102.2	101.3	81.3	94.0	**88.4**
Machakos	67.0	76.4	71.0	127.4	116.0	**121.1**	74.3	88.0	81.1

Source: Kenya National Bureau of Statistics 2018

[26] The Universities Act (2012), Technical and Vocational Training Act (2013), the Basic Education Act (2013) and the National Education Sector Plan (2014) are some of the policy milestones in the education sector that have triggered exponential growth in access despite existing and credible reservations about quality.

[27] Gross Attendance Ratio (GAR) is the total number of persons attending school regardless of their age, expressed as a percentage of the official school age population for a specific level of education.

[28] Kenya National Bureau of Statistics (KNBS). Basic Report Based on 2015/16 Kenya Integrated Household Budget Survey. Nairobi: KNBS, 2018.

[29] Ibid.

Nyandarua (95.4%) and Kiambu (92.9%) counties have the highest aggregate GAR for the pre-primary level although they also manifest the most glaring differences between boys and girls (Table 6). Machakos County reported the lowest GAR at pre-primary for both boys (67.0%) and girls (76.4%) but drastically improves to post the highest aggregate GAR at primary (121.1%). Nairobi County had a consistently higher GAR for girls at pre-primary (86.4%), primary (102.2%) and secondary (94.0%) than for boys. Overall, female pupils have the highest GAR at primary across all the four counties which is critical for significant transition to secondary.

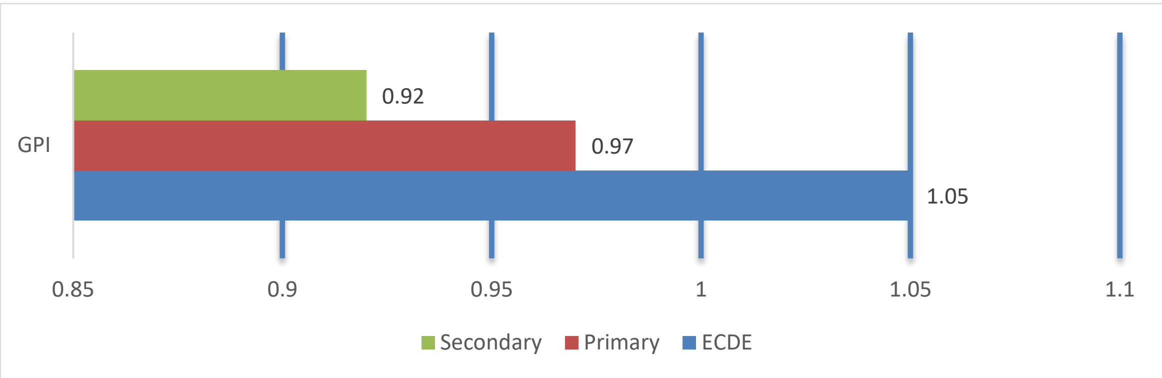
Table 6: Net Attendance Ratio (%)

County	Pre-Primary			Primary			Secondary		
	Male	Female	Total	Male	Female	Total	Male	Female	Total
Nyandarua	73.9	67.6	70.6	88.4	91.3	89.8	46.6	52.0	49.1
Kiambu	79.2	63.5	72.5	88.6	94.4	91.3	62.5	58.7	60.5
Nairobi	68.6	78.8	74.0	90.1	91.1	90.6	62.4	67.0	65.0
Machakos	58.6	64.9	61.3	95.3	95.3	95.3	41.3	58.2	49.6

Source: Kenya National Bureau of Statistics 2018

The Net Attendance Ratio[30] (NAR) shows better access to education by girls than boys across pre-primary, primary and secondary levels. The NAR for girls at primary school was consistently higher than for boys in all counties except for Machakos County where they were at par in primary school (95.3% each) and Kiambu County where boys had a higher NAR at secondary school (62.5%). Nairobi County had the highest aggregate NAR for pre-primary school (74.0%) and secondary school (65.0%) compared to the other three counties while Machakos County had the highest NAR for primary school (95.3%).

Figure 2: Gender Parity Index (GPI) in Kenya



Source: Ministry of Education, Science and Technology, 2014

The national Gender Parity Index (GPI) for pre-primary schools (1.05), primary (0.97) and secondary (0.92) is skewed in favour of boys.[31] Structural factors such as child marriage, female genital mutilation, household poverty and discrimination against girls and women expose them to low participation in education. Disability and geographical drivers of inequality in access and participation are legitimate dimensions of delivering on the education commitment in Kenya.

[30] Net Attendance Ratio (NAR) is defined as the total number of persons in the official school age group attending a specific level to the total population in that age group.

[31] Republic of Kenya. Ministry of Education, Science and Technology (MoEST). Basic Education Statistical Booklet. Nairobi: UNICEF, 2014.

Table 7: Tertiary and University Education

County	Mid-Level Colleges		University	
	Male	**Female**	**Male**	**Female**
Machakos	3.1%	4.4%	5.6%	1.8%
Nyandarua	4.1%	4.0%	1.2%	0.8%
Kiambu	7.2%	9.8%	6.3%	5.4%
Nairobi	14.9%	13.5%	10.8%	8.3%

Source: Kenya National Bureau of Statistics 2018

Almost equal number of males (5.9%) and females (5.6%) attend mid-level colleges in Kenya but the gap widens at university level where there are more male students (3.7%) than females (2.6%).[32] The number of men attending university education in Machakos County is three times (5.6%) higher than that of women (1.8%) although there are more women (4.4%) than men (3.1%) in mid-level colleges. Nyandarua County showed slight differences between men and women attending mid-level colleges (4.1% men and 4.0% women and university (1.2% men and 0.8% women). Kiambu County had the second highest proportions of women and men attending tertiary and university education as shown in Table 8. Nationally, the proportion of literate population is 84.8 per cent with a higher proportion in urban areas (93.2%) compared to their rural counterparts (78.8%). More men (89.0%) than women (80.2%) in Kenya are considered literate.[33] Overall, the national education inequality index is 61%.[34]

4.2.2 Main Sources of Livelihoods and Income for Women and Men

Key Findings

- ✓ Women recognize water as the main source of livelihood and income.
- ✓ Men recognize land as the main source of livelihood and income.

The main productive resources and social amenities that anchor livelihoods in Nyandarua, Kiambu and Machakos counties water, hospitals, schools, roads and buildings, according to women. Accordingly water is important for both domestic consumption and agricultural activities that rely on irrigation. On their part, men in the three counties recognized land as a critical source of livelihood followed by water and schools. This position was similarly expressed by leaders who ranked land as the foremost source of livelihood and income followed by water. It is therefore water that cuts across all cadres of respondents as an important source of livelihood although more preferred by women just as men would lean more towards land (Table 9).

[32] Kenya National Bureau of Statistics (KNBS). Basic Report Based on 2015/16 Kenya Integrated Household Budget Survey. Nairobi: KNBS, 2018.

[33] Kenya National Bureau of Statistics (KNBS). Basic Report Based on 2015/16 Kenya Integrated Household Budget Survey. Nairobi: KNBS, 2018.

[34] National Gender and Equality Commission (NGEC). Status of Equality and Inclusion in Kenya. Nairobi: NGEC, 2016.

Table 8: Sources of Livelihoods and Income

County	Respondent	Identification and Ranking Sources of Livelihood and Income				
		1	2	3	4	5
Nyandarua	Women	Water	Roads	Buildings	Crops	Skills
	Men	Land	Water	AGR	Forests	Infrastructure
	Leaders	N/A	N/A	N/A	N/A	N/A
Kiambu	Women	Hospital	Schools	Water	Infrastructure	Land
	Men	Land	Water	Infrastructure	Hospital	Education
	Leaders	Water	Land	Schools	Hospital	Energy
Machakos	Women	Water	Health	Education	Roads	Land
	Men	Land	Schools	Hospital	Water	Road
	Leaders	Water	Land	People	Hospitals	Schools

Key informant interview sessions generated a wide range of livelihood and income sources associated with women in Nyandarua County. These included subsistence farming, group savings and loans associations (chamas), dairy and poultry keeping, running grocery businesses, retail shops, and provision of mobile money services (Mpesa). Women also work in the hotel industry, banks and micro-finance institutions. Soap making, poultry keeping, horticulture, table banking and merry-go-rounds, working as labourers and crushing ballast for sale were cite as sources of income for women in Machakos County. They also worked for the county government, banks, SACCOs and MFIs. Women in Kiambu participate in formal employment, small and micro-enterprises, agricultural activities (green houses, dairy farming, pig keeping, crop farming (maize, beans, sweet potatoes, and arrowroots), and poultry keeping, according to key informants. Various livelihood and income generation activities that men in Nyandarua, Kiambu and Machakos engage in are presented in Table 9.

Table 9: Livelihood and Income Activities for Men

County	Livelihood and Income Activities
Nyandarua	✓ Dairy keeping ✓ Crop farming peas, irish potatoes, carrots and maize) ✓ Men's saving groups (buy and sell property) ✓ Boda boda, retail shops (electronic equipment), Mpesa shops. ✓ Work for county government, banks, SACCOs and micro-finance institutions.
Machakos	✓ Sand harvesting and general quarry business ✓ Boda boda ✓ Mpesa outlets and retail shops (electronics) ✓ County government employment ✓ Agro-forest nurseries ✓ Sell of coffee (coffee is a man's crop) ✓ Horticulture
Kiambu	✓ Real estate business ✓ Dairy farming ✓ Informal employment commonly referred to as casual jobs ✓ Cash crop farming (tea, coffee, pineapple, French beans, avocados, macadamia, bananas) ✓ Boda boda and Mpesa and retail shops (electronics) ✓ Work for county government, banks, SACCOs and micro-finance institutions

4.2.3 Land as a Productive Resource

Land as a productive resource was consistently highly ranked by men and community leaders in all the three counties. The men explained the primacy of land as residing in its high value when sold and as a factor of production. Women in Kiambu reported that land is valuable because of farming but still ranked it as the last item among their sources of income and livelihood. They observed that both men and women have access to land but control (title deed and selling family land) was solely by men. Contrary views were stated by men in the county who asserted that women had access to land, and both men and women enjoyed equal levels of control and benefit. Low ranking of land as a source of livelihood by women in Machakos was attributed to the small sizes of the parcels that are owned which they have to be shared with their sons. They reported that women and men had access to land but only men controlled and benefitted from it. Their male counterparts intimated that women and men had access to and benefitted from land but only men controlled it. Women in Nyandarua had access to land which they also owned although they said the parcels are small (50X100 square metres). They ranked land as the last item after livestock though both did not feature among their top 5 sources of livelihoods and income. They acknowledged benefitting from the land which they said was controlled by men who held title deeds to the land. Men in the same county agreed with their female counterparts on access and benefit. They also argued that women controlled the land on the basis of provisions in the Constitution of Kenya (2010). Constraints and coping mechanisms associated with land were reported as shown in Table 10.

Table 10: Land Related Constraints and Coping Mechanisms

County	Gender	Constraint	Coping Mechanisms
Nyandarua	Women	• Small in size due to dense population	• Practice mixed farming, allocating each crop a small portion.
	Men	• Women have no control over land since men are the owners of title deeds.	• Be loyal to men to use the land and maximize the opportunity they get to use the land.
Kiambu	Women	• Small size of land parcels	• Mainly grow crops for subsistence farming
	Men	• Small in size, sub-division for children • Garbage disposal at the residential homes (sanitation)	• Get used to it and survive
Machakos	Women	• It is small and unproductive	• Accept and move on
	Men	• Rains are scarce and unreliable	• Planting in the long rain seasons (October).

Leaders in Kiambu intimated that women are culturally barred from owning land but as a coping initiative women are buying land for individual ownership. They alluded to provisions in the Constitution that allow women and girls to inherit land from their parents and spouses. While acknowledging that women are not allowed to own land, leaders in Machakos County reported that land buying and having to endure living with mean spouses were coping strategies used by women. They vouched for sensitization of women on matrimonial property laws and the Constitution as a way of addressing this manifestation of discrimination and inequality.

Land acquisition in Nyandarua County is largely through inheritance (succession) which disadvantages women because of its patriarchal bias for men and boys, according to community leaders. This has led to women losing control of farm produce (carrots, cabbage and potatoes) despite providing much of the farm labour. The leaders reported that women had resorted to buying their own land and drifted to roadside markets (Soko Mjinga) to eke out a living through petty trade. They expressed concern that the roadside business has led to dysfunctional families as women gain economic traction and withdraw from relationships with men who cannot provide for the family adequately. Participation in table banking activities ad embracing family planning were also cited as women's copying mechanisms. The leaders underscored expose to diseases such as HIV, malnutrition, drug abuse, and sexual violence (rape) as common repercussion of women lacking access and control over land in the county.

Public officers in Kiambu and Machakos counties pointed out that freehold is the dominant for of land tenure which provided opportunities for credit acquisition and investment if used as collateral, construction of business premises (rental houses), or leasing out. This form of land tenure guaranteed full control of one's land and forestalled conflicts inherent in communal ownership practices, according to the public officers. Some of the drawbacks that they associated with the freehold tenure had to do with men unilaterally deciding to dispose off the land without seeking consent from their spouses. Leasehold was also cited as a form of land tenure system in Machakos County. The public officers concurred that most women in the county do not own land while those who are married can only access it but not exercise any form of control. To alleviate this, the officers proposed empowerment of women through financial products to acquire and own land alongside enforcing legal frameworks that recognize women's land rights.[35]

Freehold tenure was similarly practiced in Nyandarua, according to expert views. This form of land ownership created opportunity for women in the county to own land through purchase, decide what to plant, have control over the farm produce, and access credit from banks and SACCOs using the land as collateral. Public officers in the county observed that most women do not own land but a trend was emerging whereby women are increasingly buying land and having title deeds in their name. Generally, female landholders in the county have less land than male landholders. Even when women own land, their husbands are still perceived as de facto owners by virtue of being household heads.

35

4.2.3.1 Ownership and Control of Other Productive Resources and Assets

Water is a crucial resource for development across all counties encompassed in the gender analysis. Women respondents in Nyandarua indicated that they had access, controlled and benefitted from water since they are the ones who fetch and decide on its use. They stated that men did not access, control or benefit from water. Similarly, women in the county said they accessed, controlled and benefitted from crops unlike their male counterparts who did not have access to and control of crops but benefitted. Notably livestock was accessed and controlled by men without the involvement of women. The men controlled livestock since they owned the land and could decide when to dispose of livestock and how proceeds would be expended. On the contrary, men in the county asserted that both men and women accessed, controlled and benefitted from water. They agreed with the women on control of livestock as a socially sanctioned preserve of men but expressed divergent sentiments on access and benefit which they associated with both men and women. The men further asserted that they accessed and controlled machinery within the homestead but both women and men benefitted from their use.

Female community respondents in Kiambu County explained that water, schools, hospitals and infrastructure as sources of livelihood are accessed, controlled and benefit men and women equally. The women stated that water is a basic need for all and schools enrolled boys and girls without discrimination as a public policy. They further underscored that the poor state of roads especially during the rainy season and shortage of drugs in public hospitals affected both men and women. Male community respondents in the same county stated that women accessed and benefitted from water but it was the men who controlled it. They cited services such as road infrastructure, schools and security (police) as being equally accessed, controlled and beneficial to men and women. Healthcare was entirely a responsibility of women, according to male respondents.

Female community respondents in Machakos County presented the following views (Table 13) on access and control of resources. They also indicated who benefits when resources are appropriated.

Table 11: Access to Resources by Men and Women in Machakos County

Resource	Access to		Control of		Benefit	
	Men	Women	Men	Women	Men	Women
Land	✓	✓	✓	X	✓	x
Water	x	✓	X	✓	X	✓
Livestock	✓	✓	✓	X	✓	x
Crops	x	✓	✓	X	✓	x
Roads	✓	✓	✓	✓	X	✓

On their part, male community respondents in the same county reported that women accessed and controlled water as resource but both men and women benefitted. They further indicated that men and women accessed and benefitted from public amenities such as schools, hospitals, electricity and infrastructure (roads) although they were controlled by men.

The leading sources of drinking water for residents of Nairobi County are water being piped into plot / yard (33.1%), into dwelling (32.2%), and into pubic tap/ stand tap (25.9%). The largest proportion of households in Nairobi County (53.3%) consumes between 1001-2000 litres of water followed by 23.1% of households who consume between 0-1000 litres followed by 15.8% of households who consume between 2001-3000 litres. Majority of households in Nairobi County (50.9%) take less than 30 minutes to access the source of drinking water while 46.5% take neglible time (zero) to fetch water because the source is located within premises.

Constraints and coping strategies associated with access to water in Machakos, Kiambu and Nyandarua counties are presented in Table 12.

Table 12: Access to Water Constraints and Coping Strategies

County	Sex of Respondents	Constraint	Coping Strategy
Machakos	Women	➲ Distance to water source ➲ Contaminated water ➲ High cost at KShs. per 20 litres	➲ Ration use in the house hold. ➲ Fetch from wells
	Men	➲ Water scarcity	➲ Water management strategies such as roof harvesting. ➲ Boreholes also provide water which is sold at 5/= per 20 litre Jerican
	Leaders (male and female)	➲ Long distance to water source ➲ Insecurity due to long distance	➲ Dig wells. ➲ Buy water tanks (some)
Kiambu	Women	➲ Floods in the region when it rains heavily, leading to shortage of water due to blocked pipes	➲ Store water in tanks
	Men	➲ Water on the roads when it rains	➲ Get used to it.
	Leaders	➲ Long distances ➲ Water shortage	➲ Rationing and buying water
Nyandarua	Women	➲ Dirty / contaminated water ➲ Lack of tap water	➲ Use rain water for drinking ➲ Buy drinking water from shops ➲ Use water from wells and boreholes
	Men	➲ Accessing water points which are very far from residential areas.	➲ Transport water using donkeys that are not common.
	Leaders	➲ Poor quality water and low quantity ➲ Infiltration and contamination of wells / boreholes Flooding occurs ➲ Lack of water for irrigation	➲ Water harvesting ➲ Installed water tanks ➲ Drilling bore holes by Government. ➲ Stopped burying to go to cemetery ➲ Dependence on rain fed agriculture.

4.2.4 Crops Grown by Women and Men

Women in Kiambu County listed maize, beans, potatoes and kales as food crops that are widely grown mostly by women since men played a nominal role in farming. They stated that men only participate in farming as hired labour and therefore there is no food crop that is associated with men.

For instance all activities associated with maize production (land preparation, purchase of inputs, selection of seeds, planting, weeding, insecticide control, harvesting and storage) are done by women except marketing where men come in as middle men (Table 13). The women also receive and bank proceeds from the sell of maize.

Table 13: Participation in Maize Farming by Men and Women in Kiambu

Stage	Only Men	Only Women	Men and Women	Only Girls	Only Boys	Men and Boys	Boys and Girls	Girls and Women	Hired labour	N/A
Land preparation		√					√			
Purchase of input		√								
Selection of seeds		√								
Planting		√								
Weeding		√								
Insecticide control		√								
Harvesting		√								
Storage		√								
Marketing			√							
Receiving money		√								
Banking money		√								

While agreeing that maize, beans, kales and potatoes are the dominant food crops, men in Kiambu County stated that potatoes were considered as a woman's crop while men were associated with maize farming.

There was however neither restriction nor social stigma associated with women growing maize or men growing potatoes. The men said they participated in land preparation and purchase of inputs while women handled weeding and storage activities.
They reported that selection of seeds, planting, insecticide control and harvesting of maize is jointly done by men and women but marketing, receiving and banking payment from the sale of maize is the responsibility of men.

Maize, peas and beans define food crop farming in Machakos County as intimated by women community respondents. They said that all these crops are considered as a woman's crop because women are the only once involved in their production. The women stated that men only get involved during harvesting so that they can sell the produce and misappropriate income. Social norms that would determine crops for men and those that are associated with men are therefore none-existent. Men's participation in maize farming was said to be at the point of marketing, receiving and expending money (they don't bank proceeds). The women expressed fear of being subjected to violence should they ask about income from the sale of farm produce. To avoid this, women sell a little maize in the absence if the man and keep the money in discreet places in the house. "***We sell a few kilos like 5-20 kgs which we put it in our water containers on our way to the streams when the man is not around. Also, when going to the posho mill you carry more maize than is needed so that you can sell the extra***," the women concurred with each other.

"Once the man receives money and you ask about it, you end up sleeping out, or you are sent back to your parents due to the kind of beating you receive. They brag around the market place that they are great farmers," said a female community respondent in Machakos County.

Men in Machakos County listed cassava, arrow roots, sorghum, maize, beans, wheat and vegetables as crops that are widely grown. They said that women grow vegetables whereas men grow cassava. The men explained that women are not allowed to grow cassava lest the produce become bitter or is poor. They explained that women who grow crops perceived to be men's crops would be doing so with the consent and guidance of their spouses where as a man who grows a crop considered a women's crop is viewed as weak and controlled by the wife. The men asserted that both men and women would take part in land preparation and harvesting of maize. The tasks related to maize farming that are performed by men only included purchase of input, insecticide control, storage, and marketing, receiving and banking money, according to the male community respondents. They acknowledged that women had the sole responsibility of selecting of seed, planting and weeding maize.

Men in Nyandarua listed potatoes, beans, sukuma wiki, cabbages, tomatoes, apples, pyrethrum, maize, carrots and peas while women listed potatoes, cabbages, carrots, peas (Minji) and kales as food crops grown in the county. The men said peas and beans are considered as a woman's crop while cabbages and potatoes (grown for commercial purposes) are viewed as a man's crop. On their part, women in the county stated that peas and beans are grown by women because of small parcels of land while men grow potatoes, cabbage and carrots because of the high market prices associated with these crops. Both men and women in the county agreed that there is no restriction on crops to be grown by a particular sex. The main determining factor would be the size of the parcel of land that one has.

Table 14: Participation in Potato Farming by Men and Women in Nyandarua County

Stage	Only Men	Only Women	Men and Women	Only Girls	Only Boys	Men and Boys	Boys and Girls	Girls and Women	Hired labour	N/A
Land preparation			✓							
Purchase of input			✓							
Selection of seeds			✓							
Planting			✓							
Weeding			✓							
Insecticide control	✓									
Harvesting			✓							
Storage			✓							
Marketing			✓							
Receiving money			✓							
Banking money			✓							
Any other:										

The only aspect of potato farming that is solely undertaken by men is insecticide control. All other tasks, including marketing, receiving and banking money, are jointly done by men and women, according to male respondents. Women respondents reported that purchase of inputs, marketing, receiving and banking money accruing from cabbage farming is done by men while women carry out weeding and storage. They indicated that land preparation for cabbage farming, selection of seed, planting and insecticide control are done by both men and women. The women underscored their exclusion from proceeds generated through cabbage farming. ***"If you have a good husband he will inform you about the agreed prices though he is the one who receives money from brokers. Some of us don't have a clue about the selling prices of cabbage. We only rely on the little money that we are given for shopping," lamented a female community respondent.***

4.2.5 Access to Extension Services and Training

Men in Machakos County reported having access to and control of extension services and training unlike their female counterparts. Women in the same county expressed contrary views, stating that they had access to and control of extension services which men lacked. They also stated that both men and women accessed and controlled training services. Men in Kiambu County and Nyandarua said that both women and men had access to and control of extension and training services but women in Kiambu contradicted this position by asserting that whereas both men and women had access to extension services and training, only women exercised control. Women in Nyandarua stated that they had access to extension services and training but only men maintained control despite having no access to the same.

4.2.6 Access to Credit

Both women and men in the three counties have access to credit although women access credit through a wider variety of platforms than men. For instance men in Nyandarua accessed credit products through banks and SACCOs while women had numerous options such as women's chamas (table banking groups), Uwezo Fund, Women Enterprise Fund, Banks, MFIs, and SACCOs. Similarly, women in Machakos could access credit through MFIs (KWFT), SACCOs, churches, women's chamas, banks, and online credit outlets such as Tala and Mshwari while the men secured such financial products through banks, chamas, SACCOs (boda boad and sand harvesting associations) and work place cooperatives. It is however noticeable that unlike men, women in Kiambu County access credit through individual lenders (Shylocks). On the other hand, men can access credit through online outlets such as Tala, Branch and Mshwari which were not reported as providing similar services to women. Both men and women in Kiambu County stated that they access credit through banks, SACCOs, and self-help groups. The Women in Nyandarua pointed out that men in the county are disinterested in credit products while their counterparts in Machakos complained of harassment by some banks when they defaulted and unwillingness to refund savings by some MFIs. This could be the reason why only 18% of households in Nyandarua sought credit (Table 16).

Nationally, the most preferred sources of credit are merchants or shops (28.2%); self-help groups/chamas (19.4%); relatives, friends and neighbours (14.0%), and SACCOs at 11.2%.[36] Other significant sources of credit include commercial banks (8.8%), mobile phone platform (7.6%), and micro-finance institutions at 5.3%.[37] Employers (1.3%), Shylocks (1.2%), NGOs (1.2%) and Government funds (1.2%) are lowly rated as sources of credit.[38] Male-headed households (34.6%) compared to female headed households (31.9%) are majority of those who sought credit but slightly more female headed households (90.7%) than male headed households (89.8%)received credit.

[36] Kenya National Bureau of Statistics (KNBS). Basic Report Based on 2015/16 Kenya Integrated Household Budget Survey. Nairobi: KNBS, 2018.

[37] Ibid.

[38] Ibid.

Table 15: Access to Credit

County	Access to Credit	
	Proportion of households that sought credit (%)	Proportion of households that sought and accessed credit (%)
Nyandarua	18.2	88.9%
Kiambu	31.5	94.4%
Nairobi	33.9	89.2%
Machakos	38.5%	96.9%

Source: Kenya National Bureau of Statistics 2018

4.2.7 Access to Information and Technology

The survey noted that there was limited access to Climate Information by residents, and more so women yet they are the major partakers in search for water. Both men and women in Kiambu, Nyandarua and Machakos counties stated that they have access to and control of mobile phones. Nairobi County has the highest proportion of residents who have access to mobile phones (85.8%), followed by Kiambu County (83.7%) and Machakos (65.1%).[39] Nyandarua has the least proportion of the population that has access to mobile phones (61.5%) which brings to the fore disparities in access to ICT between urban and rural counties. Overall, radio (79.1%), mobile phone (62.8%), and television (47.8%) are the most prominent ICT equipment used for populations aged 3 years and above. Internet (16.6%) and computer (9.5%) are the less patronized ICT equipment. The leading consumers of ICT equipment and services fall within the 18-35 years age group; mobile phone (90.1%), radio (83.5%), television (60.5 %), internet (36.8%) and computer (18.4%).[40] It was also established that a high proportion of the population aged 70 years and above used radio (66.2%) and mobile phone (59.2%) while their use of computer and internet was negligible.[41] Table 16 presents use of ICT equipment in Kiambu, Nyandarua, Nairobi and Machakos counties.

Table 16 Access to Information and Technology

County	Use of ICT Equipment				
	Television	**Radio**	**Mobile Phone**	**Computer**	**Internet**
Machakos	33.4%	77.1%	65.1%	5.7%	14.8%
Nyandarua	26.0%	83.9%	61.5%	3.2%	11.4%
Kiambu	81.0%	91.1%	83.7%	14.4%	24.3%
Nairobi	88.0%	89.1%	85.8%	29.0%	43.8%

39 Ibid.
40 Ibid.
41 Ibid.

Source: Kenya National Bureau of Statistics 2018

Except radio where Kiambu Couty leads, Nairobi County is ahead of the rest in other 4 major ICT equipment used (television, 88.0%; mobile phone, 85.8%; computer, 29.0% at 43.8%). Access to and utilisation of computers across all four counts still lags behind all other ICT equipment.

4.2.8 Levels of Income and Wages for Men and Women

More men than women in Kenya have access to regular household income[42] but women would have slightly higher access to non-regular income.[43] The trends manifest in both rural and urban areas. Nationally, the highest monthly regular income (KShs. 12, 284) is rental income followed by pension (KShs. 2,106), and investment income at KShs. 1,265. Non-regular income is the lowest at KShs. 486.

The average monthly rental income received by households in urban areas is three times higher (KShs. 19,976) than it is in rural areas (KShs. 6,088). Female headed households received lower amounts of regular income in all categories except for rental income in rural areas (Table 17).

Table 17: National Income Levels

Residence/ Household headship	Regular Income (in KSh)				Non-regular income (in KSh)
	Saving, interest and investment income	Pension	Rental	Other Types	
National	**1,265**	**2,106**	**12,284**	**925**	**486**
Male headed	1,430	2,285	13,174	993	423
Female headed	687	1,480	9,155	688	707
Rural	**662**	**2,027**	**6,088**	**1,165**	**691**
Male headed	766	2,140	5,532	1,405	600
Female headed	359	1,696	7,715	464	956
Urban	**2,014**	**2,205**	**19,976**	**627**	**232**
Male headed	2,179	2,448	21,804	527	224
Female headed	1,262	1,102	11,677	11,078	271

Source: Kenya National Bureau of Statistics 2018

A key informant in Kiambu County could not estimate income and wages earned by men and women, attributing it to potential inaccuracies while another indicated that on average both men and women earned a monthly income above $100 from sectors such as dairy keeping, crop farming, business, formal employment, informal employment and information technology.

[42] Household income is the aggregate earnings of all household members. It includes all forms of income arising from employment, household enterprises, agricultural produce, rent, pension and financial investment.

[43] Kenya National Bureau of Statistics (KNBS). Basic Report Based on 2015/16 Kenya Integrated Household Budget Survey. Nairobi: KNBS, 2018.

Business which generated over $100 monthly for residents of Machakos County was cited by key informants as the sector that generated the highest levels of income equally for both men and women followed by informal employment ($30), and crop farming and the dairy sector at $10 each.

4.2.9 Main Areas of Household Spending

Female community respondents in Kiambu ranked food, education and housing as three leading areas of household expenditure in a descending order. Food was reported to be costly because it is recurrent while education was said to have remained high despite capitalization by government. They reported that the cost of housing was generally high in the area. Expenditure on healthcare was ranked as the next costly item for households due to irregular payment of NHIF charges and perennial shortages of drugs in public health facilities. Transport to and fro work was placed in the fifth position by the female respondents followed by water which they said was supplied only twice a week, forcing them to buy from vendors at Kshs.10 per 20 litre container, or Kshs. 700 per month for water pumped from boreholes. Gas, charcoal and kerosene were ranked in the 7th position as being equally expensive due to the drastic rise in the cost of fuel. The women regarded expenditure on electricity, clothing and social functions as relatively affordable, depending on one's individual prudence.

Male community members in Kiambu County similarly ranked food as the leading component of household expenditure followed by housing and education. They attributed the high cost of food to its recurrent nature while rent was said to be generally expensive. The men ranked expenditure on transport in the fourth position while water was considered as the fifth most expensive item despite stating that rarely do they experience water related challenges. They ranked electricity as the 6th most expensive item due to the cost of tokens. The men pointed out that clothing was a pressing cost since it is mandatory while health care (which they ranked in position eight) would be a heavy drain on household resources when illness occurs. The cost of social functions was dismissed by the men as discretionary.

Table 18: Main Areas of Household Spending

Household Expenditure Item	Ranking			
	Women	Why?	Men	Why?
Food	1	It's a daily purchase.	1	It is a basic need which is a recurrent expenditure.
Education	2	Our children have to get education.	2	Paying of schools fees for good institution is expensive.
Healthcare	4	Buy drugs or private hospital.	3	Health care is expensive because we purchase drugs.
Housing	8	For those renting it's expensive.	9	Most people have their own houses.
Transport	5	When going to other places especially Machakos hospital.	6	Transport to and from work though it is not much.
Water	3	Buying in most cases.	4	Buying is around 5/= per 20 liter Jerican and it's also scarce.
Fuel	6	To light our houses.	7	Gas, charcoal, kerosene has become expensive.
Electricity			5	Tokens get depleted very quickly.
Clothing	7	We need to look attractive.	8	This is not recurrent so not very expensive.
Social functions (visitors, funerals, church activities, etc.)	9	Not compulsory.	9	Not expensive because these are done by many people.

Food and education were reported by both men and women in Machakos as being the leading components of household expenditure followed by water for women and healthcare for men Table 18. Expenditure on housing and social functions is rated as manageable in the county because people reside in their own houses while social functions are collectively addressed by the community.

Women in Nyandarua County identified healthcare as the most costly aspect of household spending because of purchasing drugs from private chemists which is expensive. They ranked spending on education as the second most expensive because of fees and having to spend on construction of buildings. The women pointed out that food would be the third most expensive item which they attributed to cold whether that spoiled farm produce, forcing them to procure from retail shops at exorbitant prices.

The women explained that the high cost of electricity (4th position) was baffling since they only used it in the evening when they returned from farms. They explained that the cost of transport rises significantly (5th position) during the rainy season when roads become impassable. Those who live in rental houses reported that this would be the 6th component when they rank household expenditure because of the high cost of rent. The women attributed the high cost of fuel (7th position) to frequent price increases while social responsibilities were said to be costly (8th position) due to preparatory processes. They cited the cold weather in the county as the reason why the cost of clothing is high since they have to purchase warm clothes. Water was ranked last (10th position) as a household expenditure item but they lamented that it was mostly contaminated by sewage infiltration. Men in the same county ranked education as the most pressing aspect of household expenditure followed by food, water and healthcare. Clothing, fuel and transport were ranked next in a descending order that concluded with social functions, housing and electricity as the least costly items.

4.2.10 Needs and Priorities

Male and female community respondents in Nyandarua, Kiambu and Machakos counties listed their development needs and priorities as indicated in Table 19.

Table 19: Needs and Priorities

County	Sex	Pairwise Ranking of Needs and Priorities				
		1	2	3	4	5
Nyandarua	Women	Clean water & hospital	Education and roads	Police station	Electricity	Markets
	Men	Water	Skills	Schools	Infrastructure	Employment / Capital
Kiambu	Women	Hospital	Education	Water	Roads	Electricity
	Men	Roads	Sewerage / Garbage	Health	Education	Water
Machakos	Women	Water	Roads	Employment	Education	Healthcare
	Men	Land	Schools	Hospital	Water	Roads

Leaders in Machakos County identified education, water, income generation, roads and maternity units as women's priority needs in a descending hierarchy while capacity building and access to capital were prioritized for men. On their part, leaders in Kiambu stated the foremost needs of women as water, hospitals and schools while men would need schools, industries and an affirmative action fund. Clean water for domestic consumption was equally ranked by leaders in Nyandarua as the highest priority for the community followed by roads and improved drainage /sewerage system. The leaders ranked water for irrigation and access to technical training as the fourth and fifth priorities respectively.

Machakos stakeholders sequentially identified water, roads, employment, education and healthcare, mainly maternity units as women's priorities; while men stressed on land, schools, hospital, water and roads, including capacity building and access to capital for productive businesses. On the other hand, women in Kiambu reiterated emphasis on water, hospitals, schools and roads; whereas men gave more value to roads, sewerage system, health, education and water. Nyandarua marked a unifying interest on safe accessible water by both women and men. But hospitals, education, roads, police station and electricity are within women's priorities; while men also focus on skills, schools and infrastructure.

The society has socialized women, more so those from Kiambu and Nyandarua to invest more of their time in reproductive roles, ranging from all domestic chores, communal work and unpaid agricultural Labour. In Nyandarua however, household farm repairs and works are done by both women and men. Cross-cuttingly, control over water resources is dominantly a men's preserve. The cost of water too varies from county to county with Machakos experiencing a slight average deviation from the rest. Women remain the key collectors of water for regular domestic use.

4.3. Practices and Participation

4.3.1 Division of Labour between Men and Women

Women in Nyandarua County reported that household tasks such as fetching water, cooking, routine shopping, and cleaning fall within their purview of domestic duties. They are also responsible for taking care of livestock as productive work. They indicated that working on the household farm and repair works around the compound are jointly done by men and women. Men in the same county acknowledged that women are solely responsible for cooking and cleaning while men undertake repair works around the compound and provide security. According to the male participants, both men and women are involved in routine shopping but fetching water for the household, working on the household farm and taking care of livestock is mostly done by men with the help of women.

Women in Machakos and Kiambu counties asserted that they mind all reproductive and productive work within the household that includes cleaning, fetching water, cooking, routine shopping, taking care of livestock, working on the farm, and repair works around the compound. Those in Machakos County also intimated responsibility for provision of fees, security for the family and fulfilment of conjugal rights.

In both counties, women with resources hire labour for taking care of livestock.

Women in Machakos County argued that they should be compensated for running the family which is supposed to be done by men. They attributed men's nominal participation in securing the well-being of the family to alcoholism which they said should be curbed by the government raising the cost of alcoholic and lower the cost of fuel so that men can find alcohol unaffordable and elect to stay at home. The women admitted willingness to engage in transactional sex despite the risk of HIV acquisition to cushion their households against extreme indigence. They advised that the government should improve access to affirmative action funds by young women so that they can venture into business and provide for their families. They explained that marriage was only important because it bestowed parental identity on the offspring and earned societal respect for the woman otherwise they would not hesitate to withdraw from relationships that are resource deprived. Men in Machakos County largely concurred with their female counterparts on the gender division of labour as shown in Table 3.

Table 20: Gender Division of Labour

Task	Woman	Man	Woman and Man	Mostly woman, with help of man	Mostly man, with help of woman	Hired labour (House girl / House boy)	Others (Male child, female child)
i. Fetching water for the household?	√						

ii. Cleaning work for the household?	√						
iii. Cooking for the household members?	√						
iv. Routine shopping for the household?	√						
v. Working on the household farm?			√				
vi. Takes care of livestock?			√			√	
vii. Repair works around the homestead?			√				

The only tasks that are solely undertaken by men in Kiambu County, according to male community respondents are repair works around the homestead and counselling young men. They confirmed that women in the community are responsible for cleaning, cooking, fetching water and taking care of livestock. The men said routine shopping is done by both men and women but working on the household farm is the responsibility of hired labour.

4.3.2 Household Care Work

Table 21: Household Care Work

Household Care Task	Nyandarua		Kiambu		Machakos	
	Man	Woman	Man	Woman	Man	Woman
i. Taking care of child aged 0-5 years?	X	√	x	√	x	√
ii. Taking children to school in the morning?	X	√	x	√	x	√
iii. Picking child from school in the afternoon?	x	√	x	√	x	√
iv. Caring for elderly family members?	x	√	x	√	x	√
v. Taking care of the disabled in the family	x	√	-	-	-	-

Female community respondents in Nyandarua, Kiambu and Machakos counties asserted that they were responsible for all the care work in their households which included care of children aged 0-5 years of age, taking and picking children from school and caring for elderly members of society. Women in Nyandarua County also indicated that they are responsible for the well-being of PWDs in the society. Men in Kiambu were affirmative about the care roles played by women (Table 21) but also reported that although taking care of children aged 0-5 years was mostly done by women hired labour (house girl / house boy) would also be used to provide support. On their part men in Machakos County intimated that taking children to school in the morning is done both men and women, and hired labour. Similar was submitted by men in Nyandarua County who admitted that it is mostly women who took and picked children from school but both the man and woman took care of sick elderly members of family.

4.3.3 Participation in Formal and Informal Economy

The analysis of wage employment by industry and sex is presented in Table 22. The only sectors where employment of females was above that of males in 2015 and 2016 was in human health and social work activities, and activities of households as employers; undifferentiated goods - and services-producing activities of households for own use. Agriculture, fishing and forestry slightly above 30% of female labour in 2015 (33.9%) and 2016 (33.1%). Women's representation in the manufacturing industry remained dismal in 2015 (16.7%) and 2016 (16.1%) just the way they marginally participated in water supply; sewerage, waste management and remediation activities in 2015 (20.6%) and 2016 (22.0%). The education reported narrow margins of disparity in numbers of male and female employees in 2015 and 2016 at 47% females and 53% males for both years. Casual employment registered a 7.9 per cent growth with the proportion of casuals to total employees rising from 21.8 per cent in 2015 to 22.8 per cent in 2016.[44]

Table 22: Wage Employment by Industry and Sex, 2015 and 2016 (`000)

Industry	2015		2016	
	Male	Female	Male	Female
Agriculture, forestry and fishing	222.6	114.4	225.1	111.6
Mining and quarrying	12.2	2.2	12.7	2.5
Manufacturing	246.1	49.4	252.1	48.7
Electricity, gas, steam and air conditioning supply	12.6	4.3	12.8	4.5
Water supply; sewerage, waste management and remediation activities	9.2	2.4	9.9	2.8
Construction	103.6	44.4	114.9	48.1
Wholesale and retail trade; repair of motor vehicles and motorcycles	179.3	52.7	185.7	54.2
Transportation and storage	62.5	20.1	64.8	20.8
Accommodation and food service activities	53.7	22.4	54.6	22.8
Information and communication	69.9	35.7	72.9	37.7
Financial and insurance activities	46.3	27.0	47.5	28.2
Real estate activities	3.1	0.9	3.1	1.0
Professional, scientific and technical activities	49.7	18.9	51.4	19.3
Administrative and support service activities	4.7	0.5	4.9	0.5
Public administration and defence; compulsory social security	142.5	79.5	148.3	83.0
Education	267.9	239.5	275.6	249.0
Human health and social work activities	59.9	64.1	61.3	66.3
Arts, entertainment and recreation	4.9	2.0	4.8	2.2
Other service activities	21.3	10.4	22.9	10.3

[44] Kenya National Bureau of Statistics. Economic Survey Report 2017. Nairobi: (KNBS), 2017

Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use	48.4	65.7	48.6	65.8
Activities of extraterritorial organizations and bodies	0.8	0.3	0.9	0.3
TOTAL	1,621.2	856.8	1,674.8	879.6
Total of which casual employees	375.3	164.9	406.3	176.5

Source: Kenya National Bureau of Statistics, 2017

4.3.4 Climate Risks and Impact

Nationally, 61.9% of households reported experiencing a shock of some kind.[45] The proportion of households that reported at least one shock was 68.7 per cent in the rural areas compared to 53.4 per cent in urban areas. A higher proportion of households that experienced shocks were in Kitui (96.0%), Migori (92.0%) and Baringo (91.0%).[46] Among counties included in the gender analysis, Machakos County recorded the highest proportion of households that had experienced at least one shock (70.5%) followed by Nairobi County (58.6%) and Nyandarua County (45.3%). Kiambu County (16.8%) is apparently the least prone to shocks. The specific types of shocks and their different levels of magnitude in each of the counties are shown in Table 23.

Kiambu (22.7%) and Nairobi (10.9%) counties present the highest proportion of households that are lileky to experience robbery / burglary and assault cases as a result of shocks. Machakos (2.2%) and Nyandarua (2.8%) would experience lower levels of similar violence in the wake of shocks.[47]

Table 23: Proportion of households by the first severe shock and County

County	Droughts or Floods	Crop disease or crop pests	Livestock died	Large rise in agricultural input prices	Severe water shortage	Household business failure, non-agricultural	Loss of salaried employment or non-payment of salary	Large fall in sale prices for crops	Large rise in price of food
Machakos	11.5	0.2	5.1	0.2	12.2	4.3	12.4	0.6	8.9
Nyandarua	2.5	15.6	12.6	2.3	0.3	4.5	1.9	1.9	15.0
Kiambu	0.0	0.8	6.5	1.3	0.0	1.6	7.9	2.5	14.6
Nairobi	2.5	1.8	3.2	0.7	4.2	7.3	9.5	1.0	29.9

Leaders in Machakos County identified prolonged dry spells that affect farming activities and flooding that leads to destruction of houses and displacement of families as some of the climate risks impact experienced in the County.

[45] A shock is an event that may trigger a decline in the well-being of an individual, a community, a region, or even a nation. It is worth noting that the same shock could affect different households differently.

[46]

[47]

They underscored that floods destroyed roads, making them impassable. Disease outbreaks due to lack of clean drinking water and emergence of pests that destroy crops during the dry season were also cited by the leaders as adverse manifestations of climatic changes.

The Agriculture Officer interviewed in the County stated that water shortages occasioned by drought forced women to walk long distances in search of water which consumed a lot of their productive time. The officer also indicated that the shrinking forest cover had constrained access to wood fuel which forced women to opt for the more expensive cooking gas. The high cost of farm inputs due to emergence of pests caused by harsh climatic conditions, and inadequate water for irrigation affected men, according to the agriculture officer since they have to divert money meant for family subsistence to purchase pesticide and address crop diseases associated with aridity.

The impact of climate risks on communities in Machakos County were further explained by the Chief Officer Water and Irrigation as leading to decline in the productivity of crops, migration of male labor and decreased household food supply with little or no surplus produce for commercial use. As an equally significant climate risk in the area that affected both men and women, the Chief Officer submitted that floods interfered with water pathways and have adverse effects like sweeping animals and children away.

Generally, floods would lead to loss of human life, damage to property, destruction of crops, loss of livestock, and deterioration of health conditions owing to waterborne diseases.

The Officer also observed that floods cause decreased household food supply and reduced margins of profitability accruing from commercial farming.

Leaders in Kiambu County associated soil erosion as a result dust bowls attendant to drought with vulnerability to disease especially water borne ailments among children and due to desperation for water. A gender expert in the county intimated that drought in the area affected both men and women due to reduced food production and supply in the household, compelling household members to seek work as hired labour that constrained family relationships. The expert also observed that the area experienced floods which precipitated low household food production and intake, disease outbreaks, and children dropping out of school due to lack of fees. Heavy floods would also sweep away animals and affect household livelihood. A water, sanitation and hygiene (WSH) expert in Kiambu County agreed that drought and floods are the major climate related risks in the county whose impact included a decline in crop and animal productivity due to aridity and erosion. Loss of human life, damage to property, destruction of crops, loss of livestock, and deterioration of health conditions owing to waterborne diseases were also cited by the WASH Officer as consequences of floods in the area. Leaders in Nyandarua County identified climate risks and impacts shown in Table 24.

Table 24: Climate risks and their impact

Sex	Specific Climate Risks	Impact
Women	❖ Drought ❖ Floods	❖ The most immediate consequence of drought is a fall in crop production, due to inadequate and poorly distributed rainfall. ❖ Covering long distances to fetch water and look for food ❖ Low rainfall causes poor pasture growth and may also lead to a decline in fodder supplies from crop residues. ❖ Plants and animals depend on water, so when drought occurs, their food supply can shrink and their habitat can be damaged. ❖ Floods damage standing agricultural crops and in some instances also carry away the top soil making the land barren. ❖ Floods lead to loss of human life, damage to property, destruction of crops, loss of livestock, and deterioration of health conditions owing to waterborne diseases. ❖ Diseases outbreaks due to drainage challenges associated with flooding.
Men	❖ Drought ❖ Floods	❖ Plants and animals depend on water, so when drought occurs, their food supply can shrink and their habitat can be damaged. ❖ Drought and famine leads to Male labor migration and may become permanent in some instances. ❖ Drought contributes to decreased household food supply and little or no crops surplus for sale ❖ Floods lead to Male labor migration and may become permanent in some instances. ❖ Floods contributes to decreased household food supply and little or no crops surplus for sale

Expert views in Nyandarua County asserted that floods and drought occurred in the area, leading to low productivity that affects women's and men's capacity to provide for their families, loss of property and increased vulnerability to animal and human diseases.

The expert pointed out that women are compelled to espouse negative coping mechanisms like sex work while men drift away from homes in search of employment opportunities. As significant 30.5% of households would do nothing when floods or drought are experienced due to limited capacity.[48]

4.4 Balance of Power and Decision Making

4.4.1 Decisions on Household Resources, Assets and Finances

The participation of women in Nyandarua County in making decisions related to household resources, assets and finances is dismal, as reported by female community respondents (Table 25). Apparently, the only decisions autonomously made by women are those that relate to the number of children and seeking medical care while decisions to do with management of credit, expenditure of money acquired through credit and institutions that children should attend are jointly decided by men and women. The women said that they had the latitude to decide on the number of children to bear depending on the income of the family and attributed their inordinate role in healthcare to the broad spectrum of childcare responsibilities. They observed that in most cases it is women who attend school meetings. On the contrary, male community participants submitted that men and women play an equal role in making all decisions (Table 25) except deciding on working away from hone which is the prerogative of men.

Table 25: Decision Making

Decision	Only men	Only women	Both the same	Women, with the help of men	Men, with the help of women
Who saves money /bank	✓				
Who keeps money	✓				
Who receives credit / loans	✓				
Who manages credit / loans			✓		
What to spend on			✓		
What to rear / crop to grow	✓				
What to sell and when to do it	✓				
Who participates in extension services?	✓				
Who joins membership and participates in producer organizations?	✓				
Who works outside the community	✓				
How many children to have?		✓			
Where the children attend school			✓		
Who participates in community work	✓				
Who participates in courses and workshops / training activities	✓				
Look for medical help		✓			

According to female community respondents, the visible role of women in decision making in Kiambu County was because men have abdicated their responsibilities.

The women pointed out that the only decision solely made by men had to do with working outside the home while women decided on who receives credit / loans, who manages credit / loans, what to spend on, what to rear / crop to grow, what to sell and when to do it, how many children to have, and who participates in community work.

They said that decision jointly made by men and women included who saves money /bank, who keeps money, who participates in extension services, who joins membership and participates in producer organizations, where the children attend school, who participates in courses and workshops / training activities, and who looks for medical help.

Men in the same county asserted that they unilaterally made decisions to do with what to sell and when to do it, who participates in extension services, who joins membership and participates in producer organizations, who participates in community work, and who participates in courses and workshops / training activities. They reported that women had absolute autonomy to decide on who keeps money, what to spend on, how many children to have, where the children attend school, and seeking medical care. Decisions that the men said are jointly made by both men and women included who saves money /bank, who receives credit / loans, who manages credit / loans, what to rear / crop to grow and who works outside the community. The men said that they recognized women as custodians of family money. However they stated that women are discreet when they acquire credit products. ***"When it comes to loans in most cases you won't know if she took one unless you learn from a third party,"*** a male respondent said during a focus group discussion. The men showed disdain for community work, saying it is for lazy people who loiter around with nothing to do in the community except passing time.

Patriarchal dominance in decision-making in Machakos count was underscore by male community participants indicated that they made virtually all decisions (Table 25) except decisions involving what to rear / crop to grow, how many children to have, and who participates in community work.

Similarly women in the county stated that they were responsible for most of the decisions made in the household except what to sell and when to do it, and where the children attend school. They acknowledged that decisions involving saving / banking money are jointly made by men and women as equals. The women explained that should they cannot afford to contribute towards schools fees needs due their limited resources.

4.4.2 Participation in Community Decision-Making

Most decisions at the community level in Nyandarua County are apparently made by men as reported by community leaders. These include decisions concerning roads and construction of schools. Women attend public participation meetings but it is men who exercise de facto authority when deciding on key actions to be implemented. Broader political participation in the county is also dominated by men and male youth. Women's role is restricted to voting since they cannot join politics without the consent of the spouse.

The participation of women and men in decision-making in Kiambu County was affirmed by leaders at the community level who said that although men make most of the decisions the involvement of women is deliberately sought and meaningfully harnessed.

The leaders further stated that the broader political platform manifests the heavy involvement of men but women remain influential and are significantly engaged due to their numerical strength as a voting bloc.

Men in Machakos County contribute to community development projects and participate in social forums to discuss development matters while women are involved in women committees in markets and development projects, according to community leaders. Both men and women participate in broader political activities. The level of men's and women's involvement in broader political leadership at the county and national levels is presented in Table 26.

Table 26: Participation in Community Decision-Making

Position	Elected		Nominated		Total	% women Elected + Nominated
	Male	Female	Male	Female		
President	1	0	-	-	1	0%
Deputy President	1	0	-	-	1	0%
Senator	44	3	2	18	67	31%
MNA	267	23	7	5	302	9%
WMNA	0	47	-	-	47	100%
Governor	44	3	-	-	47	6%
Deputy Governor	40	7	-	-	47	15%
MCA	1334	96	97	650	2177	34%
Total	**1731**	**179**	106	673	2689	32%

All governors and senators for Kiambu, Nairobi, Machakos and Nyandarua counties are male. Machakos County has 34% female Members of County Assembly (MCAs), while Nyandarua (13), Kiambu (30) and Nairobi County (41) has 33% each. The comparatively better representation of women at the County Assembly level and Senate is attributed to the provision in the Constitution that not more than two-thirds of the members of elective public bodies shall be of the same gender (Article 81) and that Parliament shall enact legislation to promote the representation in Parliament of (a) women; (b) persons with disabilities; (c) youth; (d) ethnic and other minorities; and (e) marginalized communities (Article 100).[49]

4.4.2.1 Institutional Membership

Women in Nyandarua County stated that unlike men who only belonged to political formations and social groupings, women are members of cooperatives, economic empowerment entities such as chamas, political parties and church based groups. They gave examples of cooperatives such as the Moki Dairy Cooperative and the Kinangop Dairy Cooperative. The women reported that men are reluctant to establish or join chamas (table banking groups) and even those that are church based have very few men who are members. On the contrary, men in the county asserted that both men and women belonged to cooperatives, economic empowerment entities, political formations, and social groupings.

In Kiambu County, female community respondents said both men and women belonged to cooperatives, economic empowerment entities, political formations, and social groupings except chamas that only women established and supported. A divergent view by men indicated that membership of cooperatives and political parties was a preserve of men while both men and women participated in economic empowerment initiatives such as chamas and social groupings.

Cooperative societies are not an aspect of social organization in Machakos County according to female respondents who also indicated that they belonged to chamas, political parties and church based groups. The women expressed concern that most men don't permit their women to join chamas but there are also women who fear interaction with others through such gatherings.

49

4.5 Legal Rights and Status

4.5.1 Access to Legal Documents

All women in Nyandarua County have identification cards (IDs) while majority have voter's cards which illustrates their role in civic processes in the county. Some have marriage certificates but few have birth certificates, school/college certificates, title deed and hospital / NHIF card PIN number. None has a passport letter of employment, and milk supply number. They said they don't know how a passport looks like and could not get letters of employment since they were self-employed.

Only women-headed households would be expected to have title deeds to land but only men are expected to access milk supply numbers. A similar picture of women exclusion from accessing legal documents emerged in Machakos County where women said that all of them have voter's cards and IDs, few have birth, school and marriage certificates but none has a passport, title deed, Hospital / NHIF card, and PIN number. They said that title deeds belong to men while they use the husband's NHIF card. ***"My father took mine to the bar in exchange for alcohol. Going for another one is very expensive,"*** one female FGD participant explained the reason why she doesn't have a school certificate. Access to legal documents by women in Kiambu County is comparatively better as shown in Table 27.

Table 27: Legal Documents

Document	All have	Majority have	Some have	Few have	None has	N/A	Comment
Birth certificate		√					
School / college certificate		√					
ID Card		√					
Passport			√				
Marriage certificate			√				
Title Deed			√				
Voter's card		√					
Hospital / NHIF card		√					They have card but they dont contribute
Employment letter				√			Most are self employed
Milk supply number		√					
PIN number		√					

4.6 Knowledge, Beliefs and Perceptions

4.6.1 Traditional Knowledge of Men and Women

Leaders in Machakos and Kiambu County associated women with management of household, organizing social activities, farming and religious matters. In Nyandarua County women are associated with Harvesting of potatoes, Sukuma, transporting and selling cabbages and milking.

However, men in Kiambu County are associated with property inheritance and cultural issues. In Machakos County its construction projects and transport industry example bodaboda.Men in Nyandarua County are associated with Grazing animals, Brokerage of food crops, Packaging vegetables [cabbages, potatoes], Harvesting carrots and cabbages, Transporting milk, counting crops, receiving payment, milking and selling of products.

4.6.2 Common Beliefs and Perceptions

Some of the myths according to female community respondents in Kiambu County about women are that what a man can do a woman can while in Machakos women one must get married even if it's for the sake of the children to inherit a name. In Kiambu County the myths about their men is that they are best with home repairs.

In Kiambu County it is believed that women who are self-accomplished are too arrogant. However, the Kiambu female respondents believe that their men are better leaders than women Values of a woman according to Kiambu female respondents are a real woman takes care of her husband and children while in Machakos a woman builds her home. Men in Kiambu valued to be providers while in Machakos County they are the heads in their homes.

Some of the stereotypes according to female respondents in Kiambu County are that women are like children while in Machakos County they said that a woman's job is reproductive work. Some of the stereotypes associated with men in Kiambu County are that men eat a lot while in Machakos County men say that family planning destroys their bodies and kills their sexual desires.

According to male community respondents in Machakos County some of the myths surrounding women are Women are not to eat the tail of a sheep it makes them unfaithful in their marriages while their counterparts in Nyandarua county claim that Women keep on saying they will go back to their maternal home and they will rarely go. This happens due to domestic quarrels. Men from Nyandarua on the other hand say that Even if women say what, you don't agree immediately, you will take time to think and agree later.

Male community respondents in Machakos County believe that their women are short tempered hence cannot manage people and Politics is not safe for women while in Nyandarua What women say men don't believe immediately, it will take several days for them to believe, don't just believe on mere facial expressions. Men from Machakos County believe that Men are good leaders.

Some of the values highlighted on women by male respondents from Machakos are home keeper while in Nyandarua Being generous, Faithfulness, Honesty, Ambitious and hardworking. Men in Nyandarua were said to have this values generosity, faithfulness,honesty,ambitious and hardworking compared to their counterparts in Machakos who are providers.

Stereotypes associated with women in Machakos County were a woman's place is home while in Nyandarua County they have a mark around their legs due to constant wearing of boots.

Men in Machakos County are attested to being better decision makers than their Nyandarua men who cannot sire children.

4.7 Time and Space (Gender Daily Activity)

This baseline study analysed the major household chores that most women do and the amount of time allocated to each activity. The Female FGD discussants indicated that, apart from doing the household unpaid work, they must supplement the little income available by getting casual jobs such as washing other people's clothes or other manual jobs. With this acute time poverty, women have little time for these economic activities and therefore perpetuating their paucity state. Culture is a key definer of social roles between women and men in Africa. In most cases, it defines roles that brings inequality posing inherent gender dimensions leaving women with time poverty and therefore limited access to economic rights. The allocation of time between women and men in the household and in the economy is a major gender issue in the evolving discourse on time poverty. The baseline analysed the allocation of roles between women and men through a gender daily activity clock exercise and presented in Table 28.

Table 25: Gender Daily Activity Clock

	Activity by gender	
TIME	**MEN**	**WOMEN**
4:00am	Sleeping	Awake
5:00am	Sleeping	Preparing breakfast
6:00am	Taking shower	Escorting kids to school
7:00am	Leaving for work	Start house chores
8:00am	Already at work	Go look for casual labor
9:00am	At work	Go look for casual labor
10:00am	At work	Go look for casual labor
11:00am	At work	Start washing peoples clothes
12:00pm	At work	Washing peoples dishes and houses
1:00pm	At work	Back at home for lunch preparation
3:00pm	At work	Go pick up kids from school
4:00pm	At work	Wash kids uniform and shower them
5:00pm	At work	Go fetch food for supper
6:00pm	Back from work	Help kids with homework
7:00pm	Watching news	Preparing supper
8:00pm	Having supper	Having supper as
9:00pm	Relaxing in bed	Washing clothes and ironing their morning clothes
10:00pm	Already in bed	Polishing their shoes
11:00pm	Wondering why the wife is not coming to bed for sex.	Take a shower
12:00am	Sleeping	Sleeping, Breasting feed, Changing diapers

The study established that in the project focus areas, the patriarchal systems prevailed where men are considered to be superior to women in the society and this aspect defines the gender roles. The allocator of roles is the man who is seen as the head of household therefore controlling women's time and labor. From the gender daily activity clock exercise undertaken with men and women during the survey, it was clear that women and girls are allocated critically important and time-consuming responsibilities, which overburden them with work in the reproduction, economic, household, and community spheres. The functions of reproducing and household chores are considered low-status activities, unremunerated and unrecognized in the national accounting system. This phenomenon affects the status of women in the society where they are seen as only dependent on men.

The women indicated that, apart from doing the household unpaid work, they must supplement the little income available by getting casual jobs such as washing other people's clothes or other manual jobs. With this acute time poverty, women have little time for these economic activities and therefore perpetuating their paucity state.

The discussion indicated that, by the time women are retiring to bed at night, they are usually tired and exhausted.

"The society will be very fast to condemn a woman who leaves her children in the house and goes to look for a job but if a man does the same, it will be considered normal", said a male respondent in an FGD discussion from Mua, Machakos

Women who spend all their time performing these tasks are often considered as "not working." This is a key impediment to their economic security because they are overburdened with activities which have less economic/monetary returns while their male counterparts are working and getting income for their roles.

This has restricted women from having equal economic opportunities with men. Women's economic empowerment and the realization of women's rights to and at work are essential for the achievement of the Beijing Declaration and Platform for Action and the 2030 Agenda for Sustainable Development (United Nations, Economic and Social Council, 2016). This can allow them to pursue education and carriers.

Women dedicated most of their time on reproductive activities at home while men barely put in sufficient hours on economic activities. This was quite evident in Machakos County. Additionally, the study ascertained that women in Kiambu County were industrious than men. This was demonstrated by the number of hours spend on income generating ventures as well as domestic chores. The same pattern was reflected in Nyandarua County.

The participants cited many reasons that had precipitated the discrepancies in time spent by women and men. The following are reasons that were linked to this occurrence:

Age-old customs: Retrogressive customs laced with patriarchal patronage that burdens women with domestic chores. This denies women opportunity to successful engage in commerce and change their economic disposition. Men are expected to be the providers and consequently take up activities that either generate or are perceived to generate income for the households.

Drug –abuse: While men are supposed to assume the position of the sole provider unfortunately most of them have succumbed to rigors of drug abuse especially alcohol. This is more pronounced in Kiambu County; this has forced women in Kiambu to assume dual roles of being providers through trade and performance of domestic chores.

Women are work-horses: Participants especially women were of the view that traditionally women are work-horses and were naturally inclined to outperform men in working at home and in business. This assertion was supported by the fact that men had more hours to rest and partake in leisure activities and idle.

Duty Allocation: Both male and female FGD participants proposed that men and women are built for different duties. For instance, men are built for heavy lifting while women are specialists in performing light duties.

5.0 CHAPTER FIVE

5.1 Introduction

This chapter presents a summary of the findings of study, discussions, conclusions and recommendations.

5.2 Summary of findings

Through the gender analysis carried out it the four counties on building resilience of communities in the athi river catchment area it was noted that women and girls are majorly responsible for finding water resource that their families need to survive - for drinking, cooking, sanitation, and hygiene and by extension they are also responsible for watering the livestock. They walk long distances to collect water, they also brave long queues in water collection points and in drought seasons they are forced to pay exorbitant amounts of money to secure water which is often very time consuming and arduous.

5.3 Discussion of Findings

The discussionws of the findings in this study are based on the outputs from the analyses of the data obtained from the study in the four counties as follows:

5.4 Nyandarua County

5.4.1Water Access and Sources

Most of the households depend on water from shallow wells, roof catchments and rivers. The average distance to the nearest water point is 1.5km. Some of the households have access to piped water while a few households have access to portable water.

5.4.2 Water Resources and quality

The county has inadequate water resources and the situation has been aggravated by the degradation of water catchments leading to reduced ground water recharge. As a result, boreholes have medium to low yields. Main source of water for domestic use is dams, roof catchment and shallow wells. Most of the water used is untreated which poses great health risk to the population. Aberdare forest, providing water resource with high quality, has been destroyed by illegal harvest and deforestation activities. Consequently, the quality and quantity of water from this forest has also deteriorated

5.4.3 Climate risks

The county has witnessed a change in climate over the years. Cases of crop failure e.g. Irish potatoes have been common due to extremely low temperatures at night leading to frost bite.

Livestock farming has also been affected leading to reduced productivity especially for dairy and beef products.

The county over time has experienced climate risks due to increase in unregulated quarrying activities (especially in Olkalou, Kipipiri and Kinangop) and sand harvesting in Miharati; silting of dams and rivers and deforestation of the catchment areas.

Recommendations for Nyandarua County

NEMA in collaboration with other stakeholders to enhance mechanisms for:

- Enforcing the laws and regulations regarding encroachment of river banks, forests, wetlands and other riparian reserves by encouraging both women and men's constructive participation and opinions in decision making.
- Repossessing illegally acquired forests, riverbanks and wetlands where they have been grabbed to attain a friendly and clean environment with maximum vegetation and catchment capacity for people's use, especially women.
- Encourage active participation of men and women in tree planting in the project area to boost their ownership spirit and sustained community driven zeal for environmental conservation and management.

5.5 Machakos County

5.5.1 Water Sources and Distances

The overall average distance to the nearest water source is 6 kilometers. Climate change factor has played a major role in increasing the average distance to the nearest water source especially in rural areas. The prolonged dry season for instance, has led to drying up of rivers, springs, boreholes, wells and dams subsequently increasing the average distance to the nearest water source.

5.5.2 Water Scarcity

Water scarcity affects women and girls both in the rural and urban areas of the county since they are charged with the responsibility of ensuring the household needs for water are met. In the process, they are denied the opportunity to engage in other economic activities and schooling. Most of the existing water facilities are old and dilapidated and require rehabilitation and augmentation in order to meet the present and future demands of the fast growing population of Machakos.

The adequacy, equity and reliability of government rural water supply projects in the County have deteriorated due to; inadequate budgetary provision, facilities have not been upgraded to cope with increasing demand, and technical performance has declined with increasing age of equipment and inadequate maintenance.

5.5.3 Sand Harvesting

Rampant, Uncontrolled and unsustainable sand harvesting has led to substantial decrease in water holding capacity of rivers. This has caused drying of rivers resulting to water scarcity for both domestic and commercial use. Sand harvesting has led to severe environment degradation leading to change in the regime of some of the rivers and loss of retention capacities of some of the seasonal rivers. A good example of affected water mass is River Thwake. Mwania and Mania river which are the major sources of water for Machakos have also been affected.

5.5.4 Climate Change

Negative impacts are prolonged periods of drought, erratic rainfalls and rise in average temperatures which have led to low agricultural production.

Most of the County residents use unsustainable cooking methods such as firewood and charcoal as the major source of fuel. This has resulted to deforestation in most areas leading to rampant and expansive soil erosion.

Recommendation for Machakos County

- Proper Licenses for Sand harvesting governed by the Sand harvesting guidelines to address environmental concerns including those that attach on women and men's livelihoods in order to avoid haphazard scooping that would lead to distraction of the environment.
- Involve women and men through support initiatives that will increase water supply by inculcating effective harvesting and storage methods. For example, promoting rain water harvesting to guarantee an improved water source for a majority of the households.
- Support the adaptation strategies including livestock and crop zoning, and introduction of drought resistant crops and livestock breeds benched on gender sensitivity and responsiveness approach.

5.6 Kiambu County

5.6.1 Water sources and access

The average distance to the nearest water source averages between 0.5 -1km. There is major degradation of forests and Rivers. Kinale and Kieni forests have faced major deforestation due to population demand for shelter and fuel and encroachment for farming demand. In addition rivers like Athi River have experienced much pollution through dumping of waste which comprises of effluents, agricultural chemicals and industrial waste.

The major contributors to environmental degradation are; increased population leading to massive deforestation and encroachment of water catchment areas. In addition, industries have emissions that have led to lot of air and water pollution especially in Thika. Farming has also led to pollution due to the release of various agrochemicals in the water sources.

5.6.2 Climate risks impact

The County are manifested through changes in rainfall patterns, distribution and amounts. The county experiences less drought and flooding effects in comparison to other counties. In drought periods, the water levels within the rivers and wetlands are reduced resulting in ripple effects of reduced water flows hence less water supply within the county for industrial, household and agriculture. This affects women and girls because they are charged with the responsibility of collecting water. From the FGD discussions the county has several water collection points that are on average about 0.5-1km from the dwellings. They brave the long queues and pay an average amount of between Kes 5-20 for a 20litre jerican depending with the water need at that particular season.

Kiambu County is currently experiencing fast population growth with major land use changes being conversion of land from agriculture to urban centers and settlement. This has put a strain to the county water resources, e.g. riparian and wetland encroachment through construction and farming activities. There is increased environmental degradation as a result of the soil erosion, deforestation, and changes of wetlands into other landforms, through encroachment.

5.6.3 Challenges on Water Resources

Some water resources are highly polluted, making them unsafe to use. This is due to pollution of water sources from effluent waste, disposal of waste water from treatment plants, raw effluent discharges in urban and semi urban settlements, and agro-chemical in the agricultural areas. There is high depletion of vegetal cover due to deforestation within the water resources, including wetlands within the county. This is attributed to population pressure and increased demand for both environmental and forest resources especially for timber. Depleted vegetation cover therefore exposes water resources to high evapotranspiration rates, more runoff and less/low recharge of ground water aquifers.

There is high rate of depletion of ground water resources due to over extraction, low recharge and environmental degradation. Majority of the water resources have challenges of encroachment of riparian zones and wetlands. Some of the riparian lands are under farming of agricultural crops, loosening the soils and clearing the vegetation, hence soils erosion and increased levels of evapotranspiration.

Recommendation for Kiambu County

- Regulate water vendors to ensure quality and guard against exploitation of consumers while giving gender sensitive focus on women's concerns and challenges.
- Invest in sustainable access to clean and safe water through effective and gender-inclusive approaches in management of the water resources.
- Enforcement of the laws and regulations barring encroachment, and pursuant to the protection of wetlands, forests and riparian reserves to stop and reduce cases of water pollution to reduce vulnerability rate, especially among women who interact the most with water.
- Encouraging women's participation in conservation efforts and management of available water resources.
- Improve compliance on licensing of water harvesting and use considering the water needs of women.
- Awareness and capacity building of the community and stakeholders on the most appropriate techniques of water harvesting and storage to address women's concerns and needs including other specific gender issues.
- Upscaling the representation and participation of women in project planning and representation.
- Empowerment on the constitutional rights of both women and men, that any person is entitled to land ownership and utilization against retrogressive cultural beliefs and perceptions.

5.7 Nairobi County

5.7.1 Water sources and access

The main sources of water for the residents in Nairobi County are from Sasumua Dam in Nyandarua, Kikuyu Springs, Ruiru Dam, Thika Dam and Ngethu water works. Although Nairobi River is permanent, its water is unsafe for human consumption. There are residents that use borehole water, water kiosks especially those in slums, wells and roof catchments. Over 80 per cent of the residents have access to piped water. On average, 52.5 per cent and 24.7 per cent of the population takes 0 and 1-4 minutes respectively to fetch water. Only 0.9 per cent of the population takes 30-59 minutes to nearest water point. (Nairobi County Integrated Development Plan, 2018). Consequently, this is a serious setback to residents, especially women who interact more with, and use water the most for domestic and other purposes. Women have to move distances during scarcity times, at times in risky conditions thus escalating their vulnerability to insecurity.

Women's access safety in access to water services is basically limited as attributed to security fears owing to the fact that slums are densely populated with poor construction planning. Some path ways or routes to water points are narrow and act as hot spot joints for criminal activities. Time when water becomes available stands as the primary determinant to women and children's safety. Such conditions are indeed part of the contributing drivers to sexual gender based violence, for example; defilement, rape, sodomy, gang rape, attempted rape and other kinds of indecent acts, mostly against women and girls. HIV, some chronical diseases with inclusion of sexually transmitted infections are resultant health effects of the vices.

5.7.2 Water Resources

Nairobi County has no main water tower; most of the supply is from the Tana Basin and is pumped to the City from distances of around 50 Km. This bulk water-supply is not reliable during periods of drought, and is also endangered by siltation of the reservoir due to deforestation in the catchment areas. The supply problem is further aggravated by the poor state of the distribution system, which results in about 50 per cent losses due to leakage, illegal connection and inefficient and wasteful use of water by some consumers. (Nairobi County Integrated Development Plan, 2018)

5.7.3 Water and Sanitation

Nairobi County has no major water tower and relies on other neighboring counties within Tana Basin for its water supply which is around 50 Km from the city. This bulk water-supply is not reliable during periods of drought, and is also endangered by siltation of the reservoir due to deforestation in the catchment areas. The supply problem is further aggravated by the poor state of the distribution system, which results in about 38 per cent losses due to leakage, illegal connection and inefficient and wasteful use of water by some consumers. (Nairobi County Integrated Development Plan, 2018)

5.7.4 Water management

The responsibility for water supply and sewerage in Nairobi is shared between an asset holding company, Athi Water Services Board (AWSB), and an operating company, the Nairobi City Water and Sewer Company. Also some of water is managed by CBOs and NGOs. Water and sewer services in Nairobi city are provided by the Nairobi City Water & Sewerage Company Ltd. Service standards are set and monitored by a national water regulatory agency called the Water Services Regulatory Board (WASREB). There is poor quality and inadequate water supply in Nairobi. Only 40% of those with home connections receive water continuously. (Nairobi County Integrated Development Plan, 2018)

Recommendation for Nairobi County

5.8 Conclusion

Protection of Athi river catchment area project is an important establishment in conservation of water resources and ecosystems management. Sustainable and well-managed water supplies are therefore essential to achieving access to water for all, and for ensuring the maintenance of water in the interests of ecological balance and the needs of future generations.

5.9 Recommendations

- Invest in sustainable access to adequate and clean water for domestic use and commercial purposes through gender sensitive approaches.
- Invest in modern agri-business technologies that recognize and develop further the unique individual and corporate resources of men and women.
- Ensure expansion of water supply sources to keep pace with ever growing population needs of women and men with similar consideration of girls and boys.
- Embrace gender affirmative representation while strengthening water governance sector to ensure reduction of water losses by use of metered zoning system under monitored mandate of the Nairobi City County Water Company Services and other accredited agencies. .
- Strengthen Water policies and focus on the management of water and not just the provision of water
- Strengthen the agency of women and human rights consciousness of men to remedy gender and social norms and practices.
- Embrace risk informed livelihood projects that appreciate the unique vulnerabilities and coping capacities of men and women.
- Research and surveillance to monitor and document the differential impact of climatic risks on men and women.

6.0 Glossary of terms

Gender: refers to the roles, behaviors, activities, and attributes that a given society at a given time considers appropriate for men and women. In addition to the social attributes and opportunities associated with being male and female and the relationships between women and men and girls and boys, gender also refers to the relations between women and those between men. These attributes, opportunities and relationships are socially constructed and are learned through socialization processes. They are context/ time-specific and changeable. Gender determines what is expected, allowed and valued in a woman or a man in a given context.

Gender analysis: is a critical examination of how differences in gender roles, activities, needs, opportunities and rights/entitlements affect men, women, girls and boys in certain situation or contexts. Gender analysis examines the relationships between females and males and their access to and control of resources and the constraints they face relative to each other. A gender analysis should be integrated into all sector assessments or situational analyses to ensure that gender-based injustices and inequalities are not exacerbated by interventions, and that where possible, greater equality and justice in gender relations are promoted.

Gender Equality: This refers to the equal rights, responsibilities and opportunities of women and men and girls and boys. Equality does not mean that women and men will become the same but that women's and men's rights, responsibilities and opportunities will not depend on whether they are born male or female. Gender equality implies that the interests, needs and priorities of both women and men are taken into consideration, recognizing the diversity of different groups of women and men.

Gender mainstreaming refers to a strategy for implementing greater equality for women and girls in relation to men and boys.

Mainstreaming a gender perspective is the process of assessing the implications for women and men of any planned action, including legislation, policies or programs, in all areas and at all levels. It is a way to make women's as well as men's concerns and experiences an integral dimension of the design, implementation, monitoring and evaluation of policies and programs in all political, economic and societal spheres so that women and men benefit equally and inequality is not perpetuated. The ultimate goal is to achieve gender equality.

Gender roles: These often determine the traditional responsibilities and tasks assigned to men, women, boys and girls (see gender division of labor). Gender-specific roles are often conditioned by household structure, access to resources, specific impacts of the global economy, occurrence of conflict or disaster, and other locally relevant factors such as ecological conditions. Like gender itself, gender roles can evolve over time, in particular through the empowerment of women and transformation of masculinities.

Gender-responsive budgeting or GRB: is a method of determining the extent to which government expenditure has detracted from or come nearer to the goal of gender equality. A gender-responsive budget is not a separate budget for women, but rather a tool that analyzes budget allocations, public spending and taxation from a gender perspective and can be subsequently used to advocate for reallocation of budget line items to better respond to women's priorities as well as men's, making them, as the name suggests, gender-responsive.

Gender disaggregated data: refers to any data on individuals broken down by sex. Gender statistics rely on these sex-disaggregated data and reflect the realities of the lives of women and men and policy issues relating to gender. Data collected and tabulated separately for women and men. They allow for the measurement of differences between women and men on various social and economic dimensions and are one of the requirements in obtaining gender statistics.

Gender action plan: A Gender Action Plan (GAP) is a key gender mainstreaming tool and mechanism for ensuring gender-inclusive design and implementation of projects. Gender Action Plan (GAP) forms the basis for operationalizing the results and recommendations of the gender analysis. It contains specific gender elements to be considered in the project design and during the implementation of project measures and activities. Moreover, it helps to monitor implementation of these measures and activities. The existing institutional capacities on Gender of the executing entities will oversee the implementation of the gender action plan during the project cycle. NEMA has an existing gender policy that will further inform the implementation of the gender action plan.

Source: (https://trainingcentre.unwomen.org/mod/glossary/view.php?id=36)

ANNEXURE 1 GENDER ACTION PLAN FOR ENHANCING THE RESILIENCE OF COMMUNITIES AND ECOSYSTEMS IN THE ATHI RIVER CATCHMENT AREA

Output	Gender responsive actions	Performance indicator	Time frame (Financial Year)					Responsible
Component 1: Enhanced capacity of Government institutions to manage water resources			1	2	3	4	5	
1.1 Establishment of National Information Center for Integrated Water Resource Management	Constitute a gender inclusive committee to gather and disseminate relevant, accurate and timely climate information from local water catchment communities and county governments.	51% of women and 49% men involved in information gathering, design stages and communication strategies of the project development.						NEMA, KMD WRA
	Target men and women in the development of Early Warning systems that determine climate-driven vulnerabilities and effective adaptation options	51% of women and 49% men imparted with climate information on early warning systems and other risk reduction measures through strengthened engendered participation in setup of the systems.						NEMA, KMD
	Develop an integrated database for weather and water resources-related information by ensuring that men and women receive seasonal forecast in a timely manner.	51% of women and 49 men receive real time seasonal weather forecast information for collective and gender-specific adaptive mechanisms.						NEMA, KMD WRA
1.2 Establishment of modernized hydro-meteorological observation, monitoring, and testing systems and related networks	Coordinate with gender focal points of relevant ministries and institutions during the Installation and rehabilitation of surface hydro-meteorological monitoring and network systems.	50% of gender focal points engaged during the rehabilitation and installation of surface hydro-meteorological monitoring and network systems showing disaggregated information on the projected percentage level of women						NEMA, KMD WRA.

		and men’s access to and utilization of the network systems.						
1.3 Training and capacity building programs	Enhance an understanding on the link between gender and efficient water management for water sector professionals. Promote gender balanced staffing during the technical training on operation of monitoring and information systems	51% of female and 49% male government staffs trained on water sector and resources management through efficient operation of monitoring and information systems.						NEMA, WRUAs, CECs, County Adaptation Committees, CBOs.
Component 2. improving and rehabilitating prioritized water infrastructure								
2.1 Develop and upgrade water sources and infrastructure for domestic and commercial use	Ensure constructive participation and consideration of women’s employment needs in the rehabilitation and construction of water storage and supply infrastructure in the community.	Improved level of access by women to employment opportunities created by the project through rehabilitation and construction of water storage and supply infrastructure.						NEMA, National & County Govt
	Engage in consultative discussions with men and women on the rehabilitation process at every stage, with specific interventional considerations to issues affecting women as the more vulnerable gender, e.g. access to water points should be reviewed prior to rehabilitation/new construction of water points.	At least 51%of women benefiting from improved access to safe and reliable water for domestic and commercial use with reduction in distances covered to water collection points towards enabling reduced water charges.						NEMA, National & County Govt

	Ensure women have equal and equitable access to employment opportunities generated by new civil works during the rehabilitation of water conservation structures.	Increased number of women employed within the new civil works for improved livelihoods.						NEMA, National & County Govt
2.1 Support conservation of water catchment areas	Ensure public awareness and inclusive participation of men and women in the water catchment protection and management processes.	Improved awareness level and participation of women and men in the conservation of water catchment areas. Increased number of tree nurseries established and number of trees planted by women						NEMA, National & County Govt WRUAs CECs
Component 3. Strengthen institutional and regulatory framework.								
3.1 compliance of water regulations within the project area	Ensure that policies, institutions, coordination mechanisms and regulatory frameworks are gender sensitive and responsive through inclusive and active participation of women. Review and consider existing legal and regulatory frameworks in the water sector on their gender-sensitive and responsiveness on water management status to instigate policy geared enforcements and repealing.	More gender-friendly policies, institutions, coordination mechanisms and regulatory frameworks put in place and effectively implemented to enhance climate resilience capacity of the locals, more affirmatively for women. Improved compliance and adjustment level by existing water quality regulations, policies and laws to the gender responsive needs of women and men.						NEMA, National & County Govt

3.2 Development of County Environment Action Plans (CEAPs) for water catchment conservation as an adaptive action	Ensure maximum participation of women in the development of County Environment Action Plans (CEAPs) for water catchment conservation.	At least 51% of women involved in all the entire development process of CEAPs						NEMA, County Govt.
3.3 Development, review and implementation of sub-catchment management plan	Ensure maximum engagement and active participation of men and women in the development, review and implementation of sub-catchment management plans (SCMPs)	At least 51% of women and 49% men involved in the development, review and implementation of sub-catchment management plans.						NEMA, County Govt.

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ANNEX8i: GENDER ACTION PLAN FOR ENHANCING THE RESILIENCE OF COMMUNITIES AND ECOSYSTEMS IN THE ATHI RIVER CATCHMENT AREA

Output	Gender responsive actions	Performance indicator	Time frame (Financial Year)					Responsible
Component 1: Enhance hydrological and meteorological monitoring system to support decision making, planning and policy development in water and climate change sector			1	2	3	4	5	
1.1 Support establishment of National Information Centre for Integrated Water Resource Management	Constitute a gender inclusive committee to gather and disseminate relevant, accurate and timely climate information from local water catchment communities and county governments.	51% of women and 49% men involved in information gathering, design stages and communication strategies of the project development.						NEMA, KMD WRA
	Enforce 2/3 gender principle in Policy dialogue committee for developing data sharing protocols (between NEMA, WRA, KMD)	2/3 gender principle adhered to in all technical policy meetings						NEMA, KMD WRA
	Target men and women in the development of Early Warning systems that determine climate-driven vulnerabilities and effective adaptation options	51% of women and 49% men imparted with climate information on early warning systems and other risk reduction measures through strengthened engendered participation in setup of the systems.						NEMA, KMD WRA
	Develop an integrated database for weather and water resources-related information by ensuring that men and women receive seasonal forecast in a timely manner.	51% of women and 49 men receive real time seasonal weather forecast information for collective and gender-specific adaptive mechanisms.						NEMA, KMD WRA

1.2 Establish modernized hydro-meteorological observation, monitoring, and testing systems and related networks	Coordinate with gender focal points of relevant ministries and institutions during the Installation and rehabilitation of surface hydro-meteorological monitoring and network systems.	50% of gender focal points engaged during the rehabilitation and installation of surface hydro-meteorological monitoring and network systems showing disaggregated information on the projected percentage level of women and men's access to and utilization of the network systems.						NEMA, KMD WRA.
	Upscale the representation and participation of women in project planning and representation.	Increased representation and participation of women during project implementation						NEMA, KMD WRA.
1.2.4 Setting up of a RANET broad casting station in Machakos County	Air specific information/programs on water security and climate vulnerability targeting women and girls	30% of airtime dedicated to airing information/ programs on water security and climate vulnerability targeting women and girls						NEMA, KMD WRA.
1.3 Training and capacity building programs	Enhance an understanding on the link between gender and efficient water management for water sector professionals. Promote gender balanced staffing during the technical training on operation of monitoring and information systems Empowerment on the constitutional rights of both women and men on access and sustainable utilization of the water resources	51% of female and 49% male relevant project stakeholders trained on water sector, climate change and resources management towards efficient operation, monitoring and effective information systems.						NEMA, KMD WRA

1.3.1 Support capacity building and training for government institutions with water and climate change mandates within the project area.	Ensure gender parity in capacity building training for targeted government institutions	Gender parity and 2/3 gender rule achieved in training and capacity building of government institutions						NEMA, KMD WRA
1.3.2 Support capacity building and training for relevant water governance structures and communities within the project area (Water Resource Users Associations -WRUAs, County Environment Committees - CECs, County Adaptation Committees, Community Based Organizations - CBOs among others)	Ensure gender parity in identification of trainees	Gender parity and 2/3 gender rule achieved in training and capacity building of community groups, adaptation and water governance structures.						NEMA, KMD WRA
Component 2. Improve climate water resilience by building, enhancing and rehabilitating prioritized water infrastructure and implementing conservation activities in the catchment								

2.1 Develop and upgrade water sources and infrastructure for domestic use considering present and future impacts of climate change.	Ensure constructive participation and consideration of women's employment needs in the rehabilitation and construction of water storage and supply infrastructure in the community.	Improved level of access by women to employment opportunities created by the project through rehabilitation and construction of water storage and supply infrastructure.						NEMA, National & County Govt
	Engage in consultative discussions with men and women on the rehabilitation process at every stage, with specific interventional considerations to issues affecting women as the more vulnerable gender, e.g. access to water points should be reviewed prior to rehabilitation/new construction of water points.	At least 51%of women engaged in consultative discussions for improved access to safe and reliable water for domestic use with reduction in distances covered to water collection points towards enabling reduced water charges.						NEMA, National & County Govt
	Ensure women have equal and equitable access to employment opportunities generated by new civil works during the rehabilitation of water conservation structures.	Increased number of women employed within the new civil works for improved livelihoods.						NEMA, National & County Govt
	Ensure Women's safety in the access and use of the water conservation structures.	Increased number of women regularly accessing the water conservation structures leading to improved livelihoods						

2.1.1: Construct water storage and supply infrastructure	Structural design of the water storage and supply infrastructure to incorporate gender sensitive design to accommodate women and girls who are mainly involved in search of water for domestic and other uses	Gender sensitive design including draw off pipes, cattle troughs and overflow channels incorporated to improve safety, accommodate women and people with disabilities						NEMA, WRA.
	Constitute gender sensitive project supervision committees for all water assets contracted for rehabilitation	2/3 gender representation in all project management committees at project level						NEMA, WRA.
2.2 Support conservation of water catchment areas as adaptation to climate change measure.	Ensure public awareness and inclusive active participation of men and women in the water catchment protection and management processes to enhance the ownership spirit and sustained community driven zeal for environmental conservation and management.	Improved awareness level and participation of women and men in the conservation of water catchment areas. Increased number of tree nurseries established and number of trees planted by women						NEMA, WRA, National & County Govt WRUAs
Component 3. Strengthen water and adaptation planning, institutional and regulatory framework to respond to changing climatic conditions								

3.1 Enhance compliance of water regulations within project area and improve adaptation planning.	Review the existing legal and regulatory frameworks in the water sector on their gender-sensitive and responsiveness on water management status to instigate policy geared enforcements and repealing.	The number of existing legal and regulatory frameworks in the water sector reviewed on their gender-sensitivity and responsiveness on water management						NEMA, WRA, National & County Govt
	Encourage both women and men's constructive participation and opinions in decision making toward compliance to laws and regulations regarding encroachment of river banks, forests, wetlands and other riparian reserves	Increased collective compliance of both women and men to existing to laws and regulations regarding encroachment of river banks, forests, wetlands and other riparian reserves						
3.1.2: Sensitize the regulated communities to enhance compliance	Ensure 50% gender representation during sensitization forums for regulated communities	At least 50% women representation achieved during sensitization forums for regulated communities						NEMA, WRA,
3.2 Enhance compliance of water regulations within project area and improve adaptation planning.	Ensure maximum participation of women in the development of County Environment Action Plans (CEAPs) for water catchment conservation.	Increased participation of women and men in planning and implementation of CEAPs						NEMA, County Govt.

3.3 Development of 8No. Sub-Catchment Management Plans (SCMPs) including climate change adaptation	Ensure maximum engagement and active participation of men and women in the development, review and implementation of sub-catchment management plans (SCMPs)	Increased participation of women and men in planning and implementation of SCMPs.						WRA, County Govt.

GAP notes

1. A Technical Assistance budget allocation for a gender expert is requested for under the AE fee costs
2. The GAP will be developed further by the Gender Expert and should be complete within the first year of project implementation
3. The gender expert will support mainstreaming and implementation of the GAP during the project period
4. The budget for implementing the GAP will be drawn from the M&E cost as planned in the proposed detailed budget